



Ehrenfeucht-Fraïssé Games for Temporal (and) Epistemic Logics

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Abstract

This thesis exalts the celebrated Ehrenfeucht-Fraïssé (EF) game. While the original game is a tool for proving the inexpressibility of properties in first-order logic, its structure gives us a uniform framework for proving inexpressibility in various other logics. We survey games in the literature following this framework and present novel EF games for the branching time logic CTL^* , various temporal epistemic logics and transitive closure logics. Our new games are used to prove new expressibility results for temporal epistemic logics. The main appeal of EF games is being able to provide intuitive and simple proofs by abstracting complex or tedious details into the game's semantics. We give original reformulations of classical concepts and results for expressibility in terms of EF games, such as Gabbay's separation property and the equivalence of logics. Throughout this thesis, we also provide multiple original examples of games being played to prove known inexpressibility results.

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Aries' Paw Print

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Chapter 0

Introduction

‘Time is money’ – Benjamin Franklin

‘Money is the root of evil’ – Mark Twain

‘Knowledge is power (of 2)’ – Francis Bacon

‘Time and knowledge is evil’ – Ronald Chiang

Motivation

English allows us to express extremely complex ideas. It’s good enough to be used for this thesis and countless other ideas every day. Even still, we can come up with concepts that cannot be captured by English. How do we know this? If an idea is expressible, we can show this by using English to express it. How do we show that an idea cannot be perfectly described? Informally, a logic consists of a language and its associated meanings. We can ask the same questions about a logic.

In first-order logic, we have various tools that can help us prove the inexpressibility of a concept. Many of these tools, such as the Compactness Theorem and the Löwenheim-Skolem Theorem [Mal36], cannot be used in various restrictions and extensions of first-order logic, such as second-order logic [Vää24; Zac25].

The Ehrenfeucht-Fraïssé game [Fra54; Ehr61], originally made to prove inexpressibility results for first-order logic, provides a uniform framework which can be adapted to analyse other logics [Imm82; Bar77]. Adding different types of moves allows us to look at extensions of first-order logic. Such is the case with the EF game for second-order logic [Tur84]. By restricting the structures the game is played on, we can also restrict the class of structures we are proving

inexpressibility for. Even with these changes to the game, the typical inexpressibility proof follows the same style as those for first-order logic.

Similar games have been discovered for other families of logics with similar appeals. Most notably, Stirling’s bisimulation game [Sti99] for modal logic fits the uniform framework of EF games. Modal logics of interest include epistemic logic, for reasoning about the knowledge of individuals, and unary temporal logic, for reasoning about when events occur. The bisimulation game has shown its adaptability by being able to analyse each of these logics. Extensions of various modal logics can also be analysed with extensions of the game. For example, games have been created for extensions such as common knowledge [BMS98], public announcement logic [KB04], and linear temporal logic [EW00]. In each of these games, an inexpressibility proof is carried out in the same manner as that of an EF game, showing the uniformity of this framework.

We focus on temporal logics due to their key application in computer science. The general idea is that the execution of a program can be viewed as a timeline of events where each point represents a step in its execution. Temporal logics are extremely useful for specifying the properties that you would want to prove about a system or program, such as those of the form “[bad thing] never happens” or “eventually, [good thing] happens”. We can extend temporal logics to form temporal epistemic logics, which allow us to reason about how knowledge in a system evolves over time. The goal of this thesis is to create novel EF games for temporal epistemic logics and play some games.

The greatest selling point of EF games is the ability to abstract concepts into game semantics and simplify proofs. Rather than working with logics directly, we can work with their associated games. As playing games is a notion that we are intuitively familiar with, the process of coming up with and writing an inexpressibility proof also becomes more intuitive. The thought process behind coming up with a proof can be reduced to moving game pieces on a board. Classical results can also be reframed in terms of the types of moves and strategies of EF games. Concepts such as the equivalence between two logics can be characterised by showing that winning in one EF game is equivalent to winning in the other.

Outline

Chapter 1 mainly covers background in model theory, such as what is required to define a logic and what it means for a property to be expressible. We also motivate the use of Ehrenfeucht-Fraïssé (EF) games by showing that classical theorems for proving inexpressibility do not carry over from first-order logic to other classical logics.

In Chapter 2, existing EF games are introduced for first-order logic, second-order logic and first-order logic with finitely many variables. We provide various examples of each game being played and how these games are used to prove that a certain property is inexpressible.

Chapter 3 introduces a framework for defining a modal logic and its associated EF game. This framework is used to define epistemic logic. We extend epistemic logic with common knowledge to cover the background necessary for defining temporal epistemic logics. This chapter also discusses the connections between modal logic and first-order logic.

In Chapter 4, we discuss two main models of time: one where the future is predetermined, known as linear time, and another where the future branches into multiple possibilities. We introduce EF games for logics in linear time and present two original EF games for logics in branching time. Various classical concepts and results for temporal logic are given original characterisations and proofs through these games.

Chapter 5 defines several temporal epistemic logics. We further broaden the literature by providing a novel EF game for each of these logics. We use these games to explore a natural extension of expressibility questions asked about temporal logics.

Chapter 6 introduces first-order logic with transitive closure and an associated novel EF game. Using this game as inspiration, original improvements are made to existing EF games introduced in previous chapters.

Chapter 1

Classical Logic

We first introduce basic concepts from model theory, which studies the sentences of a language and how they can be interpreted. To familiarise ourselves with these definitions, this chapter includes applications to first-order logic statements in graph theory and group theory. We mainly follow the theory from [Lib04; Mar10]. These concepts give us a framework to define the other logics presented in this thesis.

1.1 First-order Logic

A *logic* is characterised by a set of valid *formulae* and a set of *structures* that assigns truth to particular *sentences*. To define what a valid formula is, we require a *signature*. A signature contains the available symbols for constants, relations and functions.

Definition 1.1. A *signature* σ is a set of the form $\{c_1, c_2, \dots, P_1, P_2, \dots, f_1, f_2, \dots\}$, where c_i are constant symbols, P_i are relation (or predicate) symbols and f_i are function symbols. Each relation and function symbol has an associated number of arguments. A symbol that takes k arguments is called *k-ary*. For 1, 2 or 3 arguments, we say *unary*, *binary* and *ternary*, respectively.

Definition 1.2. A signature σ is *relational* if there are no function symbols.

Example 1.3. When working with ordered rings, we can use the signature $\{0, 1, <, +, \times, -\}$. Our formulae would use constant symbols 0 and 1, a binary relation symbol $<$, binary function symbols $+$ and $\times$, and a unary function symbol $-$.

Given a signature, we can define the set of first-order formulae, denoted as FO. Variables are generally denoted by $x, y, z, \dots$. First-order logic uses quantifiers over variables, but other

logics may use different operators and quantifiers when defining what a valid formula is.

Definition 1.4. The set of first-order *terms* over σ are defined as follows:

- Each variable x and each constant symbol c is a term.
- If $t_1, \dots, t_k$ are terms and f is a k -ary function symbol, then $f(t_1, \dots, t_k)$ is a term.

The set of first-order *formulae* over σ are defined as follows:

- If t_1, t_2 are terms, then $t_1 = t_2$ is a formula.
- If $t_1, \dots, t_k$ are terms and P is a k -ary relation symbol, then $P(t_1, \dots, t_k)$ is a formula.
- If φ and ψ are formulae, then $\varphi \wedge \psi$ and $\neg\varphi$ are formulae.
- If φ is a formula, then $\exists x\varphi$ and $\forall x\varphi$ are formulae.

Other common boolean connectives can be defined through the given operators. In particular, we have $\varphi \vee \psi$ defined as $\neg(\neg\varphi \wedge \neg\psi)$, $\varphi \rightarrow \psi$ as $\neg\varphi \vee \psi$, and $\varphi \leftrightarrow \psi$ as $(\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)$. Formulae in logic help precisely express properties or statements from natural language. First-order logic has the existential quantifier $\exists$ and universal quantifier $\forall$ to capture the idea that properties hold for some or all members of a *universe*.

Example 1.5. When studying graphs, we can use the signature $\sigma = \{E\}$, where E is a binary relation, to represent edges. That is, we may view $E(u, v)$ to mean that there is an edge between u and v . We could then use the formula $\exists v\forall u(\neg(u = v) \rightarrow E(u, v))$ to mean “There exists a dominating vertex”.

Definition 1.6. The *quantifier rank* of a formula $\text{qr}(\varphi)$ is its depth of quantifier nesting, calculated as follows:

- $\text{qr}(t_1 = t_2) = \text{qr}(P(t_1, \dots, t_k)) = 0$.
- $\text{qr}(\varphi \wedge \psi) = \max(\text{qr}(\varphi), \text{qr}(\psi))$.
- $\text{qr}(\neg\varphi) = \text{qr}(\varphi)$.
- $\text{qr}(\exists x\varphi) = \text{qr}(\forall x\varphi) = \text{qr}(\varphi) + 1$.

We denote $\text{FO}[n]$ as the set of first-order formulae with quantifier rank at most n .

Formulae can also have variables that are not associated with any quantifier. These variables are called *free*. We can recursively find the free variables of a given term or formula.

Definition 1.7. Free variables of a term or formula are defined as follows:

- A variable x contains one free variable, x , and a constant symbol c has none.

- The free variables of $t_1 = t_2$ are the free variables of t_1 and t_2 .
- $f(t_1, \dots, t_k)$ and $P(t_1, \dots, t_k)$ have the free variables of $t_1, \dots, t_k$.
- $\varphi \wedge \psi$ has the free variables of φ and ψ ; $\neg\varphi$ has the free variables of φ .
- The free variables of $\exists x\varphi$ or $\forall x\varphi$ are the free variables of φ except x .

We say that a variable is *bound* if it is not free. A formula with no free variables is called a *sentence*. We denote formulae by $\varphi(x_1, \dots, x_n)$ where $x_1, \dots, x_n$ are the free variables in φ . More concisely, we will write $\varphi(\bar{x})$.

While we have defined a set of formulae, the truth of each sentence depends on the world we are in. For example, “There exists a dominating vertex” is only true if the graph in question has a dominating vertex. A *structure*, or model, assigns concrete values to the symbols in a signature, allowing us to evaluate sentences. In our example, we could encode the graph as a structure to evaluate “There exists a dominating vertex”.

Definition 1.8. A σ -structure is written as

$$\mathfrak{A} = (A, c_1^{\mathfrak{A}}, c_2^{\mathfrak{A}}, \dots, P_1^{\mathfrak{A}}, P_2^{\mathfrak{A}}, \dots, f_1^{\mathfrak{A}}, f_2^{\mathfrak{A}}, \dots),$$

where A is the set of all objects called the *universe* and we interpret each

- constant symbol c_i from σ as an element $c_i^{\mathfrak{A}} \in A$,
- k -ary relation (or predicate) symbol P_i from σ as a set $P_i^{\mathfrak{A}} \subseteq A^k$,
- k -ary function symbol f_i from σ as a function $f_i^{\mathfrak{A}} : A^k \rightarrow A$.

Definition 1.9. Let $\mathfrak{A}$ be a σ -structure with universe A and $\varphi(\bar{x})$ be a formula over σ , where $\bar{x} = (x_1, \dots, x_n)$. For $\bar{a} \in A^n$, we define $t^{\mathfrak{A}}(\bar{a})$, the interpretation of a term t in $\mathfrak{A}$ as follows:

- If t is a constant symbol c , then $t^{\mathfrak{A}}(\bar{a})$ is $c^{\mathfrak{A}}$.
- If t is a variable x_i , then $t^{\mathfrak{A}}(\bar{a})$ is a_i .
- If t is $f(t_1, \dots, t_k)$, then $t^{\mathfrak{A}}(\bar{a})$ is $f(t_1^{\mathfrak{A}}(\bar{a}), \dots, t_k^{\mathfrak{A}}(\bar{a}))$.

We now inductively define $\mathfrak{A} \models \varphi(\bar{a})$:

- If φ is $t_1 = t_2$, then $\mathfrak{A} \models \varphi(\bar{a})$ iff $t_1^{\mathfrak{A}}(\bar{a}) = t_2^{\mathfrak{A}}(\bar{a})$.
- If φ is $P(t_1, \dots, t_k)$, then $\mathfrak{A} \models \varphi(\bar{a})$ iff $P^{\mathfrak{A}}(t_1^{\mathfrak{A}}(\bar{a}), \dots, t_k^{\mathfrak{A}}(\bar{a}))$ holds.
- $\mathfrak{A} \models \neg\varphi(\bar{a})$ iff $\mathfrak{A} \models \varphi(\bar{a})$ does not hold.
- $\mathfrak{A} \models \varphi(\bar{a}) \wedge \psi(\bar{a})$ iff $\mathfrak{A} \models \varphi(\bar{a})$ and $\mathfrak{A} \models \psi(\bar{a})$.

- If φ is $\exists y\psi(y, \bar{x})$, then $\mathfrak{A} \models \varphi(\bar{a})$ iff $\mathfrak{A} \models \psi(b, \bar{a})$ for some $b \in A$.
- If φ is $\forall y\psi(y, \bar{x})$, then $\mathfrak{A} \models \varphi(\bar{a})$ iff $\mathfrak{A} \models \psi(b, \bar{a})$ for all $b \in A$.

If $\mathfrak{A} \models \varphi(\bar{a})$, then we say $\varphi(\bar{a})$ is true in $\mathfrak{A}$.

Example 1.10. In group theory, we could use the signature $\sigma = \{1, \cdot\}$, where the constant 1 can represent the identity element and $\cdot$ is a binary function. One such σ -structure is $\mathfrak{R} = (\mathbb{R}^\times, 1^\mathfrak{R}, \cdot^\mathfrak{R})$, the multiplicative group of non-zero real numbers. In practice, we will omit the superscript and write $\mathfrak{R} = (\mathbb{R}^\times, 1, \cdot)$.

Example 1.11. For graph theory, recall the signature $\sigma = \{E\}$. Each σ -structure has the form $G = (V, E)$, where our universe is the set of vertices in G and the relation E is a set of edges. Consider the complete graph $K_3 = (\{1, 2, 3\}, \{12, 21, 23, 32, 31, 13\})$. We can use the process in Definition 1.9 to determine that

$$K_3 \models \exists v \forall u (\neg(u = v) \rightarrow E(u, v)).$$

Some notations require a domain for the quantified variable, for example, by writing $\forall u \in V$. This is not required as the universe of our model, V , is the domain for all quantifiers.

In the previous example, we have $\sigma = \{E\}$ and we think of E as the set of edges, but note that E is simply a binary relation symbol. Translating the formula $\exists v \forall u (\neg(u = v) \rightarrow E(u, v))$ into English forces our idea of what a graph is onto the symbol. If we thought of E as a total order, our formula would instead say “There exists a maximum element”.

We define a notion of equivalence between structures.

Definition 1.12. Let $\mathfrak{A}$, with universe A , and $\mathfrak{B}$, with universe B , be σ -structures. A (*partial*) *homomorphism* from $\mathfrak{A}$ to $\mathfrak{B}$ is a (partial) function $h : A \rightarrow B$ such that:

- For all constant symbols c , $h(c^\mathfrak{A}) = c^\mathfrak{B}$.
- For all k -ary relation symbols P and $(a_1, \dots, a_k) \in A^k$,
if $(a_1, \dots, a_k) \in P^\mathfrak{A}$, then $(h(a_1), \dots, h(a_k)) \in P^\mathfrak{B}$.
- For all k -ary function symbols f and $(a_1, \dots, a_k) \in A^k$,

$$h(f^\mathfrak{A}(a_1, \dots, a_k)) = f^\mathfrak{B}(h(a_1), \dots, h(a_k)).$$

An *isomorphism* between $\mathfrak{A}$ and $\mathfrak{B}$ is a homomorphism from $\mathfrak{A}$ to $\mathfrak{B}$ whose inverse is a homomorphism from $\mathfrak{B}$ to $\mathfrak{A}$. If there is an isomorphism between $\mathfrak{A}$ and $\mathfrak{B}$, then we say that they are *isomorphic*, and we write $\mathfrak{A} \equiv \mathfrak{B}$.

Proposition 1.13 ([Mar10, Theorem 1.1.10]). *Suppose that h is an isomorphism between $\mathfrak{A}$ and $\mathfrak{B}$. Then, for all formulae $\varphi(x_1, \dots, x_n)$ and $(a_1, \dots, a_n) \in A^n$,*

$$\mathfrak{A} \models \varphi(a_1, \dots, a_n) \text{ if and only if } \mathfrak{B} \models \varphi(h(a_1), \dots, h(a_n)).$$

Example 1.14. Another graph,

$$G = (\{a, b, c\}, \{ab, ba, bc, cb, ca, ac\}),$$

satisfies the same formulae as K_3 in the previous example. This is because G and K_3 are isomorphic. given by the isomorphism that maps $1 \mapsto a$, $2 \mapsto b$ and $3 \mapsto c$.

Definition 1.15. Consider a relational signature σ . Let $\mathfrak{A}, \mathfrak{B}$ be σ -structures, $a_1, \dots, a_n$ be elements in $\mathfrak{A}$ and $b_1, \dots, b_n$ be elements in $\mathfrak{B}$. There is a *partial isomorphism* between $(a_1, \dots, a_n)$ and $(b_1, \dots, b_n)$ over $\mathfrak{A}$ and $\mathfrak{B}$ if

- For all $i, j \leq n$, $a_i = a_j$ iff $b_i = b_j$.
- For all constant symbols $c \in \sigma$ and $i \leq n$, $a_i = c^{\mathfrak{A}}$ iff $b_i = c^{\mathfrak{B}}$.
- For all k -ary relation symbols P and every sequence of indexes $(i_1, \dots, i_k)$,

$$(a_{i_1}, \dots, a_{i_k}) \in P^{\mathfrak{A}} \text{ iff } (b_{i_1}, \dots, b_{i_k}) \in P^{\mathfrak{B}}.$$

We can now determine whether a sentence is true or false in a given model, but proving properties of a particular graph or group can be tedious. Instead, we want to examine a *class* of structures with specific properties.

Definition 1.16. A *theory* T over σ is a set of sentences over σ . We say $\mathfrak{A}$ is a *model* of T , denoted as $\mathfrak{A} \models T$ if $\mathfrak{A} \models \Phi$ for all sentences $\Phi \in T$. A class of σ -structures $\mathcal{P}$ is an *elementary class* if there exists a theory T such that $\mathcal{P} = \{\mathfrak{A} : \mathfrak{A} \models T\}$.

Example 1.17. Graph theory is a set of properties that govern what a graph is. The study of graph theory examines σ -structures that follow these properties, where $\sigma = \{E\}$. If we want to study simple, undirected graphs, we could define this theory with

$$\forall u \forall v (E(u, v) \rightarrow E(v, u)) \text{ and } \forall v \neg E(v, v).$$

Example 1.18. We define group theory over $\sigma = \{1, \cdot\}$ as the set with the following sentences:

- Associativity: $\forall a \forall b \forall c (a \cdot b) \cdot c = a \cdot (b \cdot c)$,
- Identity: $\forall a (a \cdot 1 = a \wedge 1 \cdot a = a)$,
- Inverse: $\forall a \exists b (a \cdot b = 1 \wedge b \cdot a = 1)$.

1.2 Expressibility

A key question is, given a language such as the first-order formulae over σ , what *properties* can be *expressed*? To show that a property is expressible, we can provide a sentence to state it, such as with graphs being undirected in our previous example.

Definition 1.19. A *property* is a class of σ -structures $\mathcal{P}$. The property is *expressible* in $\mathcal{P}$ if there exists a sentence Φ such that $\mathcal{P} = \{\mathfrak{A} : \mathfrak{A} \models \Phi\}$.

We showcase two model-theoretic tools to help us prove inexpressibility.

1.2.1 The Compactness Theorem

The Compactness Theorem [Mal36] for first-order logic is a consequence of Gödel's Completeness Theorem, and has a wide variety of applications both in and outside of logic.

Definition 1.20. A theory T is *satisfiable* if there exists a model of T .

Theorem 1.21 (Compactness Theorem). *A theory T is satisfiable if and only if every finite subset of T is satisfiable.*

Proposition 1.22 ([Lib04, Proposition 3.1]). *Connectedness of a graph is not expressible in first-order logic.*

Proof. Let $\sigma = \{v_1, v_2, E\}$ and $\mathcal{P}$ be the set of σ -structures that are connected. Assume that there is a sentence Φ such that $\mathcal{P} = \{\mathfrak{A} : \mathfrak{A} \models \Phi\}$. Let φ_n be the sentence that there is no path of length $n + 1$ from v_1 to v_2 :

$$\neg(\exists x_1 \exists x_2 \dots \exists x_n (E(v_1, x_1) \wedge E(x_1, x_2) \wedge \dots \wedge E(x_n, v_2))).$$

Let T be the theory

$$\{\varphi_n : n \geq 1\} \cup \{\neg(v_1 = v_2), \neg E(v_1, v_2)\} \cup \{\Phi\},$$

which says there is no path of length n between v_1 and v_2 for all $n \in \mathbb{N}$ and the graph is connected. We will show that T is satisfiable. Let T' be a finite subset of T . Then there is an integer $N > 1$ such that $N \geq n$ for all $\varphi_n \in T'$. The connected graph consisting of a single path of length $N + 1$ from v_1 to v_2 is a model of T' . Therefore, T' is satisfiable.

By compactness, T must also be satisfiable. That is, there exists a connected graph that has no path from v_1 to v_2 with length n , for all $n \in \mathbb{N}$. This is a contradiction. Hence, connectedness of a graph is not expressible in first-order logic. $\square$

Proposition 1.23. *The statement “There is an element of infinite order” in a group is not expressible in first-order logic.*

Proof. Let $\sigma = \{1, g, \cdot\}$ and $\mathcal{P}$ be the set of σ -structures with an element of infinite order. Assume that there is a sentence Φ such that $\mathcal{P} = \{\mathfrak{A} : \mathfrak{A} \models \Phi\}$. Let φ_n be the sentence $\neg(g^n = 1)$. Let T be the theory

$$\{\varphi_n : n \geq 1\} \cup \{\neg\Phi\},$$

which means there is no element with infinite order, but g has no $n \in \mathbb{N}_{>0}$ such that $g^n = 1$. We will show that T is satisfiable. Let T' be a finite subset of T . There is $N > 1$ such that $N \geq n$ for all $\varphi_n \in T'$. Then, a model of T' is the cyclic group of order $N + 1$ generated by g . Therefore, T' is satisfiable. By compactness, T is also satisfiable. There is a group with no element of infinite order, but $g^n \neq 1$ for all $n \in \mathbb{N}_{>0}$. This is a contradiction, so our assumption is false. Hence, our statement is not expressible in first-order logic. $\square$

1.2.2 The Löwenheim-Skolem Theorem

Another useful property for proving inexpressibility in first-order logic is the Löwenheim-Skolem Theorem. This theorem is poorly named, as the general form was rejected by Skolem due to his disbelief in uncountable numbers. Malcev proved the general form in the same paper where he proved the Compactness Theorem [Mal36].

Theorem 1.24 (Löwenheim-Skolem Theorem). *If theory T has a model with an infinite universe, then it has a model with a countably infinite universe.*

Proposition 1.25 ([Lib04, Proposition 3.3]). *Let $\sigma = \emptyset$. The cardinality of a set being divisible by k is not expressible in first-order logic, for $k \in \mathbb{N}_{>1}$.*

Proof. Let $k \in \mathbb{N}_{>1}$ and $\mathcal{P}$ be the set of all σ -structures with a universe of cardinality divisible by k . Assume that there is a sentence Φ such that $\mathcal{P} = \{\mathfrak{A} : \mathfrak{A} \models \Phi\}$. Let φ_n be the sentence that the universe has at least n distinct elements:

$$\exists x_1 \dots \exists x_n \bigwedge_{i \neq j} \neg(x_i = x_j).$$

We define two theories:

$$T_1 = \{\Phi\} \cup \{\varphi_n : n \geq 1\}, \quad T_2 = \{\neg\Phi\} \cup \{\varphi_n : n \geq 1\}.$$

For any finite $T'_1 \subseteq T_1$, we have a model with a universe $\mathbb{Z}/kN\mathbb{Z}$ for a sufficiently large

integer N . By compactness, we have a model that satisfies T_1 . Since we know that this model has at least n distinct elements for all $n \in \mathbb{N}_{>0}$, this model has an infinite universe. By the Löwenheim-Skolem Theorem, T_1 has a model with a countably infinite universe $\mathfrak{A}$. Similarly, T_2 has a countably infinite universe $\mathfrak{B}$.

We know that $\mathfrak{A} \models \Phi$ and $\mathfrak{B} \models \neg\Phi$. Since the universes of $\mathfrak{A}$ and $\mathfrak{B}$ are countably infinite, there is a bijection between the universes. This bijection is trivially an isomorphism between $\mathfrak{A}$ and $\mathfrak{B}$ as σ has no constant, relation or function symbols. From Proposition 1.13, we find that $\mathfrak{A} \models \Phi \wedge \neg\Phi$. This is a contradiction. As $\mathfrak{A}$ and $\mathfrak{B}$ only consist of their universes, they are sets. Thus, the cardinality of sets being divisible by k is not expressible in first-order logic. $\square$

1.3 Second-order Logic

In first-order logic, we can quantify over elements of our universe with $\forall$ and $\exists$. *Second-order logic* also allows us to quantify over relations. The relations can be unary, binary, ternary, and so on. As unary relations are subsets, we can also quantify over subsets of the universe. In this section, we show that the tools to prove inexpressibility in first-order logic, the Compactness Theorem and the Löwenheim-Skolem Theorem, do not carry over to second-order logic.

Definition 1.26. *Second-order logic* (SO) can have both first-order and second-order variables. Second-order variables, generally denoted as $X, Y, Z, \dots$, are capitalised to differentiate them from first-order. The set of second-order *terms* is defined the same way as the set of first-order terms. The set of second-order *formulae* over σ are defined as follows:

- If t_1, t_2 are terms, then $t_1 = t_2$ is a formula.
- Let $t_1, \dots, t_k$ be terms.
If P is a k -ary relation symbol, then $P(t_1, \dots, t_k)$ is a formula.
If X is a k -ary second-order variable, then $X(t_1, \dots, t_k)$ is a formula.
- If φ and ψ are formulae, then $\varphi \wedge \psi$ and $\neg\varphi$ are formulae.
- If φ is a formula, then $\exists x\varphi$, $\forall x\varphi$, $\exists X\varphi$ and $\forall X\varphi$ are formulae.

Free variables of a second-order formula can be defined similarly to Definition 1.7 by adding the following additional rules:

- $X(t_1, \dots, t_k)$ has the free variables X and the free variables of $t_1, \dots, t_k$.
- The free variables of $\exists X\varphi$ and $\forall X\varphi$ are the free variables of φ except X .

Let $\mathfrak{A}$ be a σ -structure with universe A . Consider a formula $\varphi(\bar{x}, \bar{X})$, where $\bar{x} = (x_1, \dots, x_n)$ are the free first-order variables, and $\bar{X} = (X_1, \dots, X_m)$ are the free second-order variables, with each X_i being k_i -ary. Let $\bar{a} \in A^n$ and $\bar{B} = (B_1, \dots, B_m)$ such that $B_i \subseteq A^{k_i}$. We extend Definition 1.9, with the following rules, to inductively define $\mathfrak{A} \models \varphi(\bar{a}, \bar{B})$:

- If φ is $X_i(t_1, \dots, t_k)$, then $\mathfrak{A} \models \varphi(\bar{a}, \bar{B})$ iff $B_i(t_1^{\mathfrak{A}}(\bar{a}), \dots, t_k^{\mathfrak{A}}(\bar{a}))$ holds.

For k -ary second-order variables Y , we also have the following rules:

- If φ is $\exists Y \psi(\bar{x}, Y, \bar{X})$, then $\mathfrak{A} \models \varphi(\bar{a}, \bar{B})$ iff $\mathfrak{A} \models \psi(\bar{a}, C, \bar{B})$ for some $C \subseteq A^k$.
- If φ is $\forall Y \psi(\bar{x}, Y, \bar{X})$, then $\mathfrak{A} \models \varphi(\bar{a}, \bar{B})$ iff $\mathfrak{A} \models \psi(\bar{a}, C, \bar{B})$ for all $C \subseteq A^k$.

Definition 1.27. *Monadic second-order logic* (MSO) is the restriction of SO where all second-order variables are unary. *Dyadic second-order logic* (DSO) also allows for binary second-order variables.

Naturally, similar questions of expressibility derived from FO can be applied to SO and MSO. MSO is known to be strictly more expressive than FO, as seen with the following example.

Example 1.28 ([Tur84, Example 3.1]). We can express the connectivity of a graph $G = (V, E)$ with the following monadic second-order sentence:

$$\forall W[(\exists x W(x) \wedge \exists y \neg W(y)) \rightarrow \exists x \exists y (W(x) \wedge \neg W(y) \wedge E(x, y))].$$

That is, for all subsets $W \subseteq V$, if W and its complement are non-empty, then there is an edge between W and its complement.

A famous application of logic was the formalisation of arithmetic by Peano [Pea89]. Peano's theory uses the signature $\sigma = \{0, Succ\}$, where 0 is a constant symbol and $Succ$ is a unary function symbol. We use $Succ$ to denote the successor function. The induction principle cannot be expressed in first-order logic since sets that are not first-order expressible must be considered. With second-order arithmetic, we can quantify over all sets P to get the following induction principle:

$$\forall P \left((P(0) \wedge \forall n (P(n) \rightarrow P(Succ(n)))) \rightarrow \forall n P(n) \right).$$

We provide this background for a brief sketch that compactness does not hold in second-order logic based on the proof in Section 5.1 of [Vää24].

Lemma 1.29. *Any two models of monadic second-order arithmetic are isomorphic.*

Proposition 1.30. *Compactness fails for monadic second-order logic.*

Proof. (Sketch) Let $\sigma = \{0, k, Succ\}$ where k is a constant symbol and T be the theory of monadic second-order arithmetic. Let φ_n be the following sentence:

$$\neg(k = \underbrace{Succ(Succ(\dots Succ(0) \dots))}_{n \text{ times}}).$$

Suppose that compactness is true. Then, we will prove that the theory $T' = T \cup \{\varphi_n : n \geq 0\}$ has a model. For any finite subset of T' , we have a finite number of φ_n and so we can assign k to a number N such that φ_N is not in the finite subset. Then, any model for T with this assignment to k works. By compactness, T' has a model. That is, we have a model with a number where k is not in the sequence $0, Succ(0), Succ(Succ(0)), \dots$. Note that $T \subset T'$, so any model of T' is also a model of T . In our standard model of the natural numbers, every number is some amount of the successor function applied to 0. We find a contradiction, as there is no model of T with such a k since all models of T are isomorphic. $\square$

Proposition 1.31 ([Zac25, Theorem 8.25]). *The Löwenheim-Skolem Theorem fails in dyadic second-order logic.*

Proof. This comes from the fact that countability can be defined in second-order logic. The countability of a structure is characterised as follows:

$$\begin{aligned} Countable = & \exists z \exists F (\forall x \exists y (F(x, y) \wedge \forall x \forall y \forall z (F(x, y) \wedge F(x, z) \rightarrow y = z) \\ & \wedge \forall X ([X(z) \wedge \forall x (X(x) \rightarrow \exists y (F(x, y) \wedge X(y))] \rightarrow \forall x X(x)). \end{aligned}$$

In effect, we pick a zero element z . The statement says that there exists a function F such that for all subsets of the universe X , having $F^n(z) \in X$ for all $n \in \mathbb{N}$ means that X is the universe. Note that F^n means n applications of F .

Clearly, $\neg Countable$ is true for the real numbers $\mathbb{R}$, which is an infinite set. However, there is no countable model such that $\neg Countable$ is true. The Löwenheim-Skolem Theorem fails in second-order logic. $\square$

Chapter 2

Ehrenfeucht-Fraïssé Games

In this chapter, we introduce the Ehrenfeucht-Fraïssé game [Fra54; Ehr61], a technique for proving the inexpressibility of properties in first-order logic. Chapter 1 introduces key theorems used in first-order logic to prove inexpressibility. However, the final section showcases how these theorems fail in second-order logic. Unlike most other techniques for first-order logic, the Ehrenfeucht-Fraïssé game is a flexible framework that can be adapted to other logics. We highlight how the game can be adapted to other classical logics by detailing games for monadic second-order logic [Tur84] and the k -variable fragment of first-order logic [Bar77; Imm82]. With each game, we give examples of playing them and how each example can be used for an inexpressibility result. This highlights how proofs can become more intuitive as the prover only needs to work with game semantics. The main reference for this chapter is [Lib04].

2.1 The Game

Ehrenfeucht-Fraïssé (EF) games allow us to determine if all first-order sentences evaluate to the same truth value for two given structures. To show a property is inexpressible, we can then pick two structures that are equivalent under EF games, where the property holds in one but not the other. Since first-order logic cannot distinguish these structures, it cannot express the given property. Fraïssé was the first to find a necessary and sufficient condition to show that two models were indistinguishable by first-order formulae [Fra54]. Ehrenfeucht then presented this condition in the form of a game [Ehr61].

Definition 2.1 ([Ehr61]). The *Ehrenfeucht-Fraïssé (EF) game* is played by two players: the *spoiler* and the *duplicator*. We are given a pair of structures $\mathfrak{A}$ and $\mathfrak{B}$, over the same relational signature σ . The game is played over n rounds with pebbles labelled $a_1, a_2, \dots, b_1, b_2, \dots$.

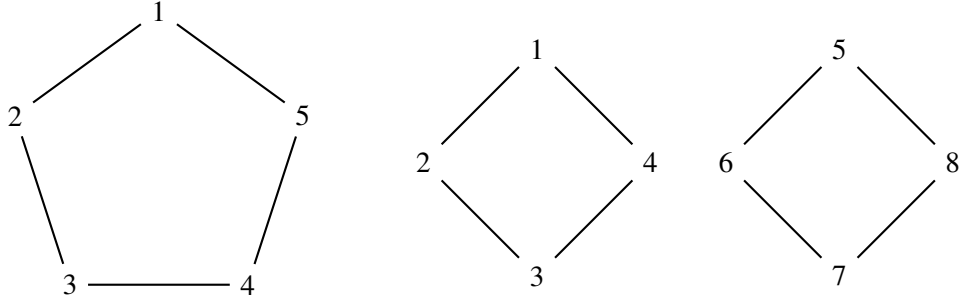


Figure 2.1: G_1^1 (Left) and G_2^1 (Right)

The goal of the spoiler is to show that the two structures are different, while the goal of the duplicator is to show they are the same. Round i goes as follows:

1. The spoiler chooses either $\mathfrak{A}$ or $\mathfrak{B}$.
2. The spoiler places a pebble on an element of that structure: either a_i for $\mathfrak{A}$ or b_i for $\mathfrak{B}$.
3. The duplicator responds by placing the other i th pebble in the other structure.

Let $c_1, \dots, c_l$ be the constant symbols in σ . The duplicator is *winning in round k* if there is a partial isomorphism between $(a_1, \dots, a_k, c_1^{\mathfrak{A}}, c_2^{\mathfrak{A}}, \dots, c_l^{\mathfrak{A}})$ and $(b_1, \dots, b_k, c_1^{\mathfrak{B}}, c_2^{\mathfrak{B}}, \dots, c_l^{\mathfrak{B}})$ over $\mathfrak{A}$ and $\mathfrak{B}$. The duplicator has an *n -round winning strategy*, denoted as $\mathfrak{A} \equiv_n^{\text{FO}} \mathfrak{B}$, if the duplicator can play in a way that guarantees a winning position no matter how the spoiler plays. Otherwise, the spoiler has an *n -round winning strategy*.

Example 2.2 (Sets). For any two sets A, B such that $|A|, |B| \geq n$, we have $A \equiv_n^{\text{FO}} B$. Note that A and B are structures with no symbols. Without loss of generality, on round i , suppose the spoiler places a_i on an element in A . The strategy for the duplicator at round i is as follows: if $a_i = a_j$ for some $j \leq i - 1$, then place b_i on b_j ; otherwise, place b_i on any element with no pebble. Since there are no symbols, the only condition for a partial isomorphism on round n is that $a_i = a_j$ iff $b_i = b_j$ for all $i, j \leq n$.

If we have $|A| \geq n$ and $|B| < n$, then $A \not\equiv_n^{\text{FO}} B$. The spoiler has a winning strategy where they can pick n different elements for each pebble $a_1, \dots, a_n$. This cannot be replicated in B as the pigeonhole principle states that at least two pebbles from $b_1, \dots, b_n$ are on the same element. This violates the condition for a partial isomorphism.

Example 2.3 (Graphs). Let $\sigma = \{E\}$ and $n \in \mathbb{N}$. Let G_1^n be a cycle of size $2^{n+1} + 1$ and G_2^n be two disjoint cycles of size 2^{n+1} . The case for $n = 1$ is shown in Figure 2.1. We define the distance between vertices $x, y \in G$, denoted as $d_G(x, y)$, as the shortest path length from x to y . If there is no such path, $d_G(x, y) = \infty$ i.e. $d_G(x, y) > n$ for all $n \in \mathbb{N}$. We want to

prove that $G_1^n \equiv_n^{\text{FO}} G_2^n$. We will fix n and induct on the number of moves thus far. We will prove that on G_1^n and G_2^n , the duplicator has a strategy where after k moves, for all $i, j \leq k$, either $d_{G_1^n}(a_i, a_j) = d_{G_2^n}(b_i, b_j)$ or $d_{G_1^n}(a_i, a_j)$ and $d_{G_2^n}(b_i, b_j)$ are both greater than or equal to 2^{n-k+1} . When this condition is true, the graphs induced by $a_1, \dots, a_n$ and $b_1, \dots, b_n$ will be isomorphic, since this process ensures that $a_i a_j$ is an edge if and only if $b_i b_j$ is too.

For $k = 1$, any vertex that the spoiler chooses in one graph, the duplicator can pick any vertex in the other. Our condition is vacuously true. Let $1 \leq k < n$ and suppose we have played k rounds so far such that for all $i, j \leq k$, either $d_{G_1^n}(a_i, a_j) = d_{G_2^n}(b_i, b_j)$ or $d_{G_1^n}(a_i, a_j)$ and $d_{G_2^n}(b_i, b_j)$ are greater than or equal to 2^{n-k+1} . We will prove that the condition holds after playing $k + 1$ moves. Suppose the spoiler places the $(k + 1)$ th pebble down. Let W be the set of pebbles with a distance of at most $2^{n-k} - 1$ from the $(k + 1)$ th pebble.

Case 1: $|W| = 0$. This means that the $(k + 1)$ th pebble is at least 2^{n-k} edges away from any other pebble. If there is a cycle with no pebbles in the other structure, the duplicator can place the $(k + 1)$ th pebble anywhere on that cycle. If there is a cycle with exactly one pebble, the duplicator can place the $(k + 1)$ th pebble opposite that pebble. Otherwise, by the pigeonhole principle, there exist two pebbles in the other structure with a distance of at least 2^{n-k+1} and no pebbles in between them. The duplicator can place a pebble in the middle of these two pebbles. Our condition still holds.

Case 2: $|W| > 0$. In the second case, the spoiler places the $(k + 1)$ th pebble. Note that any two pebbles in W have a distance of less than 2^{n-k+1} from each other, so the inductive hypothesis states that the corresponding pebbles in the other structure must be the same distance from each other. That is, there exists a copy of W with the same distances between each pebble in the other structure. The duplicator can place the $(k + 1)$ th pebble in the same relative position within the copy of W . Our condition holds, and this completes our induction. $\square$

Before moving on to the main theorem that connects the game to first-order logic, we lay out some guiding principles that we believe an EF game should follow. There should be relatively little computation outside of the game to make a move. We see this for the original game, where the spoiler and duplicator only need to place a pebble on some element. There is no computation required, as they can place a pebble on whatever element they please. Another appeal of the game is that the winning condition is based on a partial isomorphism between the pebbles currently on the board. The game is local in the sense that we do not care about anything other than the finite number of pebbles on the board and their relation with each other. The structure outside these pebbles does not matter.

The following theorem and its corollary for proving inexpressibility are hallmarks for what this thesis will refer to as an EF game.

Theorem 2.4 ([Ehr61, Ehrenfeucht-Fraïssé Theorem]). *Let $\mathfrak{A}$ and $\mathfrak{B}$ be σ -structures, where σ is finite and relational. Then, the following are equivalent:*

1. *For all sentences $\Phi \in \text{FO}[n]$, $\mathfrak{A} \models \Phi$ iff $\mathfrak{B} \models \Phi$.*
2. $\mathfrak{A} \equiv_n^{\text{FO}} \mathfrak{B}$.

Proof. We only provide the proof for $(2) \implies (1)$, as this proof is more instructive than the other direction. This is also the direction used for the corollary to prove inexpressibility. In particular, we prove the contrapositive with induction. That is, for all structures $\mathfrak{A}$ and $\mathfrak{B}$, if there exists a sentence $\Phi \in \text{FO}[n]$ such that $\mathfrak{A} \models \Phi$ but $\mathfrak{B} \not\models \Phi$, then $\mathfrak{A} \not\equiv_n^{\text{FO}} \mathfrak{B}$. Note that $\mathfrak{A} \not\equiv_n^{\text{FO}} \mathfrak{B}$ means the spoiler has a winning strategy for n rounds. For $n = 0$, let $\mathfrak{A}, \mathfrak{B}$ be σ -structures such that there is $\Phi \in \text{FO}[0]$ where $\mathfrak{A} \models \Phi$ but $\mathfrak{B} \not\models \Phi$.

To show there is no partial isomorphism, we will find some sentence of the form $t_1 = t_2$ or $P(t_1, \dots, t_k)$ which is evaluated differently by $\mathfrak{A}$ and $\mathfrak{B}$. As σ is relational and Φ is a sentence containing no free variables, any term must be a constant. The sentence Φ must be a boolean combination of sentences with the form $t_1 = t_2$ or $P(t_1, \dots, t_k)$ since $\text{qr}(\Phi) = 0$. Consider the two following rules. If $\Phi = \neg\Psi$ for some sentence Ψ , then Ψ is evaluated differently between $\mathfrak{A}$ and $\mathfrak{B}$. We continue with Ψ . If $\Phi = \Psi_1 \wedge \Psi_2$, then either Ψ_1 or Ψ_2 must be evaluated differently between $\mathfrak{A}$ and $\mathfrak{B}$. We continue with the Ψ_i , which is evaluated differently. By recursively applying these rules, we will find some sentence of the form $t_1 = t_2$ or $P(t_1, \dots, t_k)$ which is evaluated differently. There is no partial isomorphism and $\mathfrak{A} \not\equiv_0 \mathfrak{B}$.

Suppose that $(2) \implies (1)$ is the case when $n \leq k$. We prove the case for $n = k + 1$. Let $\mathfrak{A}, \mathfrak{B}$ be structures with $\Phi \in \text{FO}[k + 1]$ such that $\mathfrak{A} \models \Phi$ but $\mathfrak{B} \not\models \Phi$. We can use the two rules from the base case to find a sentence of the form $\exists x\Psi$ which is evaluated differently between the two structures. We now build a strategy for the spoiler to win. One structure has an element x such that Ψ holds, while the other structure does not. Without loss of generality, suppose x is in $\mathfrak{A}$. Let the spoiler place the pebble a_1 on x . The duplicator then responds with b_1 . We can now define two new structures: $\mathfrak{A}'$, which is $\mathfrak{A}$ with a new constant symbol x interpreted as a_1 and $\mathfrak{B}'$, which is $\mathfrak{B}$ with x interpreted as b_1 . Clearly, Ψ is a sentence that is evaluated differently between $\mathfrak{A}'$ and $\mathfrak{B}'$. Since $\Psi \in \text{FO}[k]$, we can use our inductive hypothesis to give us a k -move winning strategy for the spoiler to win. Hence, $\mathfrak{A} \not\equiv_{k+1} \mathfrak{B}$. $\square$

Corollary 2.5. *A property $\mathcal{P}$ of σ -structures is not expressible in FO if for every $n \in \mathbb{N}$, there exist two σ -structures, $\mathfrak{A}_n$ and $\mathfrak{B}_n$, such that $\mathfrak{A}_n \equiv_n^{\text{FO}} \mathfrak{B}_n$, and $\mathfrak{A}_n \in \mathcal{P}$, but $\mathfrak{B}_n \notin \mathcal{P}$.*

Proof. Suppose that for all $n \in \mathbb{N}$, there exist $\mathfrak{A}_n$ and $\mathfrak{B}_n$, such that $\mathfrak{A}_n \equiv_n^{\text{FO}} \mathfrak{B}_n$, and $\mathfrak{A}_n \in \mathcal{P}$, but $\mathfrak{B}_n \notin \mathcal{P}$. Assume that φ is a first-order sentence that expresses $\mathcal{P}$. Let $k = \text{qr}(\varphi)$. We have $\mathfrak{A}_k \equiv_k^{\text{FO}} \mathfrak{B}_k$. The Ehrenfeucht-Fraïssé Theorem tells us that $\mathfrak{A}_k$ and $\mathfrak{B}_k$ evaluate every sentence in $\text{FO}[k]$ the same. However, $\mathfrak{A}_k \models \varphi$, while $\mathfrak{B}_k \not\models \varphi$. This contradicts the fact that $\mathfrak{A}_k \equiv_k^{\text{FO}} \mathfrak{B}_k$. Our assumption must be incorrect. No first-order sentence expresses $\mathcal{P}$. $\square$

Proposition 2.6. *The cardinality of a set being divisible by k is not expressible, for $k \in \mathbb{N}_{>1}$.*

Proof. Let $n \in \mathbb{N}$. We can construct sets A, B such that $|A| = kn$ and $|B| = kn + 1$. From Example 2.2, we find that $A \equiv_n^{\text{FO}} B$, but clearly k divides $|A|$ while k does not divide $|B|$. Our proposition then follows from Corollary 2.5. $\square$

Proposition 2.7. *Connectedness of a graph is not expressible in first-order logic.*

Proof. Let $n \in \mathbb{N}$. From Example 2.3, we find that $G_1^n \equiv_n^{\text{FO}} G_2^n$, but clearly G_1^n is connected while G_2^n is not. Our proposition then follows from Corollary 2.5. These proofs can be reused for other properties that the graphs disagree on. For example, we have also proven that 2-colourability cannot be expressed in first-order logic, since G_1^n has a chromatic number of 3 and G_2^n is 2-colourable. $\square$

2.2 Extensions

In this section, we explore how changing aspects of the Ehrenfeucht-Fraïssé game changes the logic we are trying to analyse. The Ehrenfeucht-Fraïssé Theorem tells us that the number of rounds played corresponds with looking into formulae that have a quantifier rank of at most k . We also look into the types of moves and the number of pebbles we use in the game.

2.2.1 Monadic Second-Order Logic

We detail the game defined in [Tur84]. Intuitively, the proof of the EF Theorem tells us that placing a pebble corresponds to a first-order quantifier in the formula. In monadic second-order (MSO) logic, we also have second-order quantifiers which say “For all sets” or “There exists a set”. We can adapt the game by adding a new type of move when the spoiler and duplicator choose a set of elements.

Definition 2.8. The *MSO EF game* is played on a pair of structures $\mathfrak{A}$ and $\mathfrak{B}$, over the same relational signature σ with constant symbols $c_1, \dots, c_l$. The game has n rounds and round i goes as follows:

1. The spoiler chooses either $\mathfrak{A}$ or $\mathfrak{B}$.
2. The spoiler either does an element move or a set move.
 - a. Element move: Spoiler picks some element in their chosen structure.
 - b. Set move: Spoiler picks a subset of the universe of their chosen structure.
3. The duplicator responds with the same type of move in the other structure.

Suppose k rounds have been played so far. Label the i th element moves as a_i and b_i and the i th set moves as A_i and B_i . Suppose there were j element moves and $k - j$ set moves. Let $\mathfrak{A}'$ be $\mathfrak{A}$ with the extra predicates $A_1, \dots, A_{k-j}$ and define $\mathfrak{B}'$ similarly. The duplicator is winning in round k if there is a partial isomorphism between $(a_1, \dots, a_j, c_1^{\mathfrak{A}}, c_2^{\mathfrak{A}}, \dots, c_l^{\mathfrak{A}})$ and $(b_1, \dots, b_j, c_1^{\mathfrak{B}}, c_2^{\mathfrak{B}}, \dots, c_l^{\mathfrak{B}})$ over $\mathfrak{A}'$ and $\mathfrak{B}'$. We denote the duplicator having an n -round winning strategy in the MSO EF game as $\mathfrak{A} \equiv_n^{\text{MSO}} \mathfrak{B}$.

Compared to the first-order EF game, the partial isomorphism has some extra conditions:

- For all $i \leq k - j$, $c^{\mathfrak{A}} \in A_i$ iff $c^{\mathfrak{B}} \in B_i$.
- For all $p \leq j$ and $i \leq k - j$, $a_p \in A_i$ iff $b_p \in B_i$.

Note that quantifier rank is calculated the same, where if X is a set, $\text{qr}(\exists X \varphi) = \text{qr}(\forall X \varphi) = \text{qr}(\varphi) + 1$. Then, $\text{SO}[n]$ is the set of second-order formulae with a quantifier rank at most n .

Theorem 2.9. *Let $\mathfrak{A}$ and $\mathfrak{B}$ be σ -structures, where σ is finite and relational. Then, the following are equivalent:*

1. For all sentences $\Phi \in \text{SO}[n]$, $\mathfrak{A} \models \Phi$ iff $\mathfrak{B} \models \Phi$.
2. $\mathfrak{A} \equiv_n^{\text{MSO}} \mathfrak{B}$.

We have seen that strategies for the duplicator can be proven through induction. Another useful technique is the *composition* of smaller games to make a strategy for the union of these games. We follow the proof of Proposition 7.12 in [Lib04].

Definition 2.10. Let σ be relational and $\mathfrak{A}, \mathfrak{B}$ be σ structures with disjoint universes A and B , respectively. The *disjoint union* $\mathfrak{A} \oplus \mathfrak{B}$ has the universe $A \cup B$ and for all relation symbols $P \in \sigma$, we have $P^{\mathfrak{A} \oplus \mathfrak{B}} = P^{\mathfrak{A}} \cup P^{\mathfrak{B}}$.

Lemma 2.11 (Composition). *Let $\mathfrak{A}_1, \mathfrak{A}_2, \mathfrak{B}_1, \mathfrak{B}_2$ be σ -structures. Let $\mathfrak{A} = \mathfrak{A}_1 \oplus \mathfrak{A}_2$ and $\mathfrak{B} = \mathfrak{B}_1 \oplus \mathfrak{B}_2$. If $\mathfrak{A}_1 \equiv_k^{\text{MSO}} \mathfrak{B}_1$ and $\mathfrak{A}_2 \equiv_k^{\text{MSO}} \mathfrak{B}_2$, then $\mathfrak{A} \equiv_k^{\text{MSO}} \mathfrak{B}$.*

Proposition 2.12. *Let A, B be sets such that $|A|, |B| \geq 2^k$. Then, $A \equiv_k^{\text{MSO}} B$.*

Proof. We proceed with induction. The case for $k = 0$ is trivial. Suppose that sets A, B have $|A|, |B| \geq 2^{k+1}$ and assume that for all sets C, D such that $C, D \geq 2^k$, we have $C \equiv_k^{MSO} D$. Without loss of generality, we can assume that the spoiler makes their move in A . Suppose the spoiler first makes a set move, $A_1 \subseteq A$. We have the following cases:

1. $|A_1| \leq 2^k$: The duplicator can pick an arbitrary set $B_1 \subseteq B$ such that $|A_1| = |B_1|$. A set move effectively adds a new relation, which we denote as P , to the structures. Let $\mathfrak{A}$ be the structure with universe A and $P^{\mathfrak{A}} = A_1$, and $\mathfrak{B}$, which has universe B with the relation $P^{\mathfrak{B}} = B_1$. We want to show that $\mathfrak{A} \equiv_k^{MSO} \mathfrak{B}$. Let $\mathfrak{A}_1$ have the universe A_1 and relation $P_1^{\mathfrak{A}_1} = A_1$, with $\mathfrak{B}_1$ defined similarly. Clearly, $\mathfrak{A} = \mathfrak{A}_1 \oplus (A - A_1)$ and $\mathfrak{B} = \mathfrak{B}_1 \oplus (B - B_1)$.

Since $A_1 \simeq B_1$, we know $\mathfrak{A}_1 \equiv_k^{MSO} \mathfrak{B}_1$. We have $|A - A_1|, |B - B_1| \geq 2^k$ so we use our inductive hypothesis to find $A - A_1 \equiv_k^{MSO} B - B_1$. From our composition lemma, we conclude that $\mathfrak{A} \equiv_k^{MSO} \mathfrak{B}$, and hence, $A \equiv_{k+1}^{MSO} B$.

2. $|A - A_1| \leq 2^k$: We can use a symmetrical argument to case 1.
3. $|A_1|, |A - A_1| > 2^k$. Since $|B| \geq 2^{k+1}$, the duplicator can choose B_1 such that $|B_1|, |B - B_1| \geq 2^k$. From the inductive hypothesis, we have $A_1 \equiv_k^{MSO} B_1$ and $A - A_1 \equiv_k^{MSO} B - B_1$, so we can use the Composition Lemma to play for k more rounds after the first move, similar to case 1.

Suppose the spoiler first made an element move $a_1 \in A$. Then, the duplicator can pick any $b_1 \in B$. This case can be proven with the composition lemma, by showing that $\{a_1\} \equiv_k^{MSO} \{b_1\}$ and $A - \{a_1\} \equiv_k^{MSO} B - \{b_1\}$. This completes our induction. $\square$

Corollary 2.13. *The cardinality of a set being even is not expressible in MSO.*

We find that the types of moves we have can correlate with the kinds of quantifiers we allow in our formulae. The EF game associated with dyadic second-order logic would have a move where the spoiler and duplicator select a set of ordered pairs.

2.2.2 Finite Number of Variables

A *fragment logic* is a restriction on what formulae are allowed in a particular logic. We now look at the k -*variable fragment* of first-order logic, denoted as FO^k , where every formula must have at most k unique variable names.

Example 2.14. While we restrict the number of variables, this does not restrict the quantifier

rank. Consider the following 2-variable formula with a quantifier rank of 3:

$$\exists x(\forall y(x = y \vee x < y) \rightarrow \exists x\forall y(x < y + 1)).$$

Barwise [Bar77] and Immerman [Imm82] independently discovered a correspondence between the number of pebbles used to play the game and the number of variable names. As we have a finite number of pebbles, we adjust the EF game by allowing for rounds where a previously placed pebble is moved rather than placing a new pebble down.

Definition 2.15. The k -variable EF game is played on a pair of structures $\mathfrak{A}$ and $\mathfrak{B}$, over the same relational signature σ with constant symbols $c_1, \dots, c_l$. We use the pebbles labelled $a_1, \dots, a_k, b_1, \dots, b_k$. The game has n rounds, and each round goes as follows:

1. The spoiler chooses either $\mathfrak{A}$ or $\mathfrak{B}$ and some index $i \in [1, k]$.
2. The spoiler takes the i th pebble and places it on an element of their structure.
3. The duplicator responds with the i th pebble in the other structure.

Suppose the pebbles $a_{i_1}, \dots, a_{i_m}, b_{i_1}, \dots, b_{i_m}$ have been played. The current position is winning if we have a partial isomorphism between the tuples $(a_{i_1}, \dots, a_{i_m}, c_1^{\mathfrak{A}}, c_2^{\mathfrak{A}}, \dots, c_l^{\mathfrak{A}})$ and $(b_{i_1}, \dots, b_{i_m}, c_1^{\mathfrak{B}}, c_2^{\mathfrak{B}}, \dots, c_l^{\mathfrak{B}})$ over $\mathfrak{A}$ and $\mathfrak{B}$.

Theorem 2.16. Let $FO^k[n]$ be the set of formulae in $FO[n]$ with k unique variable names. Let $\mathfrak{A}$ and $\mathfrak{B}$ be σ -structures, where σ is finite and relational. Then, the following are equivalent:

1. For all sentences $\Phi \in FO^k[n]$, $\mathfrak{A} \models \Phi$ iff $\mathfrak{B} \models \Phi$.
2. The duplicator has an n -round winning strategy in the k -variable EF game of $\mathfrak{A}$ and $\mathfrak{B}$.

To simplify proofs in the following chapters, we note that it suffices only to use two types of moves: placing down a new pebble and picking up a previously placed pebble. In each round, the spoiler can choose to place down a new pebble, and the duplicator does the same in the other structure, or the spoiler can choose to pick up a pebble, and the corresponding pebble must be picked up. In effect, moving a pebble is equivalent to picking up the pebble and placing it down again. We will refer to this as the k -variable EF game from now on.

An n -round winning strategy for the duplicator in the k -variable EF game of $\mathfrak{A}$ and $\mathfrak{B}$ is denoted as $\mathfrak{A} \equiv_n^k \mathfrak{B}$. The ability to pick up pebbles also allows us to play the EF game indefinitely. We write $\mathfrak{A} \equiv_\infty^k \mathfrak{B}$ if for all $n \in \mathbb{N}$, $\mathfrak{A} \equiv_n^k \mathfrak{B}$.

Corollary 2.17. A property $\mathcal{P}$ of σ -structures is not expressible in FO^k if for every $n \in \mathbb{N}$, there exists two σ -structures such that $\mathfrak{A}_n \equiv_n^k \mathfrak{B}_n$, and $\mathfrak{A}_n \in \mathcal{P}$, but $\mathfrak{B}_n \notin \mathcal{P}$.

We know that every formula in FO^k is also in FO^{k+1} . Can every formula in FO^{k+1} be expressed in FO^k ? No! We provide a game-based proof that FO^k gives us a strict hierarchy. That is, we have the following linear order of logics in terms of expressibility:

$$\text{FO}^1 < \text{FO}^2 < \dots < \text{FO}^k < \text{FO}^{k+1} < \dots$$

Consider $\sigma = \{E\}$. A k -clique in a σ -structure is a subgraph that is isomorphic to the complete graph with k nodes K_k . For integers $k \geq 1$, we can express the existence of a k -clique in FO^k . For $k = 1$, we have $\exists v(v = v)$. For $k \geq 2$, we have

$$\exists v_1 \exists v_2 \dots \exists v_k \bigwedge_{i \neq j} (\neg(v_i = v_j) \wedge E(v_i, v_j) \wedge E(v_j, v_i)).$$

To show that the existence of a $(k + 1)$ -clique cannot be expressed in FO^k , we first note that K_{k+1} contains a $(k + 1)$ -clique while K_k does not. It suffices to prove the following:

Proposition 2.18. *For integers $k \geq 1$, we have $K_{k+1} \equiv_{\infty}^k K_k$.*

Proof. The duplicator's strategy is as follows. Suppose the pebbles $a_{i_1}, \dots, a_{i_n}$ have been placed on K_{k+1} and $b_{i_1}, \dots, b_{i_n}$ have been placed on K_k so far and we have a partial isomorphism between $(a_{i_1}, \dots, a_{i_n})$ and $(b_{i_1}, \dots, b_{i_n})$. Without loss of generality, if the spoiler chooses to remove pebble a_{i_j} , then the duplicator simply removes pebble b_{i_j} and the partial isomorphism between the remaining pebbles still holds.

Suppose that the spoiler instead places a pebble $a_{i_{n+1}}$. If $a_{i_{n+1}} = a_{i_j}$ for some $j \leq n$, then the duplicator places $b_{i_{n+1}}$ on b_{i_j} . Otherwise, the duplicator can place $b_{i_{n+1}}$ on any element in K_k with no pebbles. This is possible as there can be at most k pebbles and there are k elements in K_k . In either case, the partial isomorphism holds as for all $j, l \leq n$, we have $a_{i_j} = a_{i_l}$ iff $b_{i_j} = b_{i_l}$, and $E(a_{i_j}, a_{i_l})$ iff $E(b_{i_j}, b_{i_l})$ trivially holds as two elements have an edge between them exactly when they are distinct. $\square$

We can also continue games from a given configuration. Suppose that the pebbles $a_1, \dots, a_m$ and $b_1, \dots, b_m$ have been placed so far on $\mathfrak{A}$ and $\mathfrak{B}$. An n -round winning strategy for the k -variable EF game is denoted as $(\mathfrak{A}, a_1, \dots, a_m) \equiv_n^k (\mathfrak{B}, b_1, \dots, b_m)$. We introduce a corollary to analyse formulae with free variables, not just sentences.

Corollary 2.19 ([Lib04, Corollary 11.9]). *Let $\mathfrak{A}$ and $\mathfrak{B}$ be structures with universes A and B . Let $\bar{a} \in A^m$, $\bar{b} \in B^m$, and $m \leq k$. Then, $(\mathfrak{A}, a_1, \dots, a_m) \equiv_{2n}^k (\mathfrak{B}, b_1, \dots, b_m)$ iff for every formula $\varphi(\bar{x}) \in \text{FO}^k[n]$ with m free variables, $\mathfrak{A} \models \varphi(\bar{a}) \Leftrightarrow \mathfrak{B} \models \varphi(\bar{b})$.*

That is, to analyse formulae with m free variables, we start with m pebbles on each structure.

Chapter 3

Modal Logic

The most basic temporal and epistemic logics of interest, in this thesis, can be defined as modal logics. In particular, epistemic logic and unary temporal logic are modal logics. In this chapter, we define a general framework of modal logic and an associated EF game by highlighting that Stirling’s bisimulation game [Sti99] fits our idea of an EF game. This framework is used to generate various theorems and an EF game for a given modal logic. We then define epistemic logic and introduce common knowledge, which reasons about the knowledge of groups, to round off the epistemic background for defining temporal epistemic logics. The main references are [BRV01] for modal logic and [DHK07] for epistemic logic.

3.1 The Modal Framework

Modal logic describes a family of logics that extends propositional logic by adding different modes of truth. These modalities can be applied to propositions p to denote concepts like “someone knows p is true” and “in the future, it is the case that p is true”. We follow the semantics for modal logic defined by Kripke [Kri63].

3.1.1 Defining Modal Logic

Definition 3.1. A *modal logic (ML)* is defined over *modalities* $\Box_1, \dots, \Box_m$ and *atomic propositions* $P = \{p_1, \dots, p_n\}$. The set of *ML formulae* is defined as follows:

- The atomic propositions $p \in P$ are ML formulae.
- If φ, ψ are ML formulae, then $\varphi \wedge \psi$ and $\neg\varphi$ are ML formulae.
- For all integers $i \in [1, m]$, if φ is an ML formula, then $\Box_i\varphi$ is an ML formula.

For brevity, we use Backus-Naur form to present the definition above. The ML formulae φ can be given by the following rule:

$$\varphi := p \mid \neg\varphi \mid \varphi \wedge \psi \mid \Box_i\varphi, \text{ where } p \in P \text{ and } i \in [1, m].$$

Definition 3.2. A *Kripke model* is the structure that propositions in ML are evaluated on. The Kripke model $M = \langle S, R, V \rangle$ over $P = \{p_1, \dots, p_n\}$ and positive integer m consists of:

- A set of states S .
- A function $R : [1, m] \rightarrow S \times S$. That is, $R(i)$ is a relation on S for all $i \in [1, m]$, denoted as R_i .
- A valuation function $V : P \rightarrow 2^S$, which tells us the states a given proposition is true.

The set of Kripke models is denoted as $\mathcal{K}$.

Definition 3.3. While a sentence in first-order logic is either true or false on a given structure, an ML formula is evaluated on a particular state of a given structure. An ML formula φ is *true* in a Kripke model M at state s , denoted as $M, s \models \varphi$. We have

- $M, s \models p$ iff $s \in V(p)$.
- $M, s \models \neg\varphi$ and $M, s \models \varphi \wedge \psi$ can be defined similar to before.
- $M, s \models \Box_i\varphi$ iff for all $s' \in S$ such that $s R_i s'$, we have $M, s' \models \varphi$.

We call a pair (M, s) such that $s \in S$ a *pointed model*.

Definition 3.4. The dual of $\Box_i$, denoted $\Diamond_i$, is defined as $\Diamond_i\varphi = \neg\Box_i\neg\varphi$, for modal formulae φ . Semantically, we have

$$M, s \models \Diamond_i\varphi \text{ iff there exists } s' \in S \text{ such that } s R_i s' \text{ and } M, s' \models \varphi.$$

A modal logic can equivalently be defined by specifying the dual modalities.

Definition 3.5. Let $\mathcal{C} \subseteq \mathcal{K}$. Two formulae φ and ψ are *equivalent over $\mathcal{C}$* if for all $M \in \mathcal{C}$ and $s \in S$, we have $M, s \models \varphi$ iff $M, s \models \psi$. Two logics L_1 and L_2 are *equivalent over $\mathcal{C}$* if for all formulae in L_1 , there exists an equivalent formula in L_2 , and vice versa. Equivalence in this section refers to equivalence over $\mathcal{K}$.

Definition 3.6. The *quantifier rank* of an ML formula $\text{qr}(\varphi)$ is calculated as follows:

- $\text{qr}(p) = 0$ for all $p \in P$.
- $\text{qr}(\neg\varphi) = \text{qr}(\varphi)$.

- $\text{qr}(\varphi \wedge \psi) = \max(\text{qr}(\varphi), \text{qr}(\psi))$.
- $\text{qr}(\Box_i \varphi) = 1 + \text{qr}(\varphi)$ for all integers $i \in [1, m]$.

We denote the set of ML formulae with a quantifier rank at most n as $\text{ML}[n]$.

Definition 3.7. Let $M = \langle S, R, V \rangle$ and $M' = \langle S', R', V' \rangle$ be Kripke models. A non-empty relation $B \subseteq S \times S'$ is a *bisimulation* iff for all $s \in S$ and $s' \in S'$ with $s B s'$, we have:

atoms For all $p \in P$, $s \in V(p)$ iff $s' \in V(p)$.

forth For all $i \in [1, m]$ and $t \in S$, if $s R_i t$, then there is $t' \in S'$ such that $s' R'_i t'$ and $t B t'$.

back For all $i \in [1, m]$ and $t' \in S'$, if $s' R'_i t'$, then there is $t \in S$ such that $s R_i t$ and $t B t'$.

We say that (M, s) and (M', s') are *bisimilar* if there is a bisimulation B such that $s B s'$.

Bisimulation was independently discovered in computer science [Par81], set theory [FH83], and philosophical logic [Ben76]. We are particularly interested in the work of van Benthem, who studied the equivalence between modal logic and fragments of first-order logic. Bisimilarity is a sufficient condition for two Kripke models agreeing on all ML formulae [Ben76]. Hennessey and Milner showed this condition is also necessary for finite Kripke models [HM85]. The following is a weaker version of Theorem 2.2 in [HM85].

Theorem 3.8. Let $M = \langle S, \sim, V \rangle$ and $M' = \langle S', \sim', V' \rangle$ be Kripke models such that S, S' are finite. Let $s \in S$ and $s' \in S'$. Then, (M, s) and (M', s') are bisimilar iff for all ML formulae φ , we have $M, s \models \varphi \Leftrightarrow M', s' \models \varphi$.

3.1.2 The Bisimulation Game

Stirling reformulated bisimulation as a game [Sti99]. We present Stirling's bisimulation game and show that it fits our idea of an EF game.

Definition 3.9. The n -round *ML EF game* is played on a pair of Kripke models $M = \langle S, R, V \rangle$ and $M' = \langle S', R', V' \rangle$ over the same atomic propositions P and positive integer m . A *configuration* is any pair (s, s') such that $s \in S$ and $s' \in S'$. The game has two players, the spoiler and duplicator, with an initial configuration (s, s') . When $n = 0$, the spoiler wins if there exists $p \in P$ which is true on s but not s' or vice versa. The duplicator wins otherwise. In round k with configuration (s, s') , the spoiler will choose a type of move and play accordingly:

- $\Box_i$: The spoiler either chooses $t \in S$ such that $s R_i t$ or $t' \in S'$ such that $s' R'_i t'$. If the spoiler chooses $t \in S$, then the duplicator chooses $t' \in S'$ such that $s' R'_i t'$. Otherwise, the duplicator chooses $t \in S$ such that $s R_i t$. The new configuration is (t, t') .

If a player cannot choose a state as prescribed, that player loses. If there exists $p \in P$ which is true on t but not t' or vice versa, the spoiler wins the game. Otherwise, the game continues. If the spoiler has not won after n rounds, then the duplicator wins. We write $(M, s) \equiv_n^{\text{ML}} (M', s')$ if the duplicator has an n -round winning strategy.

As we will see, the game for ML has the same manner of theorems as the original EF game. This game also follows the guiding philosophy of an EF game that we would like to maintain. The game can be visualised as having one pebble on each Kripke model. An initial configuration (s, s') represents a pebble on s and a pebble on s' . Instead of placing down a new pebble in each round, both boards have a single pebble that gets moved along a given relation. To make a move, there is relatively little computation as we only need to check if the next element satisfies the given binary relation. The winning condition is also local in that we only need to check what atomic propositions hold at each pebble.

The following proof of Theorem 3.11 is based on the proof of Theorem 8.21 in [DHK07].

Lemma 3.10. *Let $n \in \mathbb{N}$. There are finitely many formulae in $\text{ML}[n]$, up to equivalence.*

Proof. We induct on n . The $n = 0$ case is trivial. Suppose that $\text{ML}[n]$ is finite. Every formula in $\text{ML}[n+1]$ is a boolean combination of atomic propositions or formulae of the form $\Box_i \varphi$ for some $\varphi \in \text{ML}[n]$. Since there are finitely many atomic propositions and formulae of this form, there are finitely many non-equivalent Boolean combinations. Thus, $\text{ML}[n+1]$ is finite. $\square$

We will refer to the following theorem as an *EF-style Theorem*, analogous to Theorem 2.4.

Theorem 3.11. *Let M and M' be Kripke models with the same propositions P and positive integer m . Then, the following are equivalent:*

1. *For all formulae $\varphi \in \text{ML}[n]$, $M, s \models \varphi$ iff $M', s' \models \varphi$.*
2. *$(M, s) \equiv_n^{\text{ML}} (M', s')$.*

Proof. We proceed with induction on n . The base case is trivial. We only consider the inductive case. Our inductive hypothesis states that for all pointed models (M, s) and (M', s') , $(M, s) \equiv_n^{\text{ML}} (M', s')$ iff for all formulae $\varphi \in \text{ML}[n]$, we have $M, s \models \varphi \Leftrightarrow M', s' \models \varphi$.

Suppose that for all formulae $\varphi \in \text{ML}[n+1]$, we have $M, s \models \varphi \Leftrightarrow M', s' \models \varphi$. Without loss of generality, suppose the spoiler chooses to play a $\Box_i$ move first and picks $t \in S$ such that $s R_i t$. Assume there is no t' such that $s' R'_i t'$ with (M, t) and (M', t') agreeing on all formulae in $\text{ML}[n]$. That is, for every t' such that $s' R'_i t'$, there exists $\varphi_{t'} \in \text{ML}[n]$ such that $M, t \models \varphi_{t'}$ and $M', t' \not\models \varphi_{t'}$. Let $F = \{\varphi_{t'} : s' R'_i t'\}$. Since there are finitely many formulae

in $\text{ML}[n]$, up to logical equivalence, $\varphi = \bigvee_{\psi \in F} \psi$ can be written as a finite disjunction. We find that $\text{qr}(\Box_i \varphi) \leq n + 1$ and $M, s \models \Box_i \varphi$, but $M', s' \not\models \Box_i \varphi$. This contradicts our initial assumption. Thus, we have t' such that $s' R'_i t'$ with (M, t) and (M', t') agreeing on all formulae of quantifier rank at most n . This is the first move for our duplicator. The inductive hypothesis gives us an n -round winning strategy for (M, t) and (M', t') , so we are done.

We now prove the converse. Suppose that $(M, s) \equiv_{n+1}^{\text{ML}} (M', s')$. We want to show that for all $\varphi \in \text{ML}[n+1]$, we have $M, s \models \varphi \Leftrightarrow M', s' \models \varphi$. Every formula is a boolean combination of formulae found in $\text{ML}[n]$ or formulae of the form $\Box_i \varphi$ where $\varphi \in \text{ML}[n]$. From the inductive hypothesis, (M, s) and (M', s') agree on every formula in $\text{ML}[n]$, so it suffices to show that this is also true for formulae of the form $\Box_i \varphi$. Suppose that $M, s \models \Box_i \varphi$. For any t' the spoiler chooses such that $s' R'_i t'$, there is a t such that $s R_i t$ and $(M, t) \equiv_n^{\text{ML}} (M', t')$, from our initial assumption. From our inductive hypothesis, we have $M, t \models \varphi$ iff $M', t' \models \varphi$ for all $\varphi \in \text{ML}[n]$. Since $M, s \models \Box_i \varphi$, and $s R_i t$, we have $M, t \models \varphi$. Thus, $M', t' \models \varphi$. Then, for all t' such that $s' R'_i t'$, we have $M', t' \models \varphi$. We conclude that $M', s' \models \Box_i \varphi$. $\square$

From the EF-style Theorem, we get the following corollary that shows how the game can be used for inexpressibility results. We call the following an *EF-style Corollary*. The proof of this corollary is based on the proof of Corollary 3.2 in [EW00].

Corollary 3.12. *Since formulae are evaluated on a particular state of a Kripke model, a property is a class of pointed models. A property $\mathcal{P}$ is expressible in ML iff there exists $n \in \mathbb{N}$ such that for all $(M, s) \in \mathcal{P}$ and $(M', s') \notin \mathcal{P}$, we have $(M, s) \not\equiv_n^{\text{ML}} (M', s')$.*

Proof. Suppose φ expresses the property $\mathcal{P}$. Let $n = \text{qr}(\varphi)$. For all $(M, s) \in \mathcal{P}$ and $(M', s') \notin \mathcal{P}$, we have $M, s \models \varphi$ and $M', s' \not\models \varphi$. Theorem 3.11 states that $(M, s) \not\equiv_n^{\text{ML}} (M', s')$.

For the converse, suppose that for all $(M, s) \in \mathcal{P}$ and $(M', s') \notin \mathcal{P}$, we have $(M, s) \not\equiv_n^{\text{ML}} (M', s')$. Theorem 3.11 states that there exists $\varphi_{M, s}^{M', s'}$ with $\text{qr}(\varphi_{M, s}^{M', s'}) \leq n$ that is true on (M, s) , but not (M', s') . We define φ as follows:

$$\varphi = \bigvee_{(M, s) \in \mathcal{P}} \bigwedge_{(M', s') \notin \mathcal{P}} \varphi_{M, s}^{M', s'}.$$

Since there are finitely many formulae in $\text{ML}[n]$, up to equivalence, φ can be written with finitely many terms. Thus, $\varphi \in \text{ML}$. We find that φ is true in all $(M, s) \in \mathcal{P}$, while false in all $(M', s') \notin \mathcal{P}$. We conclude that φ expresses $\mathcal{P}$. $\square$

3.1.3 The Standard Translation

In this thesis, we explore how various modal logics and their extensions relate to fragments of classical logic. We now present the *standard translation*, the first clue to what fragment of classical logic corresponds with modal logic.

Definition 3.13. We define $\text{FO}[R_1, \dots, R_m]$ as the set of first-order logic formulae over $\sigma = \{R_1, \dots, R_m, p_1, \dots, p_n\}$, with binary relation symbols $R_1, \dots, R_m$ and unary relation symbols $p_1, \dots, p_n$. For formulae $\varphi(x_1, \dots, x_k) \in \text{FO}[R_1, \dots, R_m]$ and $s_1, \dots, s_k \in S$, we write that $\varphi(s_1, \dots, s_k)$ is true in M as $M, s_1, \dots, s_k \models \varphi(x_1, \dots, x_k)$. We write $\text{FO}^k[R_1, \dots, R_m]$ for the fragment that uses at most k unique variable names.

Definition 3.14. Let x be a variable. The standard translation $\text{ST}_x : \text{ML} \rightarrow \text{FO}[R_1, \dots, R_m]$ is a function that translates a modal formula into a first-order formula, and is defined as follows:

$$\begin{aligned} \text{ST}_x(p) &= p(x), \\ \text{ST}_x(\neg\varphi) &= \neg\text{ST}_x(\varphi), \\ \text{ST}_x(\varphi \wedge \psi) &= \text{ST}_x(\varphi) \wedge \text{ST}_x(\psi), \\ \text{ST}_x(\Box_i \varphi) &= \forall y(x R_i y \rightarrow \text{ST}_y(\varphi)). \end{aligned}$$

From the standard translation, $\text{ST}_x(\Diamond_i \varphi) = \exists y(x R_i y \wedge \text{ST}_y(\varphi))$.

Proposition 3.15. *Every modal formula is equivalent to an $\text{FO}[R_1, \dots, R_m]$ formula with one free variable and at most 2 variable names.*

Proof. Every modal formula is equivalent to its standard translation. Note that the standard translation only requires two variable names, which can be achieved by alternating variable names in nested modal formulae. For example, the formula $\Box_1 \Box_1 \Diamond_2 p$ translates to

$$\forall y(x R_1 y \rightarrow \forall x(y R_1 x \rightarrow \exists y(x R_2 y \wedge p(y))).$$

□

In the next section, we see this framework in action to define epistemic logic.

3.2 Epistemic Logic

Epistemic logic (EL) is concerned with reasoning about the knowledge of individuals and groups. We describe the syntax and semantics of EL laid out by Hintikka [Hin62] using the framework in Section 3.1. In EL, individuals are referred to as agents, denoted as elements in

a finite set A . EL introduces the knowledge operator K_a , where a is some agent in A . The proposition $K_a\varphi$ is read as “Agent a knows φ ”.

Definition 3.16. We define EL as the modal logic over modalities K_a for $a \in A$ and propositions $P = \{p_1, \dots, p_n\}$. That is, the *EL formulae* φ are defined by the rule

$$\varphi := p \mid \neg\varphi \mid \varphi \wedge \varphi \mid K_a\varphi,$$

where $p \in P$ and $a \in A$. We denote the dual of K_a as $\widehat{K}_a$.

We model the world as a set of states that the world could be in. In each state, each atomic proposition is either true or false. Generally, agents are modelled with incomplete information. This means there exist states that they cannot distinguish between. If states s and s' are indistinguishable to agent a and we are in state s , agent a cannot tell if we are in state s or state s' . To represent this, we give each agent an equivalence relation where two states are equivalent if the agent cannot distinguish between them.

Definition 3.17. An *epistemic model* is a Kripke model $M = \langle S, \sim, V \rangle$, where $\sim$ assigns each agent a an equivalence relation $\sim(a)$, denoted as $\sim_a$.

From the semantics of modal logic, recall that

$$M, s \models K_a\varphi \text{ iff for all } s' \in S \text{ such that } s \sim_a s', \text{ we have } M, s' \models \varphi.$$

That is, for an agent a to know φ is true, we need to have φ be true in every state that the agent thinks we could be in. Rather than explicitly assigning each agent the facts that they know, the facts they know are a result of what states they cannot distinguish between.

Example 3.18. The following is a version of the consecutive numbers game found in Example 2.4 of [DHK07]. Two agents, Alice (a) and Bob (b), are facing each other. They know there is a number on each of their foreheads from the set $\{1, 2, 3\}$, and so they can see the other person’s number. They also know the numbers are consecutive. Alice sees a 2 on Bob’s forehead, and Bob sees 1 on Alice’s.

To represent this scenario, we can make an epistemic model. The set of agents $A = \{a, b\}$ and propositions a_n, b_n to represent that Alice has the number n on her forehead, and Bob has the number n on his forehead, respectively. A state is characterised by the number on each forehead. We use an ordered pair (m, n) , where m is Alice’s number and n is Bob’s number. The set of states is $S = \{(1, 2), (2, 3), (2, 1), (3, 2)\}$. Figure 3.1 graphically displays the epistemic model for our game. Lines labelled a connect states that Alice cannot distinguish between. For example, Alice sees a 2 on Bob’s forehead, and so she cannot tell if she has a 1 or

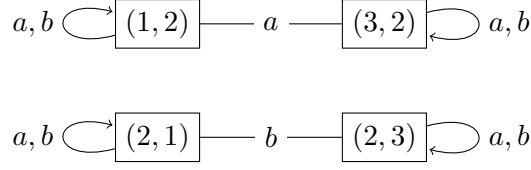


Figure 3.1: Epistemic Model for Consecutive Numbers with $\{1, 2, 3\}$

a 3. We also have loops to represent the fact that each state is indistinguishable from itself. EL formulae can be evaluated with this epistemic model, from the state $(1, 2)$. In the state $(1, 2)$, the sentence $K_a b_2$ holds as every world that Alice thinks is possible has Bob with a 2 on his forehead. Since Bob sees a 1 on Alice's forehead, the epistemic model tells us that Bob knows the only possible world is $(1, 2)$. Thus, $K_b b_2 \wedge K_b K_a b_2$ holds.

With $A = \{a_1, \dots, a_m\}$, the standard translation tells us that every EL formula is equivalent over epistemic models to a formula in $\text{FO}^2[\sim_{a_1}, \dots, \sim_{a_m}]$. We use the EL EF game from our framework to show that the converse does not hold. Recall the EL EF game.

Definition 3.19. The n -round *EL EF game* is played on a pair of epistemic models $M = \langle S, \sim, V \rangle$ and $M' = \langle S', \sim', V' \rangle$ over the same propositions P and agents A . The game is played by the spoiler and duplicator, with an initial configuration (s, s') . When $n = 0$, the spoiler wins if there exists $p \in P$ which is true on s but not s' or vice versa. The duplicator wins otherwise. In round k with configuration (s, s') , the spoiler will choose a type of move and play accordingly:

K_a : The spoiler either chooses $t \in S$ such that $s \sim_a t$ or $t' \in S'$ such that $s' \sim'_a t'$. If the spoiler chooses $t \in S$, then the duplicator chooses $t' \in S'$ such that $s' \sim'_a t'$. Otherwise, the duplicator chooses $t \in S$ such that $s \sim_a t$. The new configuration is (t, t') .

If a player cannot choose a state as prescribed, that player loses. If there exists $p \in P$ which is true on t but not t' or vice versa, the spoiler wins the game. Otherwise, the game continues. If the spoiler has not won after n rounds, then the duplicator wins. We write $(M, s) \equiv_n^{\text{EL}} (M', s')$ if the duplicator has an n -round winning strategy.

Corollary 3.20. A property $\mathcal{P}$ is *inexpressible in EL* iff for all $n \in \mathbb{N}$, there exists $(M, s) \in \mathcal{P}$ and $(M', s') \notin \mathcal{P}$ such that $(M, s) \not\equiv_n^{\text{ML}} (M', s')$.

The following proposition proves that EL is strictly less expressive than $\text{FO}^2[\sim_{a_1}, \dots, \sim_{a_m}]$.

Proposition 3.21. The formula $\exists y \neg(x \sim_a y)$ is not expressible in EL.

Proof. Let the set of agents $A = \{a\}$ and the set of propositions $P = \emptyset$. We define the epistemic models $M = \langle S, \sim, \emptyset \rangle$ and $M' = \langle S', \sim', \emptyset \rangle$, where $S = \{s\}$, $S' = \{s, s'\}$ and $\sim_a$

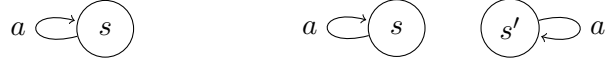


Figure 3.2: M (Left) and M' (Right)

, $\sim'_a$ are the identity relations on S and S' , respectively. The models are depicted in Figure 3.2.

We can show that (M, s) and (M', s) are bisimilar. Consider the relation $R \subseteq S \times S'$ defined as $R = \{(s, s)\}$. The **atoms** condition is vacuously true. The only t such that $s \sim_a t$ is $t = s$. We have $s \sim_a s$, $s \sim'_a s$ and $s R s$, and so the **forth** condition holds. Similarly, the back condition holds. Thus, R is a bisimulation. Then, Theorem 3.8 tells us that (M, s) and (M', s) agree on every EL formula.

Equivalently, we can show this through the EL game. We have $(M, s) \equiv_0^{\text{EL}} (M', s)$ as there are no propositions. For the inductive case, suppose that $(M, s) \equiv_n^{\text{EL}} (M', s)$. We start the game with the configuration (s, s) . When the spoiler plays a K_a move, they can only move the pebble back onto $s \in S$ on either structure. The duplicator can also do the same on the other structure. We have finished one round, ending with the configuration (s, s) . Our inductive hypothesis tells us that the duplicator has a winning strategy for the next n moves. Thus, $(M, s) \equiv_{n+1}^{\text{EL}} (M', s)$. By induction, we have shown that $(M, s) \equiv_\infty^{\text{EL}} (M', s)$. Corollary 3.20 tells us that (M, s) and (M', s) agree on every EL formula.

We see that $M, s \not\models \exists y \neg(x \sim_a y)$, but $M', s \models \exists y \neg(x \sim_a y)$, and so EL cannot express the property $\exists y \neg(x \sim_a y)$. $\square$

3.3 Common Knowledge

Common knowledge captures the idea of information that comes from social interaction. When a fact is announced, everyone who hears the announcement now knows the fact. With perfectly logical agents, everyone who hears the announcement also knows that everyone knows the fact. They also know that everyone knows everyone knows the fact. We can continue this to get an “infinite series” of knowledge. We can formalise common knowledge in two ways: defining new operators through EL [Lew67; Sch72] and creating a new relation in our epistemic models [Fri69]. These approaches are equivalent.

Definition 3.22. Let A be the set of agents and let G be a non-empty subset of A . Then, we define that “Everybody in G knows φ ” as

$$E_G \varphi = \bigwedge_{a \in G} K_a \varphi.$$

We define $E_G^0\varphi = \varphi$ and $E_G^{k+1} = E_GE_G^k\varphi$ for all $k \in \mathbb{N}$. We define “ φ is common knowledge for everyone in G ” as

$$C_G\varphi = \bigwedge_{k=0}^{\infty} E_G^k\varphi.$$

Common epistemic logic (CEL) extends EL by adding the additional rule that $E_G\varphi$ and $C_G\varphi$ are CEL formulae, if φ is a CEL formula and $G \subseteq A$.

Definition 3.23. Let $M = \langle S, \sim, V \rangle$ be an epistemic model and $s \in S$. For all $G \subseteq A$, define $\sim_{E_G} = \bigcup_{a \in G} \sim_a$ and $\sim_{C_G}$ as the transitive closure of $\sim_{E_G}$. That is, $\sim_{C_G}$ is the smallest relation such that $\sim_{E_G} \subseteq \sim_{C_G}$ and $\sim_{C_G}$ is transitive. Then,

- $M, s \models E_G\varphi$ iff for all $s' \in S$ such that $s \sim_{E_G} s'$, we have $M, s' \models \varphi$.
- $M, s \models C_G\varphi$ iff for all $s' \in S$ such that $s \sim_{C_G} s'$, we have $M, s' \models \varphi$.

Example 3.24. We extend the consecutive numbers game from Example 3.18, based on Example 2.4 in [DHK07], to illustrate the difficulty of obtaining common knowledge. Alice and Bob are facing each other. They each have a number on their foreheads, and they know the numbers are consecutive. Now, the numbers can be any number in $\mathbb{N}$. Let $A = \{a, b\}$ for Alice and Bob. Let the proposition a_n mean that Alice has the number n on her forehead and b_n mean that Bob has the number n on his forehead, for $n \in \mathbb{N}$. The epistemic model M for this scenario can be seen in Figure 3.3, where states indistinguishable to Alice are connected with an **orange** line and states indistinguishable to Bob with **blue**. The relations for indistinguishability are reflexive, so loops have been omitted from the figure for clarity.

Suppose that Alice sees a 2 on Bob’s forehead, and Bob sees 1 on Alice’s. That is, we are in state $(1, 2)$. Then, $M, (1, 2) \models \neg a_3 \wedge \neg E_A \neg a_3$. While Alice does not have a 3 on her forehead, not everyone knows this. In particular, Alice sees a 2 on Bob’s forehead, so she thinks she may have a 3. For $E_A \neg a_3$ to be true, everyone in A must know $\neg a_3$. Thus, it suffices that $\neg K_a \neg a_3$ holds to show that $E_A \neg a_3$ is not true. Graphically, $E_A \neg a_3$ is true in state $(1, 2)$ exactly when there are no lines from $(1, 2)$ to a state where a_3 is true. Since there is a line from $(1, 2)$ to $(3, 2)$, we must have $\neg E_A \neg a_3$.

We find that $M, (1, 2) \models E_A \neg b_4 \wedge \neg E_A E_A \neg b_4$. Everyone knows that b_4 is false, but not everyone knows that everyone knows b_4 is false. The statement $E_A E_A \neg b_4$ is true exactly when there is no line from $(1, 2)$ to a state where $E_A \neg b_4$. Thus, $E_A E_A \neg b_4$ is true exactly when there is no path of length 2 from $(1, 2)$ to a state where $\neg b_4$ is true. We find that $\neg E_A E_A \neg b_4$ is true at $(1, 2)$ as $(1, 2) \sim_a (3, 2)$ and $(3, 2) \sim_b (3, 4)$.

Continuing this pattern, we have $M, (1, 2) \models E_A E_A \neg a_5 \wedge \neg E_A E_A E_A \neg a_5$ and so on. We find

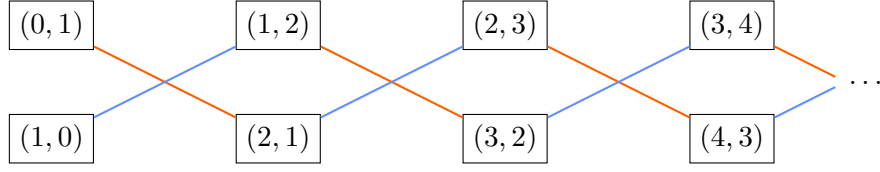


Figure 3.3: Epistemic Model for Consecutive Numbers with $\mathbb{N}$

that $M, (1, 2) \models \neg C_A \neg a_{123} \wedge C_A \neg b_{123}$. Using our definition of common knowledge, $C\varphi$ is true at $(1, 2)$ exactly when there is no path of any length from $(1, 2)$ to a state where φ is true. Thus, $C_A \neg b_{123}$ holds as we cannot reach a state of the form $(x, 123)$ from $(1, 2)$, since $(1, 2)$ is only connected to states where Bob has an even number. Similarly, $\neg C_A \neg a_{123}$ holds at $(1, 2)$ as we can get to the state $(123, 124)$. Alice seeing a 2 on Bob's head and Bob seeing a 1 on Alice's head is not enough for it to be common knowledge that Alice does not have 123.

Using these ideas, we can extend the EL game to create a game for CEL. Baltag et al. give us the following C_G move in [BMS98]. We also provide a natural move for E_G .

Definition 3.25. Consider $M = \langle S, \sim, V \rangle$ and $M' = \langle S', \sim', V' \rangle$ over the same atomic propositions P and agents A . We introduce several new moves to the EL EF game to create the CEL EF game. Let the initial configuration be (s, s') . The moves are as follows:

E_G : The spoiler either chooses $t \in S$ such that $s \sim_{E_G} t$ or $t' \in S'$ such that $s' \sim'_{E_G} t'$.
If the spoiler chooses $t \in S$, then the duplicator chooses $t' \in S'$ such that $s' \sim'_{E_G} t'$.
Otherwise, the duplicator chooses $t \in S$ such that $s \sim_{E_G} t$.

C_G : The spoiler either chooses $t \in S$ such that $s \sim_{C_G} t$ or $t' \in S'$ such that $s' \sim'_{C_G} t'$.
If the spoiler chooses $t \in S$, then the duplicator chooses $t' \in S'$ such that $s' \sim'_{C_G} t'$.
Otherwise, the duplicator chooses $t \in S$ such that $s \sim_{C_G} t$.

After each move, the new configuration is (t, t') . We write $(M, s) \equiv_n^{\text{CEL}} (M', s')$ if the duplicator has an n -round winning strategy in the CEL EF game.

In the E_G move, the spoiler can choose to go to any state that is indistinguishable by an agent in G . The duplicator then chooses a state that is indistinguishable by some (possibly different) agent in G , in the other structure. For the C_G move, the spoiler instead chooses a state that is connected to the current state by the edges of the relations of agents in G , by definition of the transitive closure. Rather than choosing a single edge, the spoiler chooses an arbitrarily long path to move the pebble along. The duplicator does the same on the other structure.

The move for C_G used in the literature is not ideal. To use the move, we require the computation of the transitive closure outside of the game. This does not follow the philosophy of EF games

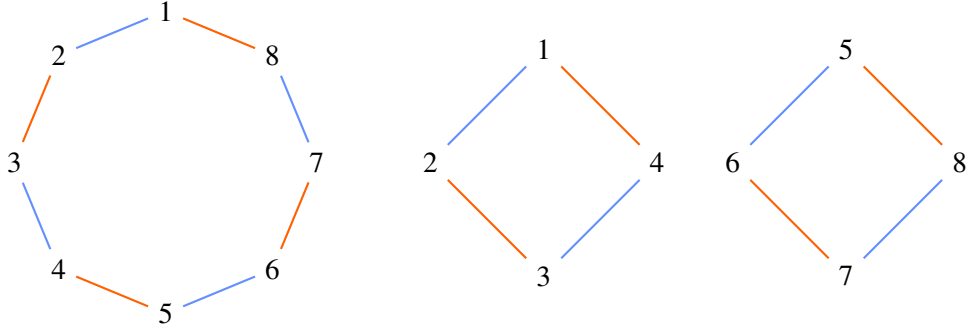


Figure 3.4: M_1^1 (Left) and M_2^1 (Right)

that we would like to follow, as it can be arbitrarily complex to verify that there exists a path between the previous state and the current state. We address this in Section 6.2, where we introduce a move that incorporates the computation of the transitive closure into the game.

Definition 3.26. The *quantifier rank* of a CEL formula $\text{qr}(\varphi)$ is calculated similarly to the quantifier rank of an EL formula. We also have

- $\text{qr}(E_G\varphi) = \text{qr}(C_G\varphi) = 1 + \text{qr}(\varphi)$ for all $G \subseteq A$.

We denote the set of CEL formulae with a quantifier rank at most n as $\text{CEL}[n]$.

Theorem 3.27. Let M and M' be epistemic models with the same propositions and agents. Then, the following are equivalent:

1. For all formulae $\varphi \in \text{CEL}[n]$, $M, s \models \varphi$ iff $M', s' \models \varphi$.
2. $(M, s) \equiv_n^{\text{CEL}} (M', s')$.

Proof. The proof is similar to that of Theorem 3.11. □

The common knowledge operator uses a conjunction with infinite terms. These conjunctions generally cannot be expressed in first-order logic. We use an EF game to show that common knowledge cannot be expressed as a first-order formula with 1 free variable. To analyse first-order formulae with 1 free variable, we play an EF game with one pebble already placed on each structure, as suggested by Corollary 2.19. Since every EL formula is equivalent to a first-order formula, we will also find that common knowledge cannot be expressed in EL.

Proposition 3.28. The formula $C_A p$ cannot be expressed in $\text{FO}[\sim_a, \sim_b]$ with 1 free variable.

Let $A = \{a, b\}$ and $P = \{p\}$. We define epistemic models as seen in Figure 3.4. Let M_1^n be a cycle of size 2^{n+2} and M_2^n be two cycles of size 2^{n+1} . Each cycle alternates between being

indistinguishable to a and b . The states connected with **orange** are indistinguishable to a and those connected with **blue** are indistinguishable to b . Note that a and b also consider each state indistinguishable from itself. We have omitted this from Figure 3.4 for clarity. The relations $\sim_a$ and $\sim_b$ are then equivalence relations. In both models, we let p be true in every state but 8.

We can use a similar proof to Example 2.3 to show that $(M_1^n, 1) \equiv_n^{\text{FO}} (M_2^n, 1)$ for all $n \in \mathbb{N}$. From Corollary 2.19, for every formula of the form $\varphi(x)$, where x is a free variable, we have $M_1^n, 1 \models \varphi(x)$ and $M_2^n, 1 \models \varphi(x)$. In M_1^n , we can reach the state 2^{n+2} from state 1, so $M_1^n \not\models C_{AP}$. In M_2^n , we cannot reach the state 2^{n+2} from state 1, so $M_2^n \models C_{AP}$. Thus, C_{AP} cannot be expressed in $\text{FO}[\sim_a, \sim_b]$ with 1 free variable.

Proposition 3.29. *The formula $C_A\varphi$ can be expressed as some $\psi(s)$ in $\text{DSO}[\sim_a, \sim_b]$.*

Proof. Suppose that φ translates to $\varphi'(x)$. Then, we have

$$\begin{aligned} \psi(s) = & \exists R \forall x \forall y \forall z ((x \sim_a y \vee x \sim_b y \rightarrow xRy) \\ & \wedge (xRy \wedge yRz \rightarrow xRz) \\ & \wedge (\neg\varphi'(x) \rightarrow \neg(sRx))). \end{aligned}$$

That is, there exists a transitive binary relation R that contains $\sim_a$ and $\sim_b$ where s is only connected to states where φ holds. We find that $\psi(s)$ is equivalent over epistemic models to $C_A\varphi$, where $A = \{a, b\}$. This translation can be extended to arbitrarily many agents. $\square$

Using a similar proof to Proposition 3.21, we get the following proposition.

Proposition 3.30. *The formula $\exists y \neg(x \sim_a y)$ is not expressible in CEL.*

We summarise how EL and CEL compare to classical logics in terms of expressibility. Notably, inequalities have been found through original examples of EF games. We find that

$$\text{EL} < \text{FO}^2[\sim_{a_1}, \dots, \sim_{a_n}] \text{ and } \text{FO}[\sim_{a_1}, \dots, \sim_{a_n}] \neq \text{CEL} < \text{DSO}[\sim_{a_1}, \dots, \sim_{a_n}].$$

First-order logic and CEL are incomparable in terms of expressibility, as there exist formulae in each logic that cannot be expressed in the other. Since $\exists y \neg(x \sim_a y)$ can be expressed in $\text{DSO}[\sim_{a_1}, \dots, \sim_{a_n}]$, but not CEL, we find that CEL is strictly less expressive than dyadic second-order logic. In fact, CEL is strictly less expressive than monadic second-order logic. We provide a translation in Example 6.4.

This concludes the background of epistemic logic needed to define temporal epistemic logic and its associated game.

3.4 The van Benthem Theorem

Before we move on to temporal logic, we return to a question first hinted at by the standard translation. What is the fragment of first-order logic that is equivalent to modal logic? In [Ben76], van Benthem provides us with the answer in his seminal theorem.

Definition 3.31. A formula $\varphi(x)$ in $\text{FO}[R_1, \dots, R_m]$ is *bisimulation-invariant* if (M, s) and (M', s') being bisimilar implies that $M, s \models \varphi(x)$ iff $M', s' \models \varphi(x)$.

Theorem 3.32 (van Benthem). *Let $\varphi(x) \in \text{FO}[R_1, \dots, R_m]$. The following are equivalent:*

1. $\varphi(x)$ is bisimulation-invariant.
2. $\varphi(x)$ is logically equivalent to an ML formula.

This discovery led to the classification of various extensions to modal logic as the bisimulation-invariant fragments of some classical logic. For example, Moller and Rabinovich found CTL^* , a temporal logic we will introduce later, to be the bisimulation invariant fragment of monadic path logic, a fragment of MSO [MR99].

Otto provides us with a proof of van Benthem's Theorem through EF games [Ott03]. However, this proof only applies to modal logics with one modality. We simplify his proof and extend the result to arbitrarily many modalities. The fact that (2) implies (1) is a consequence of Theorem 3.11. We prove that (1) implies (2).

For the proof, we introduce Gaifman's notion of locality [Gai82]. Informally, a formula is *local* if whether it holds at a state is determined by a neighbourhood around the state. We use EF games to prove that bisimulation-invariant formulae are local, and then show that local formulae can be defined in modal logic.

Definition 3.33. In a Kripke model $M = \langle S, R, V \rangle$, define the ℓ -neighbourhood of $s \in S$ as the set $N^\ell(s)$ of all states reachable from s on a directed path of length at most ℓ , using edges from $R_1, \dots, R_m$. We call these *R-paths*. Let $M|_{N^\ell(s)} = \langle N^\ell(s), R', V' \rangle$, where $V'(p) = V(p) \cap N^\ell(s)$ and $s_1 R'_i s_2$ iff $s_1, s_2 \in N^\ell(s)$ and there exists an R -path of length at most ℓ that uses $s_1 R_i s_2$ as an edge.

Definition 3.34. The *tree unravelling* of M from s , denoted $M_s^* = \langle S^*, R^*, V^* \rangle$, is defined as follows. Let S^* be the set of all R -paths from s and let $L(r)$ be the last state of $r \in S^*$. For $r, r' \in S^*$, we have $r R_i^* r'$ iff r' without $L(r')$ is r and the edge connecting r to $L(r')$ is from R_i . For all $r \in S^*$, we have $r \in V^*(p)$ iff $L(r) \in V(p)$.

Example 3.35. We see the tree unravelling of a simple Kripke model M from s and its ℓ -neighbourhood in Figure 3.5. The orange edges represents R_1 and blue edges represent R_2 .

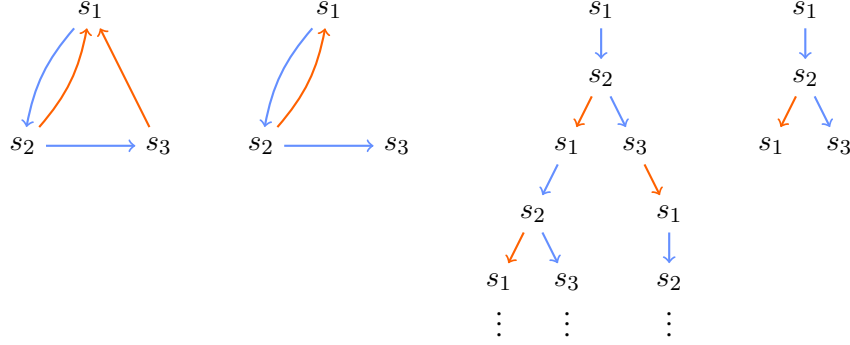


Figure 3.5: In order, we have M , $M|_{N^2(s_1)}$, $M_{s_1}^*$ and $M_{s_1}^*|_{N^2(s_1)}$

While both $s_1, s_3 \in N^2(s_1)$, there is no edge from s_3 to s_1 as there is no R -path of length at most 2 which uses this edge.

Lemma 3.36. *We have $(M, s) \equiv_{\infty}^{ML} (M_s^*, s)$. We also have $(M, s) \equiv_{\ell}^{ML} (M|_{N^{\ell}(s)}, s)$. Our example shows that $(M, s) \equiv_{\infty}^{ML} (M', s')$ does not imply that $(M|_{N^{\ell}(s)}, s) \equiv_{\infty}^{ML} (M_s^*|_{N^{\ell}(s')}, s')$. However, we have*

$$(M|_{N^{\ell}(s)}, s) \equiv_{\ell}^{ML} (M, s) \equiv_{\infty}^{ML} (M_s^*, s) \equiv_{\ell}^{ML} (M_s^*|_{N^{\ell}(s')}, s').$$

Definition 3.37. A formula $\varphi(x)$ is ℓ -local if for all Kripke modals M , we have

$$M, s \models \varphi(x) \iff M|_{N^{\ell}(s)}, s \models \varphi(x).$$

Lemma 3.38. *If $\varphi(x)$ is bisimulation-invariant, then $\varphi(x)$ is ℓ -local for some $\ell \in \mathbb{N}$.*

Proof. Let $\varphi(x)$ be bisimulation-invariant, $q = \text{qr}(\varphi(x))$ and $\ell = 2^q$. Let M be a Kripke model with $s \in S$. To show that $\varphi(x)$ is local, we will find pointed models (M', s') and (M'', s'') such that (M', s') is bisimilar to (M, s) , and (M'', s'') is bisimilar to $(M|_{N^{\ell}(s)}, s)$. We will also show that $(M', s') \equiv_q^{\text{FO}} (M'', s'')$. Then,

$$\begin{aligned} M, s \models \varphi(x) &\iff M', s' \models \varphi(x) && \text{(bisimulation-invariant)} \\ &\iff M'', s'' \models \varphi(x) && \text{(EF Theorem)} \\ &\iff M|_{N^{\ell}(s)}, s \models \varphi(x). && \text{(bisimulation-invariant)} \end{aligned}$$

We define M' to be $q + 1$ disjoint copies of M_s^* together with q disjoint copies of $(M|_{N^{\ell}(s)})_s^*$, where s' is the root of one copy of M_s^* . Since the other copies are disjoint, the back and forth conditions of bisimulation cannot access them, and so (M', s) is bisimilar to (M_s^*, s) , which

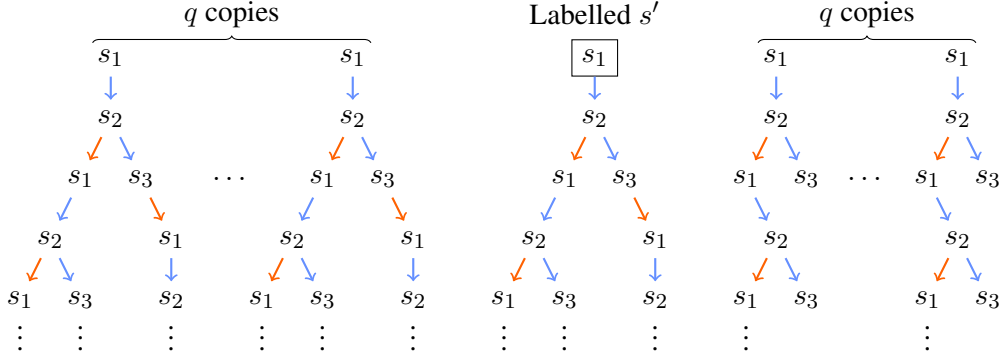


Figure 3.6: M' generated from M and $M|_{N^2(s_1)}$ in Figure 3.5

is bisimilar to (M, s) . Similarly, define M'' to be q disjoint copies of M_s^* and $q + 1$ disjoint copies of $(M|_{N^\ell(s)})_s^*$, where s'' is the root of one copy of $(M|_{N^\ell(s)})_s^*$. We have M'' bisimilar to $(M|_{N^\ell(s)}, s)$. In Figure 3.6, we depict M' for our running example M .

We now prove that $(M', s') \equiv_q^{\text{FO}} (M'', s'')$. The game starts with a pebble on s' and a pebble on s'' . The distance between two pebbles is the length of the shortest path between them. If there is no such path, the length is infinite. For round k , define the critical distance $d_k = 2^{q-k}$. We want to maintain the invariant that for all $i, j \leq k$, either $d_{M'}(s_i, s_j) = d_{M''}(s'_i, s'_j)$ or both $d_{M'}(s_i, s_j)$ and $d_{M''}(s'_i, s'_j)$ are greater than the critical distance.

Suppose that there are $k - 1$ rounds played thus far with pebbles $s_1, \dots, s_k \in M'$ and $s'_1, \dots, s'_k \in M''$. If the spoiler places a pebble within the critical distance d_k of another pebble s_i , the duplicator responds with the same element in the same structure as the corresponding pebble s'_i . By construction, s' and s'' have the same 2^q -neighbourhood. If the first pebble is placed within the critical distance 2^{q-1} of s' or s'' , then we have a matching element in the other structure. The next pebble has a critical distance of 2^{q-2} , which means that this pebble can have a finite distance of at most $2^{q-1} + 2^{q-2}$ from s' or s'' . Continuing this pattern, placing all q pebbles within critical distance will result in every pebble remaining in the 2^q -neighbourhood of s' and s'' . Thus, there is a partial isomorphism.

Otherwise, if the spoiler places a pebble in a copy of M_s^* , the duplicator responds with the same element in an isomorphic copy of M_s^* that contains no pebbles. Similarly, the duplicator does the same if the spoiler places a pebble in a copy of $(M|_{N^\ell(s)})_s^*$. There is always a copy with no pebbles available, as there are q isomorphic copies of each structure in M' and M'' . We have maintained our invariant. Similar to Example 2.3, this guarantees a partial isomorphism, and thus $(M', s') \equiv_q^{\text{FO}} (M'', s'')$. $\square$

Lemma 3.39. *Let $\ell \in \mathbb{N}$. If $\varphi(x)$ is also ℓ -local, then $\varphi(x)$ is expressible in ML.*

Proof. Let $\varphi(x)$ be ℓ -local and bisimulation-invariant. Suppose that $(M, s) \equiv_{\ell}^{\text{ML}} (M', s')$ and $M, s \models \varphi(x)$. Then, to show that $\varphi(x)$ is expressible in ML, Corollary 3.12 states that it suffices to show $M', s' \models \varphi(x)$. We first note that $(M|_{N^{\ell}(s)}, s) \equiv_{\ell}^{\text{ML}} (M'|_{N^{\ell}(s)}, s')$. This is because the duplicator's strategy only moves the pebbles to states in $N^{\ell}(s)$ as the game is only ℓ -rounds long. From Lemma 3.36, we have

$$(M_s^*|_{N^{\ell}(s)}, s) \equiv_{\ell}^{\text{ML}} (M|_{N^{\ell}(s)}, s) \equiv_{\ell}^{\text{ML}} (M'|_{N^{\ell}(s)}, s') \equiv_{\ell}^{\text{ML}} (M_{s'}'^*|_{N^{\ell}(s')}, s').$$

The EF-style Theorem implies transitivity, and so $(M_s^*|_{N^{\ell}(s)}, s) \equiv_{\ell}^{\text{ML}} (M_{s'}'^*|_{N^{\ell}(s')}, s')$. We find that both $M_s^*|_{N^{\ell}(s)}$ and $M_{s'}'^*|_{N^{\ell}(s')}$ are trees with heights of at most ℓ . There cannot be a k -round winning strategy for the spoiler where $k > \ell$ as the spoiler cannot make more than ℓ moves. Thus, the duplicator has an ∞ -round winning strategy. From the EF-style Theorem, they agree on all ML formulae. The Hennessey-Milner Theorem tells us that $(M_s^*|_{N^{\ell}(s)}, s)$ and $(M_{s'}'^*|_{N^{\ell}(s')}, s')$ are bisimilar, since the trees are finite. We conclude that $M', s' \models \varphi(x)$:

$$\begin{aligned} M, s \models \varphi(x) &\iff M_s^*, s \models \varphi(x) && \text{(bisimulation-invariant)} \\ &\iff M_s^*|_{N^{\ell}(s)}, s \models \varphi(x) && (\ell\text{-local}) \\ &\iff M_{s'}'^*|_{N^{\ell}(s')}, s' \models \varphi(x) && \text{(bisimulation-invariant)} \\ &\iff M_{s'}'^*, s' \models \varphi(x) && (\ell\text{-local}) \\ &\iff M', s' \models \varphi(x). && \text{(bisimulation-invariant)} \quad \square \end{aligned}$$

Chapter 4

Temporal Logic

Temporal logics allow us to express and reason about events and when they happen. To do so, we must agree on how time is modelled. Is time continuous or comprised of discrete points? Is time bounded or unbounded? Is time linear, or can time branch into multiple possibilities? In this chapter, we consider that time is discrete and can be bounded or unbounded. This chapter aims to cover the temporal background for defining temporal epistemic logics. Along the way, we revisit classical results and concepts, giving original proofs and reformulations in terms of EF games.

We first consider time as a linear order of points. The first section defines unary temporal logic (UTL), based on Prior's tense logic [Pri57], using our modal framework. We then introduce linear temporal logic (LTL) [Kam68], an extension of UTL, and associated EF games for each of its fragments [EW00]. We now delve into various expressibility results about UTL and LTL. This features an original proof that LTL is more expressive than UTL [Kam68]. We give a novel characterisation of Gabbay's separation property [Gab81] in LTL through EF games. This includes using games to show that a temporal logic does not have separation, and that LTL is initially equivalent to LTL without past-time operators. We also present a novel proof strategy for proving the equivalence of two logics. This strategy is used to prove that UTL is equivalent to 2-variable first-order logic [EVW02], and analyse other fragments of UTL.

We then consider time as a tree of points, with the present being the root of the tree. At any point, the present can branch into multiple possible futures. The branching temporal logic we examine is CTL^* [EH85]. We introduce an original EF game for CTL^* and use the game to extend a consequence of Gabbay's separation property: the addition of operators that reason about the past does not increase the expressibility of CTL^* [HT87]. We mainly follow the definitions in [BK08; Rab14].

4.1 Unary Temporal Logic

Time is modelled as a linearly-ordered set of points, and events are propositions that can be evaluated as true or false at any given point in time.

Example 4.1. If we model the events across a year, we could have 365 points, where each point represents one day. When playing a move-based game such as chess, the n th move can be seen as the n th point in time. Generally, we model time as the natural numbers or the integers, and we can give an external meaning for what points in time represent.

Definition 4.2. A *linear model* is a Kripke model, written as $\mathfrak{T} = (T, <, s, V)$, where T is a non-empty set, $<$ is a linear order relation on T , and s is a successor relation on T . We then have propositions $p_1, \dots, p_n$ with a valuation function $V(p) \subseteq T$. For common linear orders such as $\mathbb{N}$ or $\mathbb{Z}$, we define $s(x, y)$ to mean $x + 1 = y$.

Definition 4.3. *Unary temporal logic (UTL)* is a modal logic with modalities **X**, **Y**, **F**, **P** over propositions $P = \{p_1, \dots, p_n\}$. The *UTL formulae* φ are given by the rule

$$\varphi := p \mid \neg\varphi \mid \varphi \wedge \psi \mid \mathbf{X}\varphi \mid \mathbf{Y}\varphi \mid \mathbf{F}\varphi \mid \mathbf{P}\varphi,$$

where $p \in P$. We inductively define $\mathfrak{T}, t \models \varphi$ as follows:

- For atomic propositions p , $\mathfrak{T}, t \models p$ iff $t \in V(p)$ holds.
- $\mathfrak{T}, t \models \neg\varphi$ and $\mathfrak{T}, t \models \varphi \wedge \psi$ can be defined similar to before.
- $\mathfrak{T}, t \models \mathbf{X}\varphi$ iff there exists t' such that $s(t, t')$ and $\mathfrak{T}, t' \models \varphi$.
- $\mathfrak{T}, t \models \mathbf{Y}\varphi$ iff there exists t' such that $s(t', t)$ and $\mathfrak{T}, t' \models \varphi$.
- $\mathfrak{T}, t \models \mathbf{F}\varphi$ iff there exists $t' > t$ such that $\mathfrak{T}, t' \models \varphi$.
- $\mathfrak{T}, t \models \mathbf{P}\varphi$ iff there exists $t' < t$ such that $\mathfrak{T}, t' \models \varphi$.

UTL formulae φ can be thought of as propositions that are true or false at a given time point. Temporal operators state when these propositions occur, relative to the present point in time. The formula $\mathbf{X}\varphi$ can be read as “in the neXt time point, φ holds”. The formula $\mathbf{F}\varphi$ can be read as “eventually/in the Future, φ holds”. The dual of **F** is denoted as **G**, where $\mathbf{G}\varphi$ can be read as “Globally, in the future, φ holds”. The operators **X**, **F**, **G** express notions of future events.

The operators **Y**, **P**, **H** express notions of past events. The formula $\mathbf{Y}\varphi$ can be read as “Yesterday/in the previous time point, φ held”. The formula $\mathbf{P}\varphi$ can be read as “in the Past, φ held”. The dual of **P** is denoted as **H**, where $\mathbf{H}\varphi$ can be read as “Historically, φ has held”. We also have propositions which are false at all time points and propositions which are true at all

time points, denoted as $\perp$ and $\top$, respectively. This gives us an intuition for how the truth of a formula is determined. The operators **G** and **H** can be evaluated as follows:

- $\mathfrak{T}, t \models \mathbf{G}\varphi$ iff for all $t' > t$, we find that $\mathfrak{T}, t' \models \varphi$.
- $\mathfrak{T}, t \models \mathbf{H}\varphi$ iff for all $t' < t$, we find that $\mathfrak{T}, t' \models \varphi$.

As UTL is a modal logic, the framework also provides us with the following EF game.

Definition 4.4. The *UTL EF game* is played on a pair of linear models $\mathfrak{A}$ and $\mathfrak{B}$, over the same propositions P along with a prescribed initial configuration (x, y) . The game is played over n rounds, and each model has a single pebble. Thus, a configuration represents the current board state. In round i , starting with configuration (x, y) , the spoiler will choose a type of move and play accordingly. Each type of move relates to a particular temporal operator:

- X:** The spoiler either chooses $x' = x + 1$ or $y' = y + 1$. If the spoiler chooses $x' = x + 1$, the duplicator then chooses $y' = y + 1$. Otherwise, the duplicator chooses $x' = x + 1$.
- Y:** The spoiler either chooses $x' = x - 1$ or $y' = y - 1$. If the spoiler chooses $x' = x - 1$, the duplicator then chooses $y' = y - 1$. Otherwise, the duplicator chooses $x' = x - 1$.
- F:** The spoiler either chooses $x' > x$ or $y' > y$. If the spoiler chooses $x' > x$, the duplicator then chooses $y' > y$. Otherwise, the duplicator chooses $x' > x$.
- P:** The spoiler either chooses $x' < x$ or $y' < y$. If the spoiler chooses $x' < x$, the duplicator then chooses $y' < y$. Otherwise, the duplicator chooses $x' < x$.

If a player cannot choose x' or y' as prescribed, that player loses. Let the configuration after round i be (x', y') . If there exists an atomic proposition which is true at x' but not y' , or vice versa, then the spoiler wins the n -round UTL EF game. After n rounds, if the spoiler has not won, then the duplicator wins the n -round UTL EF game. If the duplicator has an n -round winning strategy, we denote this as $(\mathfrak{A}, x) \equiv_n^{\text{UTL}} (\mathfrak{B}, y)$.

Corollary 4.5. A property $\mathcal{K}$ is expressible in UTL iff there exists $n \in \mathbb{N}$ such that for all $(\mathfrak{A}, x) \in \mathcal{K}$ and $(\mathfrak{B}, y) \notin \mathcal{K}$ we have $(\mathfrak{A}, x) \not\equiv_n^{\text{UTL}} (\mathfrak{B}, y)$.

4.2 Linear Temporal Logic

Definition 4.6. Linear temporal logic (LTL) has the operators **X**, **Y**, **U**, **S** over propositions $P = \{p_1, \dots, p_n\}$. The LTL formulae φ are given by the rule

$$\varphi := p \mid \neg\varphi \mid \varphi \wedge \varphi \mid \mathbf{X}\varphi \mid \mathbf{Y}\varphi \mid \mathbf{F}\varphi \mid \mathbf{P}\varphi \mid \varphi \mathbf{U} \varphi \mid \varphi \mathbf{S} \varphi.$$

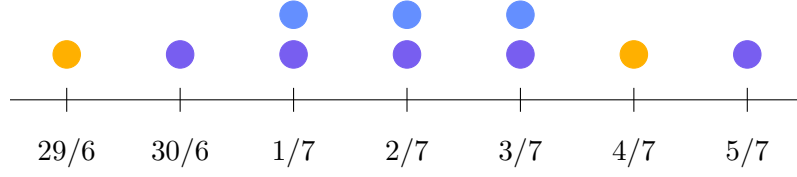


Figure 4.1: Sydney Weather Forecast from 29/6 to 5/7

We can extend the definition for UTL as follows:

- $\mathfrak{T}, t \models \varphi \text{ U } \psi$ iff there exists $t' > t$ where $\mathfrak{T}, t' \models \psi$ and $\mathfrak{T}, t'' \models \varphi$ for all $t < t'' < t'$.
- $\mathfrak{T}, t \models \varphi \text{ S } \psi$ iff there exists $t' < t$ where $\mathfrak{T}, t' \models \psi$ and $\mathfrak{T}, t'' \models \varphi$ for all $t' < t'' < t$.

The formula $\varphi \text{ U } \psi$ can be read as “ φ holds Until ψ holds”. We also have an equivalent operator for the past. The formula $\varphi \text{ S } \psi$ can be read as “ φ has held Since ψ held”.

Example 4.7. We model the weather forecast of the Sydney CBD from 29 June 2025 to 5 July 2025. Let our propositional variables be s, c, r . They denote the day being **sunny**, **cloudy** and **rainy**, respectively. We represent the propositions by circles with the colours from the previous sentence. The linear model can be visualised as depicted in Figure 4.1.

This week is not particularly sunny, with 3 days being both cloudy and rainy. The statement $r \text{ U } s$, read as “It is raining until it is sunny”, means that a (possibly empty) period of rain will start at the next time point and end when it is sunny. We have $r \text{ U } s$ hold on the 29th and 30th of June, as well as the 1st and 2nd of July. The statement does not hold true on any other day. The statement $\text{F}c$, read as “Eventually, it will be cloudy”, holds on every date except the 5th of July. This is because the final date, the 5th of July, is cloudy.

4.2.1 The Until Operator

Naturally, we ask whether U adds expressiveness to UTL. Can we really say more with U ? The answer is yes! In fact, Kamp introduced the until operator [Kam68] to combat Prior’s tense logic being unable to express certain crucial properties. From the standard translation, we know that every UTL formula is equivalent to a formula in $\text{FO}^2[<, s]$. We use the 2-variable EF game to provide an original proof of Kamp’s result that U cannot be expressed by $\text{FO}^2[<, s]$.

Proposition 4.8. *There is no UTL formula to express $a \text{ U } b$.*

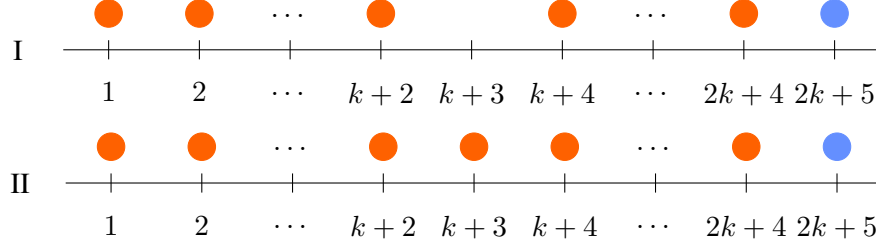


Figure 4.2: Section I and II

Let $k \in \mathbb{N}$ and $\sigma = (<, s, a, b)$. We define a pair of linear models

$$\begin{aligned} \mathfrak{A}_k &= (\mathbb{Z}, <, s, V^{\mathfrak{A}_k}) \text{ and } \mathfrak{B}_k = (\mathbb{Z}, <, s, V^{\mathfrak{B}_k}), \\ \text{where } V^{\mathfrak{A}_k}(b) &= V^{\mathfrak{B}_k}(b) = \{n(2k+5) - 1 : n \in \mathbb{Z}\}, \\ V^{\mathfrak{A}_k}(a) &= \mathbb{Z} - (V^{\mathfrak{A}_k}(b) \cup \{2n(2k+5) + (k+2) : n \in \mathbb{Z}\}), \text{ and} \\ V^{\mathfrak{B}_k}(a) &= \mathbb{Z} - (V^{\mathfrak{B}_k}(b) \cup \{(2n+1)(2k+5) + (k+2) : n \in \mathbb{Z}\}). \end{aligned}$$

We use the normal less than relation $<$ and successor relation s over $\mathbb{Z}$ in both structures.

Informally, we have two sections, which we will denote as I and II, where each section contains $2k+5$ consecutive points in time. In section I, a is true in the first $k+2$ points, then no propositions are true at the $(k+3)$ -th point, then a is true in the next $k+1$ points, and b is true at the $(2k+5)$ -th point. In section II, a is true in the first $2k+4$ points and b is true at the $(2k+5)$ -th point. This is illustrated in Figure 4.2. We represent a being true with a **orange** dot and b being true with a **blue** dot.

The timeline in $\mathfrak{A}_k$ alternates between section I and section II, starting with section I at 0. The timeline in $\mathfrak{B}_k$ also alternates between section I and section II, but starts with section II at 0. Figure 4.3 the timeline of $\mathfrak{A}_k$ and $\mathfrak{B}_k$ with $k = 1$. We find that $a \mathbf{U} b$ holds in $\mathfrak{B}_k$ but not in $\mathfrak{A}_k$. To show that $a \mathbf{U} b$ is not expressible in UTL, it suffices to show that $(\mathfrak{A}_k, 0) \equiv_k^2 (\mathfrak{B}_k, 0)$.

Lemma 4.9. *For all $k, n \in \mathbb{N}$ and $i \in \mathbb{Z}$,*

$$\text{if } |i| \leq n \leq k, \text{ then } (\mathfrak{A}_k, i) \equiv_{2(k-n)}^2 (\mathfrak{B}_k, i).$$

That is, the duplicator has a winning strategy in $2(k-n)$ moves if i is in the range $[-n, n]$.

Proof. Fix $k \in \mathbb{N}$. We proceed with induction on n . For the base case where $n = k$ and $|i| \leq n$, we can check that $(\mathfrak{A}_k, i)$ and $(\mathfrak{B}_k, i)$ are partially isomorphic, and therefore $(\mathfrak{A}_k, i) \equiv_0^2 (\mathfrak{B}_k, i)$. With only one pebble on each structure, it suffices to check that $i < i$, $S(i, i)$, $a(i)$, and $b(i)$ evaluate to the value on both $\mathfrak{A}_k$ and $\mathfrak{B}_k$. We can easily see this is the case.

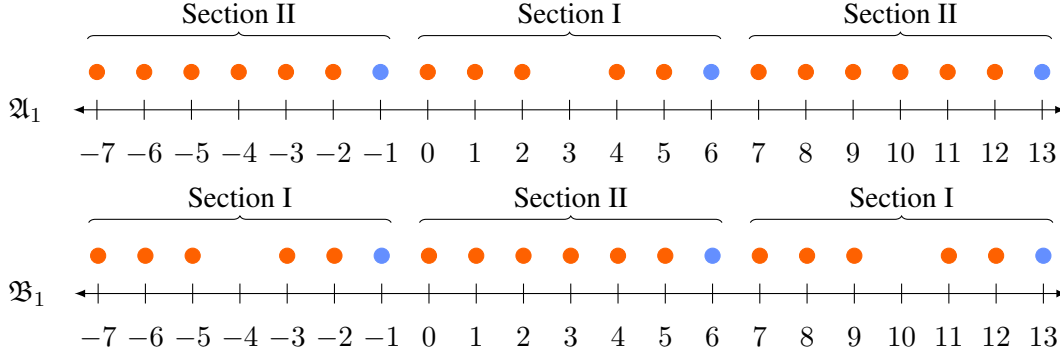


Figure 4.3: Timeline for $\mathfrak{A}_1$ and $\mathfrak{B}_1$

For the inductive case, let $j \in \mathbb{N}_{>0}$ where $j < k$. Assume that when $j \leq n$ and $|i| \leq n \leq k$, we have $(\mathfrak{A}_k, i) \equiv_{2(k-n)}^2 (\mathfrak{B}_k, i)$. We prove that when $n = j - 1$, $(\mathfrak{A}_k, i) \equiv_{2(k-n)}^2 (\mathfrak{B}_k, i)$ for all i such that $|i| \leq n$. We have the following four cases:

1. If the spoiler removes the pebble at i in one structure, then the duplicator removes the pebble in the other. We now need to show that $\mathfrak{A}_k \equiv_{2(k-n)-1}^2 \mathfrak{B}_k$. The main idea of this proof is that the timeline in $\mathfrak{A}_k$ is the timeline in $\mathfrak{B}_k$, shifted by $2k + 5$ points. We then have $\mathfrak{A}_k \equiv_N^2 \mathfrak{B}_k$ for all N . If the spoiler places a pebble at x in $\mathfrak{A}_k$, then the duplicator can place a pebble at $x + 2k + 5$ in $\mathfrak{B}_k$. If the spoiler places a pebble at y in $\mathfrak{B}_k$, then the duplicator can place a pebble at $y - 2k - 5$ in $\mathfrak{A}_k$. This is a winning strategy for the duplicator, as this shift by $2k + 5$ points defines an isomorphism between $\mathfrak{A}_k$ and $\mathfrak{B}_k$.
2. Let $x > i + 1$. If the spoiler places a second pebble in x in one structure, then the duplicator responds by placing a pebble at $x + 2k + 5$ in the other structure. As we only have 2 pebbles, the next move for the spoiler must be picking a pebble back up. If the pebble on i is picked up, the duplicator does the same, and we have the winning strategy described in case 1. If the other pebble is picked up, we have the starting position, but only $2(k - n) - 2$ moves, so we can use the strategy from our inductive hypothesis.
3. Let $x < i - 1$. We have a similar argument to case 2, if the spoiler places the second pebble on x , where the duplicator responds by placing a pebble on $x - 2k - 5$ in the other structure.
4. If the spoiler places a pebble on $i + 1$ or $i - 1$ in one structure, the duplicator must also place a pebble on $i + 1$ or $i - 1$, respectively, in the other. This is because we need to have the successor relation be evaluated the same on either structure. Both pebbles are on the board, so the next move by the spoiler must be a pebble being picked up. If either pebble is picked up, we win from the strategy in our inductive hypothesis.

We have proven our statement by induction. $\square$

Corollary 4.10. *For all $k \in \mathbb{N}$, $(\mathfrak{A}_k, 0) \equiv_{2k}^2 (\mathfrak{B}_k, 0)$.*

This corollary follows directly from Lemma 4.9, where $i = n = 0$. Since $a \mathbf{U} b$ holds in $\mathfrak{B}_k$ but not in $\mathfrak{A}_k$, we find that $a \mathbf{U} b$ is not expressible as a 2-variable formula with one free variable, and therefore, in UTL. Thus, our proof for Proposition 4.8 is complete.

The until operator is flexible in that it can be used to express $\mathbf{F}$, where $\mathbf{F}\varphi$ is equivalent to $\top \mathbf{U} \varphi$. As $\mathbf{G}$ is the dual of $\mathbf{F}$, we can also use $\mathbf{U}$ to express $\mathbf{G}$. Similarly, the statement $\mathbf{X}\varphi$ is equivalent to $\perp \mathbf{U} \varphi$. There is a symmetry between the past operators $\mathbf{Y}$, $\mathbf{P}$, $\mathbf{H}$, $\mathbf{S}$ and future operators $\mathbf{X}$, $\mathbf{F}$, $\mathbf{G}$, $\mathbf{U}$, with relations between future operators also applying to past operators. In particular, the since operator can be used to express $\mathbf{Y}$, $\mathbf{P}$ and $\mathbf{H}$. Every LTL formula, and therefore every UTL formula, can be rewritten only using the $\mathbf{U}$ and $\mathbf{S}$ operators.

4.2.2 Games for LTL Fragments

Definition 4.11. We can define several fragments of LTL by restricting the operators that can be used in formulae. Such a logic is denoted by $\text{LTL}(\mathbf{O}_1, \mathbf{O}_2, \dots, \mathbf{O}_n)$ where $\mathbf{O}_1, \mathbf{O}_2, \dots, \mathbf{O}_n$ are the operators that can be used. This fragment also has a relevant EF game, where the spoiler can choose to play moves for allowed operators.

From the argument above, LTL is equivalent to $\text{LTL}(\mathbf{U}, \mathbf{S})$. It is also clear that UTL is equivalent to $\text{LTL}(\mathbf{X}, \mathbf{Y}, \mathbf{F}, \mathbf{P})$. In the $\text{LTL}(\mathbf{F})$ game, the spoiler can only play $\mathbf{F}$ moves, and in the $\text{LTL}(\mathbf{P})$ game, the spoiler only has access to $\mathbf{P}$ moves.

Example 4.12. We define $\mathfrak{A} = (\mathbb{Z}, <, s, V_1)$ with the usual relations $<$, s , and

$$V_1(a) = \{k : k \leq 0\} \text{ and } V_1(b) = \{2k + 1 : k \in \mathbb{N}\}.$$

We also define $\mathfrak{B} = (\mathbb{Z}, <, s, V_2)$ with the usual relations $<$, s , and

$$V_2(a) = \{k : k \leq 0\} \text{ and } V_2(b) = \{2k + 2 : k \in \mathbb{N}\}.$$

The models are depicted in Figure 4.4. We represent a being true with a **orange** dot and b being true with a **blue** dot. Clearly, $\mathbf{F}(b \wedge \mathbf{H}a)$ is true at $(\mathfrak{A}, 0)$, but not $(\mathfrak{B}, 0)$.

We show that $(\mathfrak{A}, 0) \equiv_{\infty}^{\text{LTL}(\mathbf{P})} (\mathfrak{B}, 0)$. The duplicator trivially wins a 0-round $\text{LTL}(\mathbf{P})$ game. A $\mathbf{P}$ move means that the spoiler must move one pebble backwards by at least one space. In every case, the pebble is moved to a point where a is true and b is false. The duplicator can then respond by moving the pebble in the other structure back by one space. At this point, a

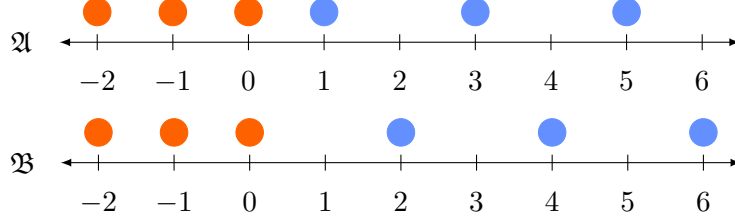


Figure 4.4: Structures for $LTL(\mathbf{F})$ and $LTL(\mathbf{P})$ Game Example

is true and b is false. Thus, the duplicator is still winning. We can repeat this strategy for an infinite number of rounds.

We also show that $(\mathfrak{A}, 0) \equiv_{\infty}^{LTL(\mathbf{F})} (\mathfrak{B}, 0)$. When playing an $\mathbf{F}$ move, the spoiler must move one pebble forward. They can either choose to move to a point where b is true, or a point where no atomic propositions hold. In either case, the duplicator can move the other pebble forward by one or two spaces to match the spoiler's choice. We can also repeat this strategy for an infinite number of rounds.

Etessami and Wilke studied and provided an EF game for $LTL(\mathbf{X}, \mathbf{F}, \mathbf{U})$ [EW00]. We modify this game to obtain moves for the $LTL(\mathbf{U}, \mathbf{S})$ game.

Definition 4.13. The setup for *LTL EF game* is the same as Definition 4.4.

U: The spoiler either chooses $\mathfrak{A}$ or $\mathfrak{B}$. We detail the move for $\mathfrak{A}$ as the move for $\mathfrak{B}$ is symmetrical. The spoiler chooses $x' > x$, and the duplicator chooses $y' > y$. Then, the spoiler can set the new configuration to (x', y') , or the spoiler chooses $y'' \in (y, y')$ and the duplicator responds with $x'' \in (x, x')$. Then, the new configuration is (x'', y'') .

S: The spoiler either chooses $\mathfrak{A}$ or $\mathfrak{B}$. We detail the move for $\mathfrak{A}$ as the move for $\mathfrak{B}$ is symmetrical. The spoiler chooses $x' < x$, and the duplicator chooses $y' < y$. Then, the spoiler can set the new configuration to (x', y') , or the spoiler chooses $y'' \in (y', y)$ and the duplicator responds with $x'' \in (x', x)$. Then, the new configuration is (x'', y'') .

Definition 4.14. We extend Definition 3.6 with $qr(\varphi \mathbf{U} \psi) = 1 + \max\{qr(\varphi), qr(\psi)\}$ and $qr(\varphi \mathbf{S} \psi) = 1 + \max\{qr(\varphi), qr(\psi)\}$.

Theorem 4.15. Let $\mathfrak{A}$ and $\mathfrak{B}$ be linear models over $\{p_1, \dots, p_n\}$. The following are equivalent:

1. For all formulae $\varphi \in LTL(\mathbf{U}, \mathbf{S})[n]$, $\mathfrak{A}, x \models \varphi$ iff $\mathfrak{B}, y \models \varphi$.
2. $(\mathfrak{A}, x) \equiv_n^{LTL(\mathbf{U}, \mathbf{S})} (\mathfrak{B}, y)$.

4.3 First-order Equivalence

Every LTL formula can be written as a first-order formula. We extend the standard translation to get the following result.

Proposition 4.16. *Every LTL formulae φ has an equivalent $FO[<, s]$ formula $\psi(x)$.*

Proof. Suppose that the LTL formulae φ_1 and φ_2 have the translations $\psi_1(x)$ and $\psi_2(x)$. Then,

- $\varphi_1 \mathbf{U} \varphi_2$ translates to $\exists y(x < y \wedge \psi_2(y) \wedge \forall z(x < z < y \rightarrow \psi_1(z)))$,
- $\varphi_1 \mathbf{S} \varphi_2$ translates to $\exists y(x > y \wedge \psi_2(y) \wedge \forall z(y < z < x \rightarrow \psi_1(z)))$.

□

In his seminal thesis [Kam68], Kamp establishes the converse. Kamp's Theorem was the first main result in the study of expressive completeness [Rab14], that is, determining if a given logic has important properties that cannot be expressed. We say that a logic over linear models is *expressively complete* if the logic is equivalent to $FO[<, s]$. Kamp shows that every property that can be expressed by an $FO[<, s]$ formula with one free variable can also be expressed by LTL. That is, LTL is expressively complete.

Theorem 4.17 (Kamp). *Every $FO[<, s]$ formulae $\psi(x)$ has an equivalent LTL formula.*

Kamp's proof was highly non-trivial, spanning over 100 pages, and consisted of a complicated translation procedure. In the next section, we introduce Gabbay's concept of separation [Gab81], a necessary and sufficient condition for proving that a temporal logic is expressively complete. We then give a characterisation of separation and Gabbay's proof of Kamp's Theorem in terms of EF games. The section that follows investigates results analogous to Kamp's Theorem for UTL and its fragments through EF games.

4.3.1 The Separation Property

A temporal logic has the separation property, first introduced by Gabbay in [Gab81], if every formula can be rewritten as a boolean combination of formulae, each only depending on the past, present or future. By *temporal logic*, we refer to any language L consisting of atomic propositions, operators $\mathbf{O}(\varphi_1, \dots, \varphi_n)$ and their boolean combinations, where each formula is equivalent to a formula in $FO[<, s]$.

Definition 4.18. Consider two linear models $\mathfrak{A} = (T, <, s, V_1)$ and $\mathfrak{B} = (T, <, s, V_2)$. We say $\mathfrak{A}, \mathfrak{B}$ *agree on* $t \in T$ if for all atomic propositions p , we have $t \in V_1(p)$ iff $t \in V_2(p)$. The models $\mathfrak{A}, \mathfrak{B}$ *agree on the past of* $t \in T$ if they agree on all $t' < t$, and they *agree on the future of* $t \in T$ if they agree on all $t' > t$.

Definition 4.19. Let L be a temporal logic. A formula $\varphi \in L$ is *pure past* if for all linear models $\mathfrak{A}, \mathfrak{B}$ and $t \in T$, such that the models agree on the past of t , we have $\mathfrak{A}, t \models \varphi \Leftrightarrow \mathfrak{B}, t \models \varphi$. We can define *pure present* and *pure future* formulae similarly. A formula is *separated* if it is a boolean combination of pure past, pure present and pure future formulae. The process of rewriting a formula as an equivalent separated formula is called *separating*.

Definition 4.20. A temporal logic L has the *separation property* if every formula in L is equivalent to a separated formula in L .

Theorem 4.21 (Gabbay’s Separation Theorem). *A temporal logic that can express $\mathbf{F}$ and $\mathbf{P}$ is expressively complete iff it has the separation property.*

Example 4.22. Clearly, $\text{LTL}(\mathbf{X}, \mathbf{Y})$ has the separation property. This is because $\mathbf{X}(\varphi \wedge \neg\psi)$ is equivalent to $\mathbf{X}\varphi \wedge \neg\mathbf{X}\psi$ and $\mathbf{Y}(\varphi \wedge \neg\psi)$ is equivalent to $\mathbf{Y}\varphi \wedge \neg\mathbf{Y}\psi$. We also have $\mathbf{XY}\varphi$ equivalent to φ . Thus, any $\text{LTL}(\mathbf{X}, \mathbf{Y})$ formula can be separated. However, we know that $\mathbf{F}$ cannot be expressed, and so Gabbay’s Separation Theorem does not apply.

Example 4.23. The classic example of a formula in LTL being separated is $\mathbf{F}(b \wedge \mathbf{H}a)$, where a and b are atomic propositions. The formula says that there is a point t in the future where b holds and a holds at every point before t . An equivalent formula is $\mathbf{H}a \wedge a \wedge a \mathbf{U} b$. This example is special as it also shows that $\text{LTL}(\mathbf{F}, \mathbf{P})$ does not have the separation property. That is, we cannot rewrite $\mathbf{F}(a \wedge \mathbf{H}b)$ as a separated formula without an operator outside of $\mathbf{F}, \mathbf{G}, \mathbf{P}, \mathbf{H}$. We provide an original proof through EF games.

Lemma 4.24. *Every pure future formula in $\text{LTL}(\mathbf{F}, \mathbf{P})$ is equivalent to a formula in $\text{LTL}(\mathbf{F})$ and every pure past formula is equivalent to a formula in $\text{LTL}(\mathbf{P})$.*

Proposition 4.25. *The temporal logic $\text{LTL}(\mathbf{F}, \mathbf{P})$ does not have the separation property.*

Proof. Assume that $\text{LTL}(\mathbf{F}, \mathbf{P})$ has the separation property. Let $\sigma = \{<, s, a, b\}$. Then, $\mathbf{F}(b \wedge \mathbf{H}a)$ can be written as a separated formula. Consider the structures $\mathfrak{A}, \mathfrak{B}$ in Example 4.12. We know that $\mathfrak{A}, 0 \models \mathbf{F}(b \wedge \mathbf{H}a)$, while $\mathfrak{B}, 0 \not\models \mathbf{F}(b \wedge \mathbf{H}a)$. Separation implies there must exist a pure past, pure present or pure future formula which is true in $(\mathfrak{A}, 0)$, but false in $(\mathfrak{B}, 0)$.

Consider the structures in Example 4.12. As $(\mathfrak{A}, 0) \equiv_{\infty}^{\text{LTL}(\mathbf{P})} (\mathfrak{B}, 0)$, Lemma 4.24 tells us there is no pure past formula which is true at $(\mathfrak{A}, 0)$, but false at $(\mathfrak{B}, 0)$. Similarly, $(\mathfrak{A}, 0) \equiv_{\infty}^{\text{LTL}(\mathbf{F})} (\mathfrak{B}, 0)$, and so there is no pure future formula true at $(\mathfrak{A}, 0)$, but false at $(\mathfrak{B}, 0)$. We also know that $(\mathfrak{A}, 0)$ and $(\mathfrak{B}, 0)$ agree on all atomic propositions. As pure present formulae are simply boolean combinations of atomic propositions, they agree on all pure present formulae. From this contradiction, we conclude that $\text{LTL}(\mathbf{F}, \mathbf{P})$ does not have the separation property. $\square$

Gabbay actually shows that LTL has *syntactic separation*, which is stronger than the separation property above. A pure future formula could contain both past and future operators. In LTL, we can rewrite each formula such that there are no nested occurrences of **U** and **S**. That is, a formula such that terms of the form $\varphi \mathbf{U} \psi$ have no usage of **S** in φ, ψ and vice versa.

Theorem 4.26 ([Gab89, Theorem 2.4]). *Every formula in $LTL(\mathbf{U}, \mathbf{S})$ is equivalent to a boolean combination of formulae in $LTL(\mathbf{U})$ and formulae in $LTL(\mathbf{S})$.*

We give an overview of the proof in [Gab89], and provide a novel reformulation through EF games that implies the separation property for LTL.

Gabbay's proof considers each way that **U** and **S** can be nested, and shows how to separate each one. Let $\varphi_1, \varphi_2, \varphi_3, \varphi_4$ be $LTL(\mathbf{U}, \mathbf{S})$ formulae. Consider the following tautologies:

$$\begin{aligned} (\varphi_1 \wedge \varphi_2) \mathbf{U} \varphi_3 & \leftrightarrow (\varphi_1 \mathbf{U} \varphi_3 \wedge \varphi_2 \mathbf{U} \varphi_3), \\ \varphi_1 \mathbf{U} (\varphi_2 \vee \varphi_3) & \leftrightarrow (\varphi_1 \mathbf{U} \varphi_2 \vee \varphi_1 \mathbf{U} \varphi_3), \\ \neg(\varphi_1 \mathbf{U} \varphi_2) & \leftrightarrow [\mathbf{G}\neg\varphi_1 \vee (\neg\varphi_2) \mathbf{U} (\neg\varphi_1 \wedge \neg\varphi_2)]. \end{aligned}$$

The statements hold when replacing **U** with **S** and **G** with **H**. The remaining cases are:

$$\begin{aligned} \varphi_1 \mathbf{U} (\varphi_2 \wedge (\varphi_3 \mathbf{S} \varphi_4)), & \quad (\varphi_1 \vee (\varphi_3 \mathbf{S} \varphi_4)) \mathbf{U} (\varphi_2 \wedge (\varphi_3 \mathbf{S} \varphi_4)), \\ \varphi_1 \mathbf{U} (\varphi_2 \wedge \neg(\varphi_3 \mathbf{S} \varphi_4)), & \quad (\varphi_1 \vee \neg(\varphi_3 \mathbf{S} \varphi_4)) \mathbf{U} (\varphi_2 \wedge (\varphi_3 \mathbf{S} \varphi_4)), \\ (\varphi_1 \vee (\varphi_3 \mathbf{S} \varphi_4)) \mathbf{U} \varphi_2, & \quad (\varphi_1 \vee (\varphi_3 \mathbf{S} \varphi_4)) \mathbf{U} (\varphi_2 \wedge \neg(\varphi_3 \mathbf{S} \varphi_4)), \\ (\varphi_1 \vee \neg(\varphi_3 \mathbf{S} \varphi_4)) \mathbf{U} \varphi_2, & \quad (\varphi_1 \vee \neg(\varphi_3 \mathbf{S} \varphi_4)) \mathbf{U} (\varphi_2 \wedge \neg(\varphi_3 \mathbf{S} \varphi_4)). \end{aligned}$$

The cases where **U** operators are nested inside **S** operators are identical to the ones above. Gabbay rewrites each case as a separated formula. We provide the explicit separation for the first case, labelled as Elimination 1 in [Gab89]. The formula $\psi = \varphi_1 \mathbf{U} (\varphi_2 \wedge (\varphi_3 \mathbf{S} \varphi_4))$ is equivalent to $\psi_1 \vee \psi_2 \vee \psi_3$, where

$$\begin{aligned} \psi_1 &= \varphi_1 \mathbf{U} \varphi_2 \wedge \varphi_3 \mathbf{U} \varphi_2 \wedge \varphi_3 \wedge \varphi_3 \mathbf{S} \varphi_4, \\ \psi_2 &= \varphi_4 \wedge (\varphi_1 \wedge \varphi_3) \mathbf{U} \varphi_2, \text{ and} \\ \psi_3 &= \varphi_1 \mathbf{U} (\varphi_1 \wedge \varphi_4 \wedge \varphi_1 \mathbf{U} \varphi_2 \wedge \varphi_3 \mathbf{U} \varphi_2). \end{aligned}$$

Note that $\text{qr}(\psi) = \text{qr}(\psi_1 \vee \psi_2 \vee \psi_3)$. We find that for every other case, the quantifier rank of the original formula and the rewritten formula is the same. Thus, a formula $\varphi \in LTL(\mathbf{U}, \mathbf{S})[n]$ can be rewritten as a boolean combination of formulae from $LTL(\mathbf{U})[n]$ and $LTL(\mathbf{S})[n]$. Let $\mathfrak{A}$ and $\mathfrak{B}$ be linear models. If φ is true in one, but not the other, there exists a formula either in $LTL(\mathbf{U})[n]$ or $LTL(\mathbf{S})[n]$ which is true in one, but not the other. In terms of EF games, we

have the following novel characterisation.

Lemma 4.27. *Let $\mathfrak{A}, \mathfrak{B}$ be linear models. For all $n \in \mathbb{N}$, if $(\mathfrak{A}, x) \not\equiv_n^{LTL(\mathbf{U}, \mathbf{S})} (\mathfrak{B}, y)$, then $(\mathfrak{A}, x) \not\equiv_n^{LTL(\mathbf{U})} (\mathfrak{B}, y)$ or $(\mathfrak{A}, x) \not\equiv_n^{LTL(\mathbf{S})} (\mathfrak{B}, y)$.*

Proposition 4.28. *Lemma 4.27 is sufficient to show that LTL has syntactic separation.*

Proof. Define $\mathcal{P}_{B,y}^{\mathbf{U}}$ as the class of pointed linear models $(\mathfrak{A}, x)$ such that $(\mathfrak{A}, x) \not\equiv_n^{LTL(\mathbf{U})} (\mathfrak{B}, y)$. Let $(C, a) \in \mathcal{P}_{B,y}^{\mathbf{U}}$ and $(D, b) \notin \mathcal{P}_{B,y}^{\mathbf{U}}$. By definition, we have $(C, a) \not\equiv_n^{LTL(\mathbf{U})} (B, y) \equiv_n^{LTL(\mathbf{U})} (D, b)$. Thus, $(C, a) \not\equiv_n^{LTL(\mathbf{U})} (D, b)$. From the EF-style corollary, we have $\varphi_{B,y}^{\mathbf{U}}$ which expresses $\mathcal{P}_{B,y}^{\mathbf{U}}$. We can define $\varphi_{B,y}^{\mathbf{S}}$ similarly.

Let φ be an LTL formula with quantifier rank n . Then, let $\mathcal{P}$ be the class of pointed linear models such that φ is true. From the EF-style Corollary, for all $(\mathfrak{A}, x) \in \mathcal{P}$ and $(\mathfrak{B}, y) \notin \mathcal{P}$, we have $(\mathfrak{A}, x) \not\equiv_n^{LTL(\mathbf{U}, \mathbf{S})} (\mathfrak{B}, y)$. From Lemma 4.27, either $(\mathfrak{A}, x) \not\equiv_n^{LTL(\mathbf{U})} (\mathfrak{B}, y)$ or $(\mathfrak{A}, x) \not\equiv_n^{LTL(\mathbf{S})} (\mathfrak{B}, y)$.

Then, φ is equivalent to

$$\bigwedge_{(B,y) \notin \mathcal{P}} (\varphi_{B,y}^{\mathbf{U}} \vee \varphi_{B,y}^{\mathbf{S}}), \quad \text{as} \quad \mathcal{P} = \bigcap_{(B,y) \in \mathcal{P}} (\mathcal{P}_{B,y}^{\mathbf{U}} \cup \mathcal{P}_{B,y}^{\mathbf{S}}).$$

Since there are finitely many formulae in LTL of quantifier rank at most n , up to equivalence, we can rewrite the formula above as a finite conjunction of separated formulae. $\square$

A well-known consequence of separation is that the addition of $\mathbf{S}$ to $LTL(\mathbf{U})$ does not increase its expressibility over the natural numbers.

Corollary 4.29. *Let ω be the class of linear models $\mathfrak{T}$ over $\mathbb{N}$. Every $LTL(\mathbf{U}, \mathbf{S})$ formula φ has an $LTL(\mathbf{U})$ formula ψ , which is equivalent over ω at time 0. That is, for all $\mathfrak{T} \in \omega$, we have $\mathfrak{T}, 0 \models \varphi$ iff $\mathfrak{T}, 0 \models \psi$.*

This is because we can write any φ as $\varphi^{\mathbf{U}} \vee \varphi^{\mathbf{S}}$. As our timeline is the natural numbers, there are no points before 0. Thus, any pure past formula is trivially evaluated to true or false. We can then write φ as a formula in $LTL(\mathbf{U})$. In terms of EF games, we find that if $\mathfrak{A}, \mathfrak{B} \in \omega$, $(\mathfrak{A}, 0) \equiv_n^{LTL(\mathbf{S})} (\mathfrak{B}, 0)$ for all n . This is because an $\mathbf{S}$ move needs the pebble to move backwards, but this isn't possible with the natural numbers.

Corollary 4.30. *For all $\mathfrak{A}, \mathfrak{B} \in \omega$, if $(\mathfrak{A}, 0) \not\equiv_n^{LTL(\mathbf{U}, \mathbf{S})} (\mathfrak{B}, 0)$, then $(\mathfrak{A}, 0) \not\equiv_n^{LTL(\mathbf{U})} (\mathfrak{B}, 0)$.*

We now showcase how this reformulation greatly aids in proofs using the LTL EF game. The proposition $\text{even}(\varphi)$ means that the proposition φ holds at every point in the future which is an even number of points away from now.

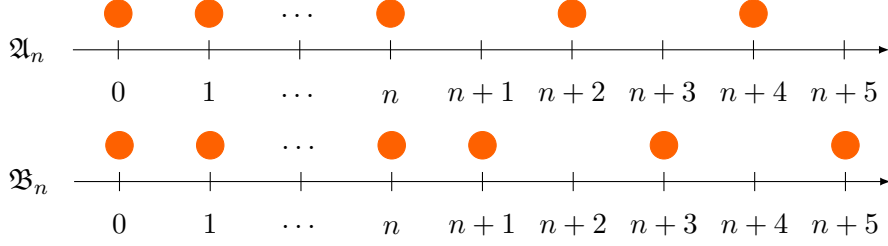


Figure 4.5: Linear Models $\mathfrak{A}_n$ and $\mathfrak{B}_n$

Proposition 4.31. *The formula $\text{even}(p)$ cannot be expressed in LTL.*

Proof. The contrapositive of Lemma 4.27 tells us if $(\mathfrak{A}, x) \equiv_n^{\text{LTL}(\mathbf{U})} (\mathfrak{B}, y)$ and $(\mathfrak{A}, x) \equiv_n^{\text{LTL}(\mathbf{S})} (\mathfrak{B}, y)$, then $(\mathfrak{A}, x) \equiv_n^{\text{LTL}(\mathbf{U}, \mathbf{S})} (\mathfrak{B}, y)$, for all $n \in \mathbb{N}$. Rather than using the $\text{LTL}(\mathbf{U}, \mathbf{S})$ game, separation means that we instead look at two simpler games: one where the spoiler only uses $\mathbf{U}$ moves, and another where the spoiler only uses $\mathbf{S}$ moves.

The only atomic proposition is p . Let $n \in \mathbb{N}$. We define $\mathfrak{A}_n = (\mathbb{N}, <, s, V_1)$ with the usual relations $<$, s , and

$$V_1(p) = [0, n] \cup \{n + 2k : k \in \mathbb{N}\}.$$

We also define $\mathfrak{B}_n = (\mathbb{N}, <, s, V_2)$ with the usual relations $<$, s , and

$$V_2(p) = [0, n + 1] \cup \{n + 2k + 1 : k \in \mathbb{N}\}.$$

The models can be seen in Figure 4.5, where an orange circle means that p is true at the given point. It is clear that $\text{even}(p)$ is true in one structure, but not the other, when evaluated at time 0. We note that $(\mathfrak{A}_n, 0) \equiv_n^{\text{LTL}(\mathbf{S})} (\mathfrak{B}_n, 0)$ holds as the spoiler can never play an $\mathbf{S}$ move. We now show that $(\mathfrak{A}_n, 0) \equiv_n^{\text{LTL}(\mathbf{U})} (\mathfrak{B}_n, 0)$ to complete the proof. This is done by inducting on n . The base case $n = 0$ is trivial.

For $n > 0$, assume that $(\mathfrak{A}_{n-1}, 0) \equiv_{n-1}^{\text{LTL}(\mathbf{U})} (\mathfrak{B}_{n-1}, 0)$. The present and future of $(\mathfrak{A}_{n-1}, 0)$ is identical to that of $(\mathfrak{A}_n, 1)$. The same is true with $(\mathfrak{B}_{n-1}, 0)$ and $(\mathfrak{B}_n, 1)$. As the spoiler only has access to the $\mathbf{U}$ move, they can only move forward. This means that $(\mathfrak{A}_n, 1) \equiv_{n-1}^{\text{LTL}(\mathbf{U})} (\mathfrak{B}_n, 1)$ since the spoiler can never use the past to differentiate the two linear models. If the spoiler moves a pebble forward by one space, the duplicator can do the same. From our inductive hypothesis, we have a strategy for the remaining $n - 1$ rounds.

Otherwise, the spoiler plays a $\mathbf{U}$ move by moving the pebble at least 2 spaces forward. Without loss of generality, consider the case where the spoiler plays on $\mathfrak{A}_n$ and moves the pebble to k . Then, the duplicator can respond by moving the pebble on $\mathfrak{B}_n$ to $k + 1$. If the spoiler continues

by moving the pebble on $\mathfrak{B}_n$ to $k' + 1$ for some k' , the duplicator can respond by moving the pebble on $\mathfrak{A}_n$ to k' . Either way, the configuration ends up having the form $(k, k + 1)$. The duplicator has a winning strategy from this position as the present and future of $(\mathfrak{A}_n, k)$ is identical to that of $(\mathfrak{B}_n, k + 1)$.

If the spoiler plays a **U** move by moving the pebble to $k + 1$ on $\mathfrak{B}_n$, the duplicator can respond by moving the pebble on $\mathfrak{A}_n$ to k . For the case that $k = 1$, the spoiler cannot continue playing. Otherwise, if $k > 1$, the spoiler can choose to move the pebble on $\mathfrak{A}_n$ to k' . If so, the duplicator can respond by moving the pebble on $\mathfrak{B}_n$ to $k' + 1$. Again, the duplicator has a winning strategy from either configuration. We have shown that $(\mathfrak{A}_n, 0) \equiv_n^{\text{LTL}(\mathbf{U})} (\mathfrak{B}_n, 0)$. $\square$

4.3.2 2-Variable First-order Logic

From the standard translation, every UTL formula can be written as a formula in $\text{FO}^2[<, s]$. Etessami et al. give us a result for UTL analogous to Kamp's Theorem [EVW02]. In their paper, they provide a procedure for translating $\text{FO}^2[<, s]$ formulae into UTL formulae. We present a proof of Theorem 1 in [EVW02] based on EF games.

Theorem 4.32. *Every $\text{FO}^2[<, s]$ formula $\psi(x)$ is equivalent to a UTL formula φ .*

In particular, we prove that a property not expressible in UTL is also not expressible in 2-variable $\text{FO}[<, s]$. Corollary 4.5 tells us that it suffices to show that the UTL EF game is enough to prove that a property is not expressible in $\text{FO}^2[<, s]$. That is, if the duplicator has a winning strategy in the UTL EF game, then the duplicator has a winning strategy in the 2-variable EF game. This is a novel proof strategy where the equivalence of two logics is abstracted to the duplicator winning in the game of one logic exactly when the duplicator wins in the game of the other logic.

Lemma 4.33. *Let $\mathfrak{A}, \mathfrak{B}$ be linear models. If $(\mathfrak{A}, x) \equiv_{2n}^{\text{UTL}} (\mathfrak{B}, y)$, then $(\mathfrak{A}, x) \equiv_{2n}^2 (\mathfrak{B}, y)$.*

Proof. We induct on n . For $n = 0$, this is immediate. Assume that for all $x' \in \mathfrak{A}$ and $y' \in \mathfrak{B}$, $(\mathfrak{A}, x') \equiv_{2n}^{\text{UTL}} (\mathfrak{B}, y')$ implies that $(\mathfrak{A}, x') \equiv_{2n}^2 (\mathfrak{B}, y')$. Suppose that $(\mathfrak{A}, x) \equiv_{2n+2}^{\text{UTL}} (\mathfrak{B}, y)$. We prove the duplicator wins the $(2n + 2)$ -round 2-variable EF game on $(\mathfrak{A}, x)$ and $(\mathfrak{B}, y)$. For the first round, the spoiler either removes or places a pebble on a board.

If the spoiler removes a pebble, the duplicator removes the only other pebble. As there are no pebbles to remove, the spoiler must now place a pebble in the second round. For this proof, we want to find the duplicator's response in the 2-variable game. To do so, we have the spoiler in the UTL game match the move that the spoiler makes in the 2-variable game, and use the

response that the duplicator gives us in the UTL game. Without loss of generality, suppose the spoiler places the pebble on $x' \in \mathfrak{A}$. We have the following cases:

- Let $x' = x + 1$. Let the spoiler play an **X** move in the UTL game. The duplicator then uses the responses, that is, the duplicator places a pebble on $y' = y + 1$.
- Let $x' = x - 1$. The duplicator copies the response y' to a **Y** move.
- Let $x' > x + 1$. We want a response $y' > y + 1$. To ensure this, we first let the spoiler play an **X** move and then have the spoiler play x' with an **F** move in the UTL EF game. The response by the duplicator y' , in the UTL game, satisfies $y' > y + 1$. The duplicator responds with y' in the 2-variable game.
- Let $x' < x - 1$. Take the response from playing a **Y** and then a **P** move where the spoiler plays x' . This response y' is less than $y - 1$.
- Let $x' = x$. The duplicator can respond with $y' = y$. Since the duplicator has a $(2n + 2)$ -round winning strategy, from this configuration in the UTL game, they also have a $2n$ -round winning strategy.

In each case, the duplicator still has a $2n$ -round winning strategy from the configuration (x', y') . We have detailed how the first 2 rounds of the 2-variable EF game should be played. We also know that $(\mathfrak{A}, x') \equiv_{2n}^{\text{UTL}} (\mathfrak{B}, y')$. Thus, our inductive hypothesis states that the duplicator has a winning strategy for the next $2n$ rounds.

Without loss of generality, consider the case that the spoiler places a pebble $x' \in \mathfrak{A}$ for the first round. The duplicator should respond with $y' \in \mathfrak{B}$ as detailed above. Since we must have at most 2 pebbles, the spoiler must remove a pebble next. If the spoiler removes the pebble on x or y , the duplicator removes y or x , respectively. Then, we get left with the configuration (x', y') . Similarly, if the spoiler removes the pebble on x' or y' , the resulting configuration is (x, y) . With both configurations, we have shown that the duplicator has a $2n$ -round winning strategy in the previous case.

Thus, given a $(2n + 2)$ -round winning strategy for the duplicator in the UTL EF game, we have detailed a $(2n + 2)$ -round winning strategy for the duplicator in the 2-variable EF game. $\square$

If a property $\mathcal{K}$ is inexpressible in UTL, then for all $n \in \mathbb{N}$, there exists $(\mathfrak{A}, x) \in \mathcal{K}$ and $(\mathfrak{B}, y) \notin \mathcal{K}$ such that $(\mathfrak{A}, x) \equiv_{2n}^{\text{UTL}} (\mathfrak{B}, y)$. From Lemma 4.33, we find that for all $n \in \mathbb{N}$, there exists $(\mathfrak{A}, x) \in \mathcal{K}$ and $(\mathfrak{B}, y) \notin \mathcal{K}$ such that $(\mathfrak{A}, x) \equiv_{2n}^2 (\mathfrak{B}, y)$. This is a sufficient condition to conclude that $\mathcal{K}$ is inexpressible in 2-variable $\text{FO}[<, s]$, concluding our proof of Theorem 4.32.

It is natural to see that the operators **F**, **P** correlate with the relation $<$. We can use the same

proof technique as above to show that $LTL(\mathbf{F}, \mathbf{P})$ is equivalent to 2-variable $FO[<]$. Consider the 2-variable EF game with the signature of $FO[<]$. That is, we no longer need to check the S binary relation for the win condition of the duplicator. We can adapt the proof of Lemma 4.33 to get the following result. The expressibility result is Theorem 2 in [EVW02].

Proposition 4.34. *Let $\mathfrak{A}, \mathfrak{B}$ be linear models and $x \in \mathfrak{A}, y \in \mathfrak{B}$. If $(\mathfrak{A}, x) \equiv_n^{LTL(\mathbf{F}, \mathbf{P})} (\mathfrak{B}, y)$, then $(\mathfrak{A}, x) \equiv_{2n}^2 (\mathfrak{B}, y)$. Thus, a property is expressible as an $LTL(\mathbf{F}, \mathbf{P})$ formula φ iff it is expressible as a 2-variable $FO[<]$ formula $\psi(x)$.*

Proof. The only change to the proof above is that we only need to examine the following 3 cases in the inductive step. Suppose the spoiler places a pebble on $x' \in \mathfrak{A}$ with an initial configuration of (x, y) . Consider the following:

- Let $x' > x$. Since the duplicator has a winning strategy in the $LTL(\mathbf{F}, \mathbf{P})$ EF game, the duplicator has a response $y' > y$ to the spoiler moving the pebble to x' in an $\mathbf{F}$ move. The duplicator can use the same response y' to the 2-variable EF game.
- Let $x' < x$. The duplicator copies the response y' to a $\mathbf{P}$ move.
- Let $x' = x$. The duplicator can respond with $y' = y$. Since the duplicator has a $(n + 1)$ -round winning strategy, from this configuration in the $LTL(\mathbf{F}, \mathbf{P})$ game, they also have an n -round winning strategy. $\square$

One may also see a correspondence between the operators $\mathbf{X}, \mathbf{Y}$ and the relation s . However, this adaptation does not work for the $LTL(\mathbf{X}, \mathbf{Y})$ game. This game is limited in the fact that pebbles can only move forward or backward by one in every round. There are only two places that the spoiler can move the pebble to. In the 2-variable EF game, the pebble can be moved anywhere. We cannot construct a winning strategy for the duplicator in the 2-variable game, since the duplicator in the $LTL(\mathbf{X}, \mathbf{Y})$ game does not have a response for a pebble placed outside of the 2 possible choices. For this proof method to work, it is necessary that the spoiler in the $LTL(\mathbf{O}_1, \dots, \mathbf{O}_n)$ game can move the pebble to any element in the linear order. In fact, there exist formulae in 2-variable $FO[s]$ which cannot be expressed in $LTL(\mathbf{X}, \mathbf{Y})$.

Proposition 4.35. *The formula $\exists x p(x)$ is not expressible in $LTL(\mathbf{X}, \mathbf{Y})$.*

Proof. Let $n \in \mathbb{N}$ and $\sigma = (<, s, p)$. We define a pair of linear models

$$\mathfrak{A}_n = (\mathbb{Z}, <, s, V^{\mathfrak{A}_n}), \mathfrak{B}_n = (\mathbb{Z}, <, s, V^{\mathfrak{B}_n}), \text{ where } V^{\mathfrak{A}_n}(p) = \{n + 1\}, V^{\mathfrak{B}_n}(p) = \emptyset.$$

We use the normal less than relation $<$ and successor relation S over $\mathbb{Z}$ in both structures. We have one timeline where p is true only at $n + 1$ and another timeline where p is not true

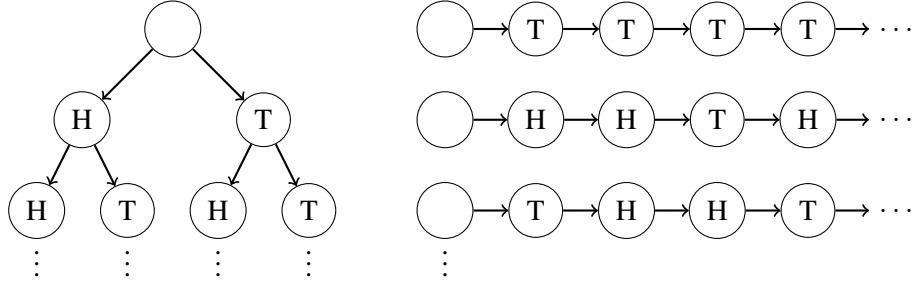


Figure 4.6: Visualising a tree in two ways

anywhere. We show $(\mathfrak{A}_n, 0) \equiv_n^{\text{LTL}(\mathbf{X}, \mathbf{Y})} (\mathfrak{B}_n, 0)$.

For the spoiler to win this game, the spoiler needs one pebble on $n+1$ in $\mathfrak{A}_n$. This is not possible since the spoiler can only move forward or backward by one in each move, and the spoiler has n moves. Thus, the duplicator has an n -round winning strategy. Clearly, $\mathfrak{A}_n, 0 \models \exists x p(x)$ and $\mathfrak{B}_n, 0 \not\models \exists x p(x)$. We conclude that $\exists x p(x)$ cannot be expressed in $\text{LTL}(\mathbf{X}, \mathbf{Y})$. $\square$

The proof above also shows that $\mathbf{F}p$ cannot be expressed in $\text{LTL}(\mathbf{X}, \mathbf{Y})$. In $\text{LTL}(\mathbf{F}, \mathbf{P})$, we find that $\exists x p(x)$ is equivalent to $\mathbf{P}p \vee p \vee \mathbf{F}p$.

4.4 Branching Time

By making time linear, the future becomes certain. Branching time allows for the current point in time to entertain multiple possible futures. In particular, time can be viewed as a directed tree starting at the root. The children of a point represent the possible next points in time. While the future at any point can be uncertain, the past is deterministic, as there is a single path from the root to that point in time.

Example 4.36. Suppose that we indefinitely flip a coin. Then, there are two possible futures at each point in time: either getting heads (H) or getting tails (T). We could model this scenario as the tree in Figure 4.6. This tree can also be represented as a set of possible timelines.

Definition 4.37. Let S be a set of states and P be a finite set of atomic propositions. A *run* $r : \mathbb{N} \rightarrow S$ represents a timeline, where $r(t)$ is the state at time $t \in \mathbb{N}$. A pair (r, t) , where r is a run and $t \in \mathbb{N}$, is called a *point*. A *tree model*, written as $\mathfrak{T} = (S, V, \mathcal{R})$, consists of a valuation function $V : S \rightarrow 2^P$ and a non-empty set of runs $\mathcal{R}$ where $r_1(0) = r_2(0)$ for all $r_1, r_2 \in \mathcal{R}$. That is, the runs have a unique first state. A pair $(\mathfrak{T}, (r, t))$ is called a *pointed tree*.

Let $r[a, b]$ be a linear model with a set of timepoints $[a, b]$ and propositions S . We have $s \in S$ be true at $t \in [a, b]$ iff $r(t) = s$. We write $r_1[a, b] = r_2[c, d]$ if $b - a = d - c$ and $r_1(a + k) =$

$r_2(c + k)$ for all $k \in [0, b - a]$.

CTL*, an extension of its predecessor Computation Tree Logic (CTL), was introduced by Emerson and Halpern in 1985 [EH85]. We have the operators from LTL to reason about time-lines, but also a new operator to quantify over the possible futures of a given point of time.

Definition 4.38. The CTL* formulae φ over propositions $p \in P$ are given by the rule

$$\varphi := p \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \mathbf{U} \psi \mid \mathbf{A}\varphi.$$

Let $\mathfrak{T} = (S, V, \mathcal{R})$ be a tree model with run r and $t \in \mathbb{N}$. We define the semantics as follows:

- For atomic propositions p , $\mathfrak{T}, (r, t) \models p$ iff $r(t) \in V(p)$ holds.
- $\mathfrak{T}, (r, t) \models \neg\varphi$ and $\mathfrak{T}, (r, t) \models \varphi \wedge \psi$ can be defined similar to before.
- $\mathfrak{T}, (r, t) \models \varphi \mathbf{U} \psi$ iff there exists $t' > t$ where $\mathfrak{T}, (r, t') \models \psi$ and we have $\mathfrak{T}, (r, t'') \models \varphi$ for all $t < t'' < t'$.
- $\mathfrak{T}, (r, t) \models \mathbf{A}\varphi$ iff $\mathfrak{T}, (r', t) \models \varphi$ for all runs $r' \in \mathcal{R}$ where $r'[0, t] = r[0, t]$.

We write the dual of $\mathbf{A}$ as $\mathbf{E}$. We then have

- $\mathfrak{T}, (r, t) \models \mathbf{E}\varphi$ iff $\mathfrak{T}, (r', t) \models \varphi$ for some run $r' \in \mathcal{R}$ where $r'[0, t] = r[0, t]$.

Definition 4.39. We extend Definition 4.14 with $\text{qr}(\mathbf{A}\varphi) = 1 + \text{qr}(\varphi)$. Let $\text{CTL}^*[n]$ be the set of CTL* formulae with quantifier rank at most n .

Example 4.40. For the tree model in Figure 4.6, we have $\mathbf{E}XH \wedge \mathbf{E}XT$ at every point. That is, it is always possible for the next state to be heads, and it is always possible for the next state to be tails. At every point, $\mathbf{A}(T \mathbf{U} H)$ does not hold, as we could flip tails indefinitely. However, $\mathbf{A}GE(T \mathbf{U} H)$ does hold, stating that at every point, there exists a path where we can flip tails until we flip heads.

Definition 4.41. To evaluate an LTL($\mathbf{U}$) formula φ on a tree model, we want all runs to satisfy φ at the unique first state. That is, φ is true, denoted as $\mathfrak{T}, (r, 0) \models^{\text{LTL}} \varphi$, if

$$\mathfrak{T}, (r, 0) \models \mathbf{A}\varphi, \text{ where } r \in \mathcal{R}.$$

Over tree models, CTL^* is at least as expressive as LTL($\mathbf{U}$).

The literature tends to define CTL^* without any past operators such as $\mathbf{Y}$, $\mathbf{P}$ or $\mathbf{S}$. One consequence of separation is that the past-time operator $\mathbf{S}$ does not add expressiveness to LTL($\mathbf{U}$) for linear models over the natural numbers. The tree model can be seen as a collection of linear models over the natural numbers. Is there an analogous property to Corollary 4.29 for CTL^* ?

Hafer and Thomas tell us that the expressiveness of CTL^* does not increase with the addition of the past-time operator S [HT87]. In the following subsections, we introduce a novel EF game for CTL^* and provide an original proof of Hafer and Thomas' result using this game, using our reformulation of separation.

4.4.1 The CTL^* EF Game

We now introduce a new EF game for CTL^* , borrowing the **U** move and its proof from the $LTL(X, F, U)$ game, first described by Etessami and Wilke [EW00]. Our contribution is the addition of the novel **A** move.

Definition 4.42. The n -round CTL^* EF game is played on tree models $\mathfrak{T}_1 = (S_1, V_1, \mathcal{R}_1)$ and $\mathfrak{T}_2 = (S_2, V_2, \mathcal{R}_2)$ over the same propositions P . A *configuration* is any pair of points $((r_1, t_1), (r_2, t_2))$ where each point belongs to $\mathfrak{T}_1$ and $\mathfrak{T}_2$, respectively. The game has two players, the spoiler and the duplicator, with an initial configuration of $((r_1, t_1), (r_2, t_2))$. When $n = 0$, the spoiler wins if there exists $p \in P$ which is true on $r_1(t_1)$ but not $r_2(t_2)$ or vice versa. The duplicator wins otherwise. In round k with configuration $((r_1, t_1), (r_2, t_2))$, the spoiler will choose a type of move and play accordingly:

U: The spoiler either chooses $\mathfrak{T}_1$ or $\mathfrak{T}_2$. We detail the move for $\mathfrak{T}_1$ as the move for $\mathfrak{T}_2$ is symmetrical. The spoiler chooses $t'_1 > t_1$, and the duplicator chooses $t'_2 > t_2$. Then, the spoiler can set the new configuration to $((r_1, t'_1), (r_2, t'_2))$, or the spoiler chooses $t''_2 \in (t_2, t'_2)$ and the duplicator responds with $t''_1 \in (t_1, t'_1)$. Then, the new configuration is $((r_1, t''_1), (r_2, t''_2))$.

A: Either the spoiler chooses $r'_1 \in \mathcal{R}_1$ such that $r'_1[0, t_1] = r_1[0, t_1]$, and the duplicator responds with $r'_2 \in \mathcal{R}_2$ such that $r'_2[0, t_2] = r_2[0, t_2]$, or the spoiler chooses $r'_2 \in \mathcal{R}_2$ such that $r'_2[0, t_2] = r_1[0, t_2]$, and the duplicator responds with $r'_1 \in \mathcal{R}_1$ such that $r'_1[0, t_1] = r_2[0, t_1]$. The new configuration is $((r'_1, t_1), (r'_2, t_2))$.

If a player cannot choose a point as prescribed, that player loses. Let the new configuration be $((r'_1, t'_1), (r'_2, t'_2))$. If there exists $p \in P$ which is true on $r'_1(t'_1)$ but not $r'_2(t'_2)$ or vice versa, the spoiler wins the game. Otherwise, the game continues. If the spoiler has not won after n rounds, then the duplicator wins. We write $(\mathfrak{T}_1, (r_1, t_1)) \equiv_n^{CTL^*} (\mathfrak{T}_2, (r_2, t_2))$ if the duplicator has an n -round winning strategy.

Lemma 4.43. *There are finitely many formulae in $CTL^*[n]$, up to equivalence over tree models.*

Theorem 4.44. *Let $\mathfrak{T}_1$ and $\mathfrak{T}_2$ be tree models over P . Then, the following are equivalent:*

1. *For all formulae $\varphi \in CTL^*[n]$, $\mathfrak{T}_1, (r_1, t_2) \models \varphi$ iff $\mathfrak{T}_2, (r_2, t_2) \models \varphi$.*

$$2. (\mathfrak{T}_1, (r_1, t_2)) \equiv_n^{CTL^*} (\mathfrak{T}_2, (r_2, t_2)).$$

Proof. We proceed with induction on n . The $n = 0$ case is trivial. Our inductive hypothesis states that for all pointed trees $(\mathfrak{T}_1, (r_1, t_1)), (\mathfrak{T}_2, (r_2, t_2))$, $(\mathfrak{T}_1, (r_1, t_1)) \equiv_n^{CTL^*} (\mathfrak{T}_2, (r_2, t_2))$ iff for all formulae $\varphi \in CTL^*$, we have $\mathfrak{T}_1, (r_1, t_1) \models \varphi \Leftrightarrow \mathfrak{T}_2, (r_2, t_2) \models \varphi$.

Suppose that $(\mathfrak{T}_1, (r_1, t_1)) \equiv_{n+1}^{CTL^*} (\mathfrak{T}_2, (r_2, t_2))$ and let $\varphi \in CTL^*[n+1]$. We consider the following cases:

- Let $\varphi = \psi_1 \mathbf{U} \psi_2$. Suppose $\mathfrak{T}_1, (r_1, t_1) \models \varphi$. Let the spoiler plays a **U** move, choosing $t'_1 > t_1$ such that $\mathfrak{T}_1, (r_1, t'_1) \models \psi_2$ and $\mathfrak{T}_1, (r_1, t''_1) \models \psi_1$ for all $t''_1 \in (t_1, t'_1)$. Let t'_2 be the response from the duplicator's winning strategy. If the spoiler chooses to set the configuration to $((r_1, t'_1), (r_2, t'_2))$, then we have $(\mathfrak{T}_1, (r_1, t'_1)) \equiv_n^{CTL^*} (\mathfrak{T}_2, (r_2, t'_2))$. As $\psi_2 \in CTL^*[n]$, our inductive hypothesis states that $\mathfrak{T}_2, (r_2, t'_2) \models \psi_2$. If the spoiler instead chooses $t''_2 \in (t_2, t'_2)$, the duplicator's winning strategy gives us $t''_1 \in (t_1, t'_1)$ such that $(\mathfrak{T}_1, (r_1, t''_1)) \equiv_n^{CTL^*} (\mathfrak{T}_2, (r_2, t''_2))$. As $\mathfrak{T}_1, (r_1, t''_1) \models \psi_1$, our inductive hypothesis states that $\mathfrak{T}_2, (r_2, t''_2) \models \psi_1$ for all $t''_2 \in (t_2, t'_2)$. Therefore, $\mathfrak{T}_2, (r_2, t_2) \models \psi_1 \mathbf{U} \psi_2$.
- Let $\varphi = \mathbf{A}\psi$, where $\varphi \in CTL^*[n]$. Suppose $\mathfrak{T}_1, (r_1, t_1) \models \varphi$. Then, for all runs r'_1 in $\mathfrak{T}_1$ such that $r'_1[0, t_1] = r_1[0, t_1]$, we have $\mathfrak{T}_1, (r'_1, t_1) \models \psi$. Let r'_2 be a run in $\mathfrak{T}_2$ such that $r'_2[0, t_2] = r_2[0, t_2]$. Suppose the spoiler plays an **A** move and chooses r'_2 . The duplicator chooses a run r'_1 in $\mathfrak{T}_1$ with $r'_1[0, t_1] = r_1[0, t_1]$, and we have $(\mathfrak{T}_1, (r'_1, t_1)) \equiv_n^{CTL^*} (\mathfrak{T}_2, (r'_2, t_2))$ from our assumption. As $\mathfrak{T}_1, (r'_1, t_1) \models \psi$, our inductive hypothesis states that $\mathfrak{T}_2, (r'_2, t_2) \models \psi$ for all runs r'_2 in $\mathfrak{T}_2$ such that $r'_2[0, t_2] = r_2[0, t_2]$. Therefore, $\mathfrak{T}_2, (r_2, t_2) \models \mathbf{A}\psi$.

Every formula $\varphi \in CTL^*[n+1]$ is a boolean combination of the cases above. Therefore, $(\mathfrak{T}_1, (r_1, t_1)) \models \varphi \Leftrightarrow (\mathfrak{T}_2, (r_2, t_2)) \models \varphi$.

Suppose that $(\mathfrak{T}_1, (r_1, t_1)) \not\equiv_{n+1}^{CTL^*} (\mathfrak{T}_2, (r_2, t_2))$. We prove there is a formula $\varphi \in CTL^*[n+1]$ which is true on one pointed model, but not the other. Consider the following cases:

- The spoiler plays a **U** move first by picking $t'_1 > t_1$. Let φ_t be the conjunction of all formulae in $CTL^*[n]$ that hold at $(\mathfrak{T}_1, (r_1, t))$. Consider the formula

$$\varphi = (\varphi_{t_1+1} \vee \varphi_{t_2+1} \vee \cdots \vee \varphi_{t'_1-1}) \mathbf{U} \varphi_{t'_1}.$$

Note that $\varphi_{t_1+1} \vee \varphi_{t_2+1} \vee \cdots \vee \varphi_{t'_1-1} = \perp$ if $t'_1 = t_1 + 1$. We can express φ_t as a formula in $CTL^*[n]$ as there are finitely many formulae in CTL^* up to equivalence. Thus, φ can be expressed as a formula in $CTL^*[n+1]$. We show that φ holds at $(\mathfrak{T}_1, (r_1, t_1))$, but not $(\mathfrak{T}_2, (r_2, t_2))$. Let t'_2 be the response of the duplicator. If $\varphi_{t'_1}$ does not hold

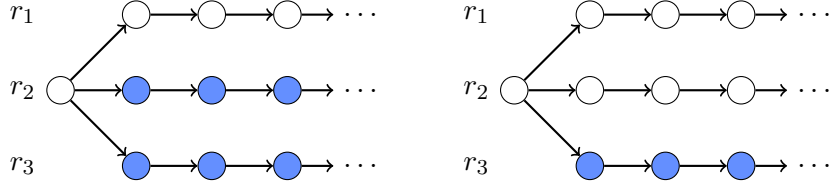


Figure 4.7: Tree models $\mathfrak{T}_1$ (Left) and $\mathfrak{T}_2$ (Right)

at $(\mathfrak{T}_2, (r_2, t'_2))$, we are done. We assume otherwise. Then, the spoiler chooses $t''_2 \in (t_2, t'_2)$. For any $t''_1 \in (t_1, t'_1)$ that the duplicator chooses, we have $(\mathfrak{T}_1, (r_1, t''_1)) \not\equiv_n^{\text{CTL}^*} (\mathfrak{T}_2, (r_2, t''_2))$. From our inductive hypothesis, there exists a formula in $\text{CTL}^*[n]$ which holds at $(\mathfrak{T}_1, (r_1, t''_1))$, but not $(\mathfrak{T}_2, (r_2, t''_2))$. Therefore, $\mathfrak{T}_2, (r_2, t'_2) \not\models \varphi_t$ for all $t \in (t_1, t'_1)$. By definition of $\mathbf{U}$, we have $\mathfrak{T}_2, (r_2, t_2) \not\models \varphi$.

- The spoiler plays an **A** move first by picking a run r'_2 in $\mathfrak{T}_2$ such that $r'_2[0, t_2] = r_2[0, t_2]$, according to their winning strategy. For any r'_1 in $\mathfrak{T}_2$, where $r'_1[0, t_1] = r_1[0, t_1]$, that the duplicator responds with, we have $(\mathfrak{T}_1, (r'_1, t_1)) \not\equiv_n^{\text{CTL}^*} (\mathfrak{T}_2, (r'_2, t_2))$. From the inductive hypothesis, we have $\varphi_{r'_1} \in \text{CTL}^*[n]$ such that $\mathfrak{T}_1, (r'_1, t_1) \models \varphi_{r'_1}$ and $\mathfrak{T}_2, (r'_2, t_2) \not\models \varphi_{r'_1}$. Let $F = \{\varphi_{r'_1} : r'_1[0, t_1] = r_1[0, t_1] \text{ where } r'_1 \text{ is a run in } \mathfrak{T}_1\}$. Since there are finitely many formulae in $\text{CTL}^*[n]$, up to equivalence, $\varphi = \bigvee_{\psi \in F} \psi$ can be written as a finite disjunction. Note that for all r'_1 such that $r'_1[0, t] = r_1[0, t]$, φ holds in $(\mathfrak{T}_1, (r'_1, t_1))$ but not in $(\mathfrak{T}_2, (r'_2, t_2))$. We find that $\mathbf{A}\varphi$ is equivalent to a formula in $\text{CTL}^*[n+1]$ and $\mathfrak{T}_1, (r_1, t_1) \models \mathbf{A}\varphi$, but $\mathfrak{T}_2, (r_2, t_2) \not\models \mathbf{A}\varphi$.

Therefore, $(\mathfrak{T}_1, (r_1, t_1))$ and $(\mathfrak{T}_2, (r_2, t_2))$ do not agree on all formulae in $\text{CTL}^*[n+1]$. $\square$

Corollary 4.45. *A property $\mathcal{P}$ is expressible in CTL^* iff there exists $n \in \mathbb{N}$ such that for all $(\mathfrak{T}_1, (r_1, t_1)) \in \mathcal{P}$ and $(\mathfrak{T}_2, (r_2, t_2)) \notin \mathcal{P}$, we have $(\mathfrak{T}_1, (r_1, t_2)) \not\equiv_n^{\text{CTL}^*} (\mathfrak{T}_2, (r_2, t_2))$.*

We now present some examples of the game being played to show off our **A** move. The first example shows that CTL^* cannot count how many possible futures have a given property.

Proposition 4.46. *The following property is not expressible in CTL^* : two runs have **Gp** hold.*

Proof. Let $S = \{s_1, s_2, s_3\}$ and $P = \{p\}$. We will play on the two tree models depicted in Figure 4.7, where blue represents that p holds. It is clear that **Gp** holds for $(r_2, 0)$ and $(r_3, 0)$ in $\mathfrak{T}_1$, while it is only true for $(r_3, 0)$ in $\mathfrak{T}_2$. The following is a formal definition of these two trees. Let $\mathcal{R} = \{r_1, r_2, r_3\}$, where $r_1(0) = r_2(0) = r_3(0) = s_1$ and $r_i(t) = s_i$ for all $t \in \mathbb{N} - \{0\}$ and $i \in [1, 3]$. We define two tree models $\mathfrak{T}_1 = (S, V_1, \mathcal{R})$ and $\mathfrak{T}_2 = (S, V_2, \mathcal{R})$ with $V_1(p) = \{s_2, s_3\}$ and $V_2(p) = \{s_3\}$.

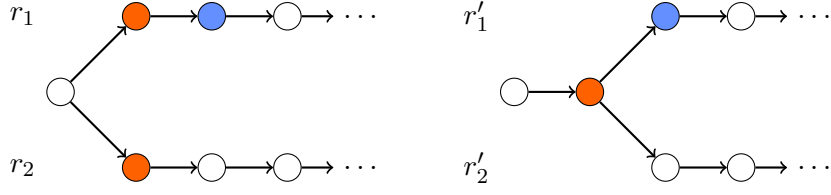


Figure 4.8: Tree models $\mathfrak{T}_1$ (Left) and $\mathfrak{T}_2$ (Right)

We now prove that $(\mathfrak{T}_1, (r_1, 0)) \equiv_{\infty}^{\text{CTL}^*} (\mathfrak{T}_2, (r_1, 0))$. The **A** move switches what branch of time the **U** move will be played on. As the only branching occurs at the root, the **A** move is only effective at the root. At any other point in time, the **A** move can only move a pebble to its current position. Thus, we only need to consider strategies where the spoiler plays some amount of **A** moves at the start and exclusively **U** moves after.

After the spoiler uses an **A** move, the duplicator can always switch branches to a run that is identical in terms of when atomic propositions are true. Suppose that the spoiler uses an **A** move in $\mathfrak{T}_1$. If the spoiler moves to r_1 , the duplicator can move to r_1 in $\mathfrak{T}_2$. If the spoiler moves to r_2 or r_3 , the duplicator can respond with r_3 in $\mathfrak{T}_2$. Otherwise, suppose spoiler uses an **A** move in $\mathfrak{T}_2$. If the spoiler moves to r_1 or r_2 , the duplicator can move to r_1 in $\mathfrak{T}_1$. If the spoiler moves to r_3 , then the duplicator can respond with r_3 in $\mathfrak{T}_1$.

The duplicator's strategy guarantees that after any number of **A** moves, our current configuration $((r_i, 0), (r_j, 0))$ has $r_i(t) \in V_1(p)$ iff $r_j(t) \in V_2(p)$ for all $t \in \mathbb{N}$. If the spoiler plays a **U** move, the duplicator can play the same way on the other tree. As the linear models of r_i and r_j are identical, no amount of **U** moves can distinguish the two runs. This gives us an ∞ -round winning strategy for the duplicator, and completes the proof of our proposition. $\square$

Example 4.47. We will show that the spoiler can distinguish between the two tree models depicted in Figure 4.8. Let $S = \{s_0, s_1, s_2, s_3, s_4\}$ and $P = \{a, b\}$. The figure uses orange to represent a and blue for b . We then define $V(a) = \{s_1, s_2\}$ and $V(b) = \{s_3\}$. We define $\mathfrak{T}_1 = (S, V, \mathcal{R}_1)$, where $\mathcal{R}_1$ contains r_1 given by the sequence $s_0, s_1, s_3, s_4, s_4, s_4, \dots$ and r_2 given by the sequence $s_0, s_2, s_4, s_4, s_4, \dots$. Now define $\mathfrak{T}_2 = (S, V, \mathcal{R}_2)$, which contains the runs r'_1 given by $s_0, s_1, s_3, s_4, s_4, s_4, \dots$ and r'_2 given by $s_0, s_1, s_4, s_4, s_4, \dots$.

We prove that $(\mathfrak{T}_1, (r_1, 0)) \not\equiv_3^{\text{CTL}^*} (\mathfrak{T}_2, (r'_1, 0))$. The spoiler first plays an **X** move such that the current configuration is $((r_1, 1), (r'_1, 1))$. The spoiler now plays an **A** move from $(r'_1, 1)$ to $(r'_2, 1)$. As there is no branching at $(r_1, 1)$, the duplicator can only remain on this run. Now, we play another **X** move to end in the configuration $((r_1, 2), (r'_2, 2))$. As b holds at $(r'_2, 2)$ but not $(r'_1, 2)$, the spoiler wins this game. In particular, we find that the formula $\mathbf{XAX}b$ is evaluated

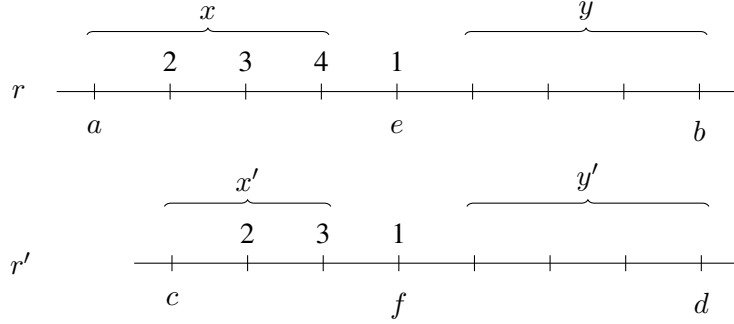


Figure 4.9: Spoiler's winning strategy for linear orders

differently on each tree model.

The introduction of an **A** move also gives rise to a novel $LTL(\mathbf{U})$ EF game. The main difference is that the original game was played over a pair of linear models. The following game is played over a pair of tree models, based on Definition 4.41.

Definition 4.48. The $LTL(\mathbf{U})$ game over tree models can be defined as follows. The setup is the same as Definition 4.42. The spoiler first plays an **A** move. Then, every round consists of the spoiler playing a **U** move. We write $(\mathfrak{T}_1, (r_1, t_1)) \equiv_n^{\mathbf{ALTL}} (\mathfrak{T}_2, (r_2, t_2))$ if the duplicator has an n -round winning strategy.

Theorem 4.49. Let $\mathfrak{T}_1$ and $\mathfrak{T}_2$ be tree models over P . Then, the following are equivalent:

1. For all formulae $\varphi \in LTL(\mathbf{U})[n]$, $\mathfrak{T}_1, (r_1, t_2) \models^{LTL} \varphi$ iff $\mathfrak{T}_2, (r_2, t_2) \models^{LTL} \varphi$.
2. $(\mathfrak{T}_1, (r_1, t_2)) \equiv_n^{\mathbf{ALTL}} (\mathfrak{T}_2, (r_2, t_2))$.

Example 4.50. We can use this game to find that CTL^* is strictly more expressive than LTL over tree models. Using the tree models in Figure 4.8, we show that $(\mathfrak{T}_1, (r_1, 0)) \equiv_\infty^{LTL} (\mathfrak{T}_2, (r'_1, 0))$, and therefore, $\mathbf{XAX}p$ is not expressible. When the spoiler plays an **A** move on $\mathfrak{T}_1$ to switch to r_i , the duplicator can switch to r'_i . If the spoiler instead plays an **A** move on $\mathfrak{T}_2$ to switch to r'_i , the duplicator can switch to r_i in $\mathfrak{T}_1$. The runs r_i and r'_i have the same atomic propositions hold at each time t . Thus, the duplicator can copy any **U** move that the spoiler makes. This gives us an ∞ -round winning strategy for the duplicator.

To play the **A** move, we must verify that $r'_1[0, t_1] = r_1[0, t_1]$ and $r'_2[0, t_2] = r_2[0, t_2]$. In the spirit of an EF game, this computation should be built into the game. The heavy lifting can be left to an EF game. In particular, we have the following proposition.

Proposition 4.51. Let r, r' be runs and $a \leq b, c \leq d$ integers. Let $n = \lceil \log_2 \max\{b-a, d-c\} \rceil$. We have $r[a, b] = r'[c, d]$ iff $r[a, b] \equiv_n^{FO} r'[c, d]$.

Proof. If $r[a, b] = r'[c, d]$, it is obvious that $r[a, b] \equiv_n^{\text{FO}} r'[c, d]$. Suppose that $r[a, b] \neq r'[c, d]$. We first consider the case where $b - a \neq d - c$. Without loss of generality, assume $b - a > d - c$. We detail the spoiler's strategy. This strategy is played out on an example in Figure 4.9, where the pebbles are labelled by the round they were played in. The spoiler only plays in $r[a, b]$.

The spoiler places a pebble e at the midpoint of $r[a, b]$. The linear order $r[a, b]$ has been split into two segments of size x and y , respectively. The duplicator's response f in $r'[c, d]$ also splits it into two segments of size x' and y' , respectively. We must either have $x' < x$ or $y' < y$. Suppose that $x' < x$. Then, the spoiler will only place pebbles before e . This means that each response the duplicator has will also be before f . If $y' < y$, then the spoiler will only place pebbles after e , and the duplicator must play after f in the other structure.

The spoiler now places a pebble at the midpoint of the half, as chosen above. The duplicator must also play in the corresponding half to maintain its winning condition. The spoiler keeps placing pebbles at the midpoint of its advantageous half. We maintain the invariant that the half that the spoiler plays in is bigger than the half that the duplicator plays in. At some point, the spoiler will place its final pebble in its chosen half, but the duplicator has no more space. Thus, the spoiler wins the EF game. This strategy will take at most $\lceil \log_2(b - a) \rceil$ moves, as the space for the duplicator to play in is cut in half after each round.

The other case is that $b - a = d - c$, but there exists $k \in [0, b - a]$ such that $r(a + k) \neq r(c + k)$. Similar to the strategy before, the spoiler places a pebble at the midpoint of $r[a, b]$, splitting the linear order into segments of size x and y . The duplicator must do the same in $r'[c, d]$, such that $x' = x$ and $y' = y$. Otherwise, we can use the strategy above for the spoiler to win. The spoiler now plays in the half where $a + k$ resides. Since we maintain that the distances between pebbles are the same, eventually, the spoiler places a pebble on $a + k$ and the duplicator places a pebble on $c + k$. As they have different states, the spoiler wins. $\square$

The revised **A** move is as follows:

A: Suppose the spoiler chooses $r'_1 \in \mathcal{R}_1$. If $r_1[0, t_1] \not\equiv_{\lceil \log_2 t_1 \rceil}^{\text{FO}} r'_1[0, t_1]$, then the duplicator wins. Otherwise, the duplicator responds with $r'_2 \in \mathcal{R}_2$. If $r_2[0, t_2] \not\equiv_{\lceil \log_2 t_2 \rceil}^{\text{FO}} r'_2[0, t_2]$, then the spoiler wins. The case for the spoiler picking $r'_2 \in \mathcal{R}_2$ is symmetric. The new configuration is $((r'_1, t_1), (r'_2, t_2))$.

4.4.2 Separation in Branching Time

Definition 4.52. Define *Past CTL** (PCTL*) as CTL* with the **S** operator. For tree models $\mathfrak{T}$, we have $\mathfrak{T}, (r, t) \models \varphi \text{ S } \psi$ iff there exists $t' < t$ where $\mathfrak{T}, (r, t') \models \psi$ and we have

$\mathfrak{T}, (r, t'') \models \varphi$ for all $t' < t'' < t$. The PCTL* EF game extends the CTL* game with the additional move:

S: The spoiler either chooses $\mathfrak{T}_1$ or $\mathfrak{T}_2$. We detail the move for $\mathfrak{T}_1$ as the move for $\mathfrak{T}_2$ is symmetrical. The spoiler chooses $t'_1 < t_1$, and the duplicator chooses $t'_2 < t_2$. Then, the spoiler can set the new configuration to $((r_1, t'_1), (r_2, t'_2))$, or the spoiler chooses $t''_2 \in (t'_2, t_2)$ and the duplicator responds with $t''_1 \in (t'_1, t_1)$. Then, the new configuration is $((r_1, t''_1), (r_2, t''_2))$.

Hafer and Thomas first showed that PCTL* was only as expressive as CTL* in [HT87].

Theorem 4.53. *Every PCTL* formula φ has an initially equivalent CTL* formula ψ . That is, for all tree models $\mathfrak{T}$ and runs $r \in \mathfrak{T}$, we have $\mathfrak{T}, (r, 0) \models \varphi$ iff $\mathfrak{T}, (r, 0) \models \psi$.*

The proof provided by Hafer and Thomas is quite complex. Kupferman, Pnueli and Vardi provide a simpler proof in [KPV12]. Their proof shows that every PCTL* formula can be written as a conjunction $\varphi \wedge \psi$, where φ is a CTL* formula and ψ starts with an S, similar to the concept of separation.

We adapt Kupferman, Pnueli and Vardi's proof to use game semantics. In the following proof, we will describe the spoiler's strategy in terms of what moves they play. We note that the spoiler's play is dependent on the duplicator's response. A different response by the duplicator can mean that the spoiler plays a different move. We can then view the spoiler's strategy as a tree of moves where each node is A, U or S, depending on what the spoiler would choose. Rather than specifying the entire tree, we instead describe the shape that each branch takes. We will use a notation based on regular expressions.

Definition 4.54. Let E be a regular expression with the alphabet $\Sigma = \{\mathbf{A}, \mathbf{U}, \mathbf{S}\}$. The *spoiler's strategy has the form E* if every branch of the strategy satisfies E .

Example 4.55. Suppose the spoiler's strategy has the following form: the first move is A, followed by some (possibly zero) amount of U and/or S moves, then only A and/or U moves. This can be written as the spoiler's strategy having the form $\mathbf{A}(\mathbf{U} \mid \mathbf{S})^*(\mathbf{A} \mid \mathbf{U})^*$.

Lemma 4.56. *Let $n \in \mathbb{N}$. Suppose that $(\mathfrak{T}_1, (r_1, t_1)) \not\equiv_n^{\text{PCTL}^*} (\mathfrak{T}_2, (r_2, t_2))$ such that the spoiler has a winning strategy of the form $(\mathbf{U} \mid \mathbf{S})^*(\mathbf{A} \mid \mathbf{U})^*$. Then, the spoiler either has a winning strategy of the form $(\mathbf{A} \mid \mathbf{U})^*$, or the form $(\mathbf{S}^* \mid \mathbf{S}^* \mathbf{A}(\mathbf{A} \mid \mathbf{U})^*)$.*

Proof. The spoiler's winning strategy has the form $(\mathbf{U} \mid \mathbf{S})^*(\mathbf{A} \mid \mathbf{U})^*$. As the prefix of this strategy consists of U and S moves, we are playing the LTL(U, S) game on r_1 and r_2 until a CTL* formula is evaluated differently on the two runs. We then use the separability of LTL to

simplify the form of the spoiler's strategy.

We now define the linear models that we will play on. Let F be the set of formulae in $\text{CTL}^*[n]$ that are either atomic propositions or have the form $\mathbf{A}\varphi \in \text{CTL}^*[n]$. We define the linear model $\mathfrak{R}_1 = (\mathbb{N}, <, s, V_1)$, where we treat the formulae in F as atomic propositions. For all formulae $\varphi \in F$, we have $t \in V_1(\varphi)$ iff $\mathfrak{T}_1, (r_1, t) \models \varphi$. We define $\mathfrak{R}_2 = (\mathbb{N}, <, s, V_2)$ similarly, where for all formulae $\varphi \in F$, we have $t \in V_2(\varphi)$ iff $\mathfrak{T}_2, (r_2, t) \models \varphi$.

The spoiler's strategy either contains an $\mathbf{A}$ move or it does not. That is, it has the form $(\mathbf{U} \mid \mathbf{S})^*$ or $(\mathbf{U} \mid \mathbf{S})^* \mathbf{A}(\mathbf{A} \mid \mathbf{U})^*$. In the first case, an atomic proposition is evaluated differently on r_1 and r_2 . In the second case, the CTL^* formula evaluated differently on the two runs has the form $\mathbf{A}\varphi \in \text{CTL}^*[n]$. Therefore, $(\mathfrak{R}_1, t_1) \not\equiv_n^{\text{LTL}(\mathbf{U}, \mathbf{S})} (\mathfrak{R}_2, t_2)$. From Lemma 4.27, we either have $(\mathfrak{R}_1, t_1) \not\equiv_n^{\text{LTL}(\mathbf{U})} (\mathfrak{R}_2, t_2)$ or $(\mathfrak{R}_1, t_1) \not\equiv_n^{\text{LTL}(\mathbf{S})} (\mathfrak{R}_2, t_2)$.

If $(\mathfrak{R}_1, t_1) \not\equiv_n^{\text{LTL}(\mathbf{U})} (\mathfrak{R}_2, t_2)$, then we can directly recover a strategy of the form $(\mathbf{A} \mid \mathbf{U})^*$. We assume that $(\mathfrak{R}_1, t_1) \not\equiv_n^{\text{LTL}(\mathbf{S})} (\mathfrak{R}_2, t_2)$. The spoiler's strategy can be translated into a strategy for the CTL^* game. For each branch, consider the formula in F that was different between the two pebbles in our final configuration. If the formula was an atomic proposition p , then the branch is unchanged and satisfies $\mathbf{S}^*$. Otherwise, the formula had the form $\mathbf{A}\varphi \in \text{CTL}^*[n]$, and we append the strategy for $\mathbf{A}\varphi$ to the branch. This gives us a branch that satisfies $\mathbf{S}^* \mathbf{A}(\mathbf{A} \mid \mathbf{U})^*$. Thus, we have a winning strategy for the spoiler of the form $(\mathbf{S}^* \mid \mathbf{S}^* \mathbf{A}(\mathbf{A} \mid \mathbf{U})^*)$. $\square$

Lemma 4.57. *Let $n \in \mathbb{N}$. Suppose that $(\mathfrak{T}_1, (r_1, t_1)) \not\equiv_n^{\text{PCTL}^*} (\mathfrak{T}_2, (r_2, t_2))$ such that the spoiler has a winning strategy of the form $\mathbf{A}(\mathbf{U} \mid \mathbf{S})^*(\mathbf{A} \mid \mathbf{U})^*$. Then, the spoiler has a winning strategy of the form $(\mathbf{S}^* \mid \mathbf{S}^* \mathbf{A}(\mathbf{A} \mid \mathbf{U})^*)$.*

Proof. Consider the spoiler's strategy as a tree with a root of $\mathbf{A}$. Each child of $\mathbf{A}$ is the root of a strategy with the form $(\mathbf{U} \mid \mathbf{S})^*(\mathbf{A} \mid \mathbf{U})^*$. From Lemma 4.56, we can replace each child of $\mathbf{A}$ with a strategy of the form $(\mathbf{A} \mid \mathbf{U})^*$ or $(\mathbf{S}^* \mid \mathbf{S}^* \mathbf{A}(\mathbf{A} \mid \mathbf{U})^*)$. If every child is replaced with a strategy of the form $(\mathbf{A} \mid \mathbf{U})^*$, we are done, as this fits the desired form of $(\mathbf{S}^* \mid \mathbf{S}^* \mathbf{A}(\mathbf{A} \mid \mathbf{U})^*)$.

Otherwise, suppose there exists a child t which is the root of a strategy with the form $(\mathbf{S}^* \mid \mathbf{S}^* \mathbf{A}(\mathbf{A} \mid \mathbf{U})^*)$. We claim that the first $\mathbf{A}$ move can be ignored, and the spoiler can directly play the strategy that t is the root of. Either t is an $\mathbf{A}$ move or an $\mathbf{S}$ move. If t is an $\mathbf{A}$ move, then the choice that would have been made at t can also be made before playing the first $\mathbf{A}$ move. The spoiler is guaranteed a winning strategy for any response from the duplicator by playing from t onwards.

In the other case, t is an **S** move. Any branch with root t satisfies $\mathbf{S}^*$ or $\mathbf{S}^* \mathbf{A}(\mathbf{A} \mid \mathbf{U})$. In the first case, the first **A** move does not change the past, and so the **A** move is inconsequential to branches of the form $\mathbf{S}^*$. In the second case, we play the **A** move, some amount of **S** moves, then another **A** move. Again, the first **A** move is inconsequential to the **S** moves. The second **A** move then overwrites the choice that the first **A** move made. Thus, the spoiler has a winning strategy by ignoring the first **A** move and instead starting their play at t . $\square$

Proposition 4.58. *Let $n \in \mathbb{N}$. If $(\mathfrak{T}_1, (r_1, t_1)) \not\equiv_n^{PCTL^*} (\mathfrak{T}_2, (r_2, t_2))$, then the spoiler's strategy has the form $\mathbf{S}^*(\mathbf{A} \mid \mathbf{U})^*$.*

Proof. We induct on the number of **A** moves that are possible in the spoiler's strategy. Suppose that we have a strategy with no **A** moves. This base case is trivial through an application of Lemma 4.56. Suppose that $(\mathfrak{T}_1, (r_1, t_1)) \not\equiv_n^{PCTL^*} (\mathfrak{T}_2, (r_2, t_2))$, where the spoiler will play the **A** move at most $k+1$ times. The spoiler's strategy will have the form $(\mathbf{U} \mid \mathbf{S})^* \mathbf{A}(\mathbf{A} \mid \mathbf{U} \mid \mathbf{S})^*$. In the final section $(\mathbf{A} \mid \mathbf{U} \mid \mathbf{S})^*$, the spoiler can play the **A** move at most k times. From our inductive hypothesis, we can replace this section with $\mathbf{S}^*(\mathbf{A} \mid \mathbf{U})^*$. We now have a strategy of the form $(\mathbf{U} \mid \mathbf{S})^* \mathbf{A} \mathbf{S}^*(\mathbf{A} \mid \mathbf{U})^*$. We now use Lemma 4.57 to replace the section of the form $\mathbf{A} \mathbf{S}^*(\mathbf{A} \mid \mathbf{U})^*$ with $(\mathbf{S}^* | \mathbf{S}^* \mathbf{A}(\mathbf{A} \mid \mathbf{U})^*)$. We have a new strategy for the spoiler of the form $(\mathbf{U} \mid \mathbf{S})^* \mathbf{S}^*(\mathbf{A} \mid \mathbf{U})^*$. We now apply Lemma 4.56 to complete the proof. $\square$

Corollary 4.59. *Let $n \in \mathbb{N}$. If $(\mathfrak{T}_1, (r_1, 0)) \not\equiv_n^{PCTL^*} (\mathfrak{T}_2, (r_2, 0))$, then $(\mathfrak{T}_1, (r_1, 0)) \not\equiv_m^{CTL^*} (\mathfrak{T}_2, (r_2, 0))$ for some $m \in \mathbb{N}$.*

Proof. We note that the construction in Lemma 4.56 gives us a $2n$ -round winning strategy for the spoiler rather than an n -round winning strategy. From Proposition 4.58, the spoiler has an m -round winning strategy of the form $\mathbf{S}^*(\mathbf{A} \mid \mathbf{U})^*$ for some $m \in \mathbb{N}$. Since there is no past at time 0, it is not possible to play any **S** moves. Thus, the strategy must have the form $(\mathbf{A} \mid \mathbf{U})^*$, which gives the spoiler a winning strategy for the CTL^* game. $\square$

Theorem 4.53. *Every $PCTL^*$ formula φ has an initially equivalent CTL^* formula ψ . That is, for all tree models $\mathfrak{T}$ and runs $r \in \mathfrak{T}$, we have $\mathfrak{T}, (r, 0) \models \varphi$ iff $\mathfrak{T}, (r, 0) \models \psi$.*

Proof. Let φ be a $PCTL^*$ formula and $\mathcal{P}$ be the class of pointed tree models at time 0 such that φ holds. From the EF-style Corollary of the $PCTL^*$ game, for all $(\mathfrak{T}_1, (r_1, 0)) \in \mathcal{P}$ and $(\mathfrak{T}_2, (r_2, 0)) \notin \mathcal{P}$, we have $(\mathfrak{T}_1, (r_1, 0)) \not\equiv_n^{PCTL^*} (\mathfrak{T}_2, (r_2, 0))$ for some $n \in \mathbb{N}$. From Corollary 4.59, $(\mathfrak{T}_1, (r_1, 0)) \not\equiv_m^{CTL^*} (\mathfrak{T}_2, (r_2, 0))$. The EF-style Corollary of the CTL^* game tells us that there exists a CTL^* formula ψ that expresses $\mathcal{P}$. $\square$

Chapter 5

Temporal Epistemic Logic

We have now covered the background required to define temporal epistemic logics. In this chapter, we introduce several temporal epistemic logics based on the conventions in [HV89]. We then introduce original EF games for each of these logics. The games are then used to explore the open question of whether past-time operators allow for greater expressibility.

5.1 Syntax, Semantics and Games

Epistemic logic can model and reason about knowledge at a single point in time. The information known to each agent is static, which contrasts with real life. One way to remedy this is temporal epistemic logic, that is, modelling how knowledge changes over time. We can construct a temporal epistemic logic by taking a temporal logic such as LTL or CTL* and allowing for the use of knowledge operators. The preferred choice of structure for temporal epistemic logics is an *interpreted system*, a generalisation of tree models. Given agents in A , we extend tree models by adding the equivalence relations $\sim_a$ for all $a \in A$, as can be seen in an epistemic model. There is also no guarantee of a unique first state.

Definition 5.1. Let S be a set of states, P be a finite set of atomic propositions, and A be a set of agents. An *interpreted system* $\mathfrak{I} = (S, V, \mathcal{R}, \sim)$ has a valuation function $V : S \rightarrow 2^P$, a non-empty set of runs $\mathcal{R}$ and a function $\sim : A \rightarrow S \times S$ such that $\sim_a$ is an equivalence relation.

Halpern and Vardi found that dozens of logicians conceived the idea of combining temporal operators with epistemic ones. However, each paper referenced a different logic due to different choices in the language and structures used. In total, 96 different temporal epistemic logics were categorised and analysed by Halpern and Vardi [HV89]. The following section will go through some of the choices that can be made to form a unique temporal epistemic logic. We

must choose the class of interpreted systems we use. The choices are as follows:

- Is there a unique first state?
- Can the agents forget information?
- Can the agents learn information?
- Is the system synchronous?

Definition 5.2. Let $\mathcal{I} = (S, V, \mathcal{R}, \sim)$ be an interpreted system. The system has a *unique first state* if $r(0) = r'(0)$ for all $r, r' \in \mathcal{R}$. An agent a *does not forget* in $\mathcal{I}$ if for all runs $r, r' \in \mathcal{R}$, times $t, t' \in \mathbb{N}$ and $u \leq t$, if $r(t) \sim_a r'(t')$, then there exists $u' \leq t'$ such that $(r, u) \sim_a (r', u')$. We also define the dual notion. An agent a *does not learn* if for all runs $r, r' \in \mathcal{R}$, times $t, t' \in \mathbb{N}$ and $u \geq n$, if $r(t) \sim_a r'(t')$, then there exists $u' \geq n'$ such that $(r, u) \sim_a (r', u')$. An interpreted system is *synchronous* if for all agents a , runs $r, r' \in \mathcal{R}$, and times $t, t' \in \mathbb{N}$, $r(t) \sim_a r'(t')$ implies that $t = t'$.

An interpreted system *does not forget* if for all agents a and runs r , agent a does not forget in run r . We define an interpreted system that *does not learn* similarly. We denote $\mathcal{I}$ as the class of interpreted systems. Subscripts denote the properties that the class has. The class of synchronous interpreted systems with a unique first state that does not forget and does not learn is written as $\mathcal{I}_{ufs,nf,nl,syn}$.

A unique first state means that an interpreted system has a tree structure. A synchronous system means that all agents know the current time. States that cannot be told apart must occur at the same time. Equivalently, agents can distinguish between states that happen at different times. We introduce the idea of a history to motivate the definition of an agent that does not forget.

Definition 5.3. Suppose that a run r is given by the sequence of state $s_0, s_1, \dots, s_n, \dots$. For $s \in S$, we define $[s]_a$ as the equivalence class of s with respect to $\sim_a$. An agent a 's *history* at (r, t) is the sequence $[s_0]_a, [s_1]_a, \dots, [s_t]_a$ with consecutive repetitions omitted.

Example 5.4. The information known to an agent is characterised by the equivalence class of states that an agent finds indistinguishable. Suppose that r starts with the sequence of states s_0, s_1, s_2, s_1, s_1 such that $s_0 \sim_a s_1$ and $s_0 \not\sim_a s_2$ for some agent a . Then, agent a 's history at $(r, 4)$ is $[s_0]_a, [s_2]_a, [s_1]_a$.

Proposition 5.5 ([HV89, Lemma 2.1]). *Agent a does not forget in $\mathcal{I}$ iff for all $r, r' \in \mathcal{R}$ and $t, t' \in \mathbb{N}$, $r(t) \sim_a r'(t')$ implies that agent a 's history is the same at (r, t) and (r', t') .*

The following choices have to be made about the language:

- Should the logic use linear or branching time?
- Should common knowledge be included as an operator?

The following semantics is for the most general syntax CTL^*CK , which uses branching time and includes common knowledge. Appropriate adjustments to the definition can be made to define CTL^*K , which does not include common knowledge.

Definition 5.6. The CTL^*CK formulae φ over propositions $p \in P$ are given by the rule

$$\varphi := p \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \mathbf{U} \psi \mid \mathbf{A}\varphi \mid K_a\varphi \mid C_G\varphi,$$

where $a \in A$ is some agent and non-empty subsets $G \subseteq A$. Let $\mathcal{I} = (S, V, \mathcal{R}, \sim)$ be an interpreted system with run r and $t \in \mathbb{N}$. We define the semantics as follows:

- For atomic propositions p , $\mathcal{I}, (r, t) \models p$ iff $r(t) \in V(p)$ holds.
- $\mathcal{I}, (r, t) \models \neg\varphi$ and $\mathcal{I}, (r, t) \models \varphi \wedge \psi$ can be defined similar to before.
- $\mathcal{I}, (r, t) \models \varphi \mathbf{U} \psi$ iff there exists $t' > t$ where $\mathcal{I}, (r, t') \models \psi$ and we have $\mathcal{I}, (r, t'') \models \varphi$ for all $t < t'' < t'$.
- $\mathcal{I}, (r, t) \models \mathbf{A}\varphi$ iff $\mathcal{I}, (r', t) \models \varphi$ for all runs $r' \in \mathcal{R}$ where $r'[0, t] = r[0, t]$.
- $\mathcal{I}, (r, t) \models K_a\varphi$ iff $\mathcal{I}, (r', t') \models \varphi$ for all runs $r' \in \mathcal{R}$ and $t \in \mathbb{N}$ where $r(t) \sim_a r'(t')$.

For all $G \subseteq A$, define $\sim_{E_G} = \bigcup_{a \in G} \sim_a$ and $\sim_{C_G}$ as the transitive closure of $\sim_{E_G}$.

- $\mathcal{I}, (r, t) \models C_G\varphi$ iff $\mathcal{I}, (r', t') \models \varphi$ for all runs $r' \in \mathcal{R}$ and $t \in \mathbb{N}$ where $r(t) \sim_{C_G} r'(t')$.

Definition 5.7. The CTL^*CK EF game is an extension of the CTL^* EF game, instead being played on a pair of interpreted systems $\mathcal{I}_1, \mathcal{I}_2$. The game also features two additional moves:

- K_a : The spoiler either chooses $r'_1 \in \mathcal{R}_1$ and $t'_1 \in \mathbb{N}$ such that $r_1(t_1) \sim_a r'_1(t'_1)$ and the duplicator responds with $r'_2 \in \mathcal{R}_2$ and $t'_2 \in \mathbb{N}$ such that $r_2(t_2) \sim_a r'_2(t'_2)$, or the spoiler chooses $r'_2 \in \mathcal{R}_2$ and $t'_2 \in \mathbb{N}$ such that $r_2(t_2) \sim_a r'_2(t'_2)$ and the duplicator responds with $r'_1 \in \mathcal{R}_1$ and $t'_1 \in \mathbb{N}$ such that $r_1(t_1) \sim_a r'_1(t'_1)$.
- C_G : The spoiler either chooses $r'_1 \in \mathcal{R}_1$ and $t'_1 \in \mathbb{N}$ such that $r_1(t_1) \sim_{C_G} r'_1(t'_1)$ and the duplicator responds with $r'_2 \in \mathcal{R}_2$ and $t'_2 \in \mathbb{N}$ such that $r_2(t_2) \sim_{C_G} r'_2(t'_2)$, or the spoiler chooses $r'_2 \in \mathcal{R}_2$ and $t'_2 \in \mathbb{N}$ such that $r_2(t_2) \sim_{C_G} r'_2(t'_2)$ and the duplicator responds with $r'_1 \in \mathcal{R}_1$ and $t'_1 \in \mathbb{N}$ such that $r_1(t_1) \sim_{C_G} r'_1(t'_1)$.

The new configuration is $((r'_1, t'_1), (r'_2, t'_2))$. We write $(\mathcal{I}_1, (r_1, t_1)) \equiv_n^{CTL^*CK} (\mathcal{I}_2, (r_2, t_2))$ if the duplicator has an n -round winning strategy.

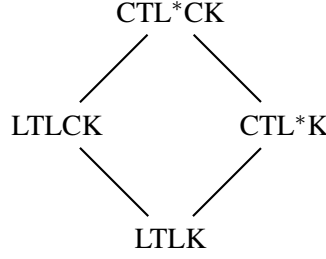


Figure 5.1: Hierarchy of Temporal Epistemic Logics

Corollary 5.8. *A property $\mathcal{K}$ is inexpressible in CTL^*CK iff for all $n \in \mathbb{N}$, there exists $(\mathfrak{J}, (r, 0)) \in \mathcal{K}$ and $(\mathfrak{J}', (r', 0)) \notin \mathcal{K}$ such that $(\mathfrak{J}, (r, 0)) \equiv_n^{CTL^*CK} (\mathfrak{J}', (r', 0))$.*

We can analyse a different logic by changing the structures we play our game on or changing the moves. By removing the C_G move from the CTL^*CK EF game, we can make the CTL^*K EF game. We can also restrict the kinds of interpreted systems we play over. To show that a property is inexpressible over $\mathcal{I}_{nl}$, we must ensure that we only play games over a pair of systems in $\mathcal{I}_{nl}$.

We now consider the case where the logic uses linear time.

Definition 5.9. The syntax of $LTLCK$ is given by

$$\varphi := p \mid \neg\varphi \mid \varphi \wedge \varphi \mid \varphi \mathbf{U} \varphi \mid K_a\varphi \mid C_G\varphi,$$

where $a \in A$ is some agent and non-empty subsets $G \subseteq A$. We define $LTLK$ as the fragment of $LTLCK$ that does not use the common knowledge operator. The $LTLCK$ formula φ is true, denoted as $\mathfrak{J}, (r, t) \models^{LTL} \varphi$, if $\mathfrak{J}, (r, t) \models \mathbf{A}\varphi$.

In terms of expressibility, we have a partial order for the temporal epistemic logics defined in this thesis. We can see a Hasse diagram in Figure 5.1.

Definition 5.10. The *LTLCK game* can be defined as follows. The setup is the same as Definition 5.7. The spoiler first plays an $\mathbf{A}$ move. Every round, the spoiler can choose to play a $\mathbf{U}$, K_a or C_G move. We write $(\mathfrak{J}_1, (r_1, t_1)) \equiv_n^{LTLCK} (\mathfrak{J}_2, (r_2, t_2))$ if the duplicator has an n -round winning strategy. We get the *LTLK game* by removing the option of a C_G move.

5.2 Past-Time Operators

As we have seen, the past-time operator $\mathbf{S}$ does not add any expressive power to LTL and CTL^* . Extrapolating these results, most of the literature on temporal epistemic logics chooses not to

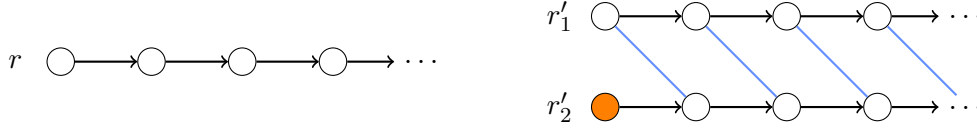


Figure 5.2: $\mathcal{J}_1$ (Left) and $\mathcal{J}'_1$ (Right)

have the past-time operator **S** either. However, we find that the addition of **S** does increase expressibility for a variety of cases. We define *Past LTLK*, *Past LTLCK*, *Past CTL*K* and *Past CTL*CK* similarly to Definition 4.52. Recall that $\widehat{K}_a$ is the dual of K_a .

Proposition 5.11. *The formula $\mathbf{A}\widehat{K}_a\mathbf{Y}p$ is not expressible in CTL^*CK over $\mathcal{I}_{nl}$.*

We first define $\mathcal{J}_1$ and $\mathcal{J}'_1$, as seen in Figure 5.2. Let $A = \{a\}$, $P = \{p\}$ and $S = \{s_0, s_1, \dots\}$. We have $\mathcal{J}_1 = (S, V, \mathcal{R}, \sim)$, where $V(p) = \{s_0\}$ and $\sim_a$ is the identity relation on S . In $\mathcal{R}$, we have one run r , where $r(t) = s_{t+1}$ for all $t \in \mathbb{N}$. We define $\mathcal{J}'_1 = (S, V, \mathcal{R}', \sim)$, where $\mathcal{R}'$ contains two runs r'_1 and r'_2 . Let $r'_1(t) = s_{t+1}$ for all $t \in \mathbb{N}$ and let $r'_2(t) = s_t$ for all $t \in \mathbb{N}$.

In Figure 5.2, the states coloured orange have p hold, while there are no atomic propositions true in uncoloured states. The blue lines represent the equivalence relation $\sim_a$. It is clear that $\mathbf{A}\widehat{K}_a\mathbf{Y}p$ holds at $(\mathcal{J}'_1, (r'_1, 0))$, but not $(\mathcal{J}_1, (r, 0))$. We also note that both structures are interpreted systems that do not learn. Thus, it suffices to prove the following lemma.

Lemma 5.12. *We have $(\mathcal{J}_1, (r_1, 0)) \equiv_{\infty}^{\text{CTL}^*\text{CK}} (\mathcal{J}'_1, (r'_1, 0))$.*

Proof. The main idea of the proof is that the moves in the CTL^*CK game can never access the point $(r'_2, 0)$. Since every other point has no atomic propositions that hold, the duplicator will trivially win. We induct on the number of moves n . When $n = 0$, the current configuration is $((r, 0), (r'_1, 0))$. We maintain the invariant that the duplicator can play such that the configuration always has the form $((r, t), (r'_1, t))$ or $((r, t), (r'_2, t + 1))$ for some $t \in \mathbb{N}$.

Suppose that the current configuration is $((r, t), (r'_1, t))$. The **A** move does not allow us to move to a different point, as there is no other run with the same past. When playing the **U** move, the spoiler will pick some $t' > t$, in which the duplicator can do the same. They can then choose to continue to the next move, or pick some $t'' \in (t, t')$. Either way, the duplicator can do the same, and we end up at a configuration with the form $((r, t'), (r'_1, t'))$ for some $t' \in \mathbb{N}$. If the spoiler plays a K_a move, then they can choose to stay at the same point, which the duplicator can also do, or move to the other run. In this case, the duplicator does the same, and we reach the configuration $((r, t), (r'_2, t + 1))$. Note that the C_G move is the same as the K_a move, as there is only one agent.



Figure 5.3: $\mathcal{J}_2$ (Left) and $\mathcal{J}'_2$ (Right)

We can do a similar analysis for when the current configuration is $((r, t), (r'_2, t + 1))$. Since we maintain this invariant, the atomic propositions true on the pebble in one interpreted system are the exact propositions true on the pebble in the other. We can play this game indefinitely. $\square$

Since $\mathbf{A}\widehat{K}_a Yp$ is expressible in Past LTLK, the formula can also be expressed in the past counterparts of LTLCK, CTL^{*}K and CTL^{*}CK. As the formula cannot be expressed in CTL^{*}CK, it cannot be expressed in LTLK, LTLCK, nor CTL^{*}K. We prove this over the class of interpreted systems where agents never learn, denoted $\mathcal{I}_{nl}$. Every interpreted system in $\mathcal{I}_{nl}$ is also in $\mathcal{I}$, and so this proof also carries over to the general class of interpreted systems. This proposition shows that in 8 different temporal epistemic logics, the first two columns of Figure 5.5, the addition of **S** does increase expressibility.

Proposition 5.13. *The formula $\mathbf{A}\widehat{K}_a Yp$ is not expressible in CTL^{*}CK over $\mathcal{I}_{ufs,nf,nl}$.*

We first define $\mathcal{J}_2$ and $\mathcal{J}'_2$, as seen in Figure 5.3. Let $A = \{a\}$, $P = \{p\}$ and $S = \{s_0, s_1, \dots\}$. We have $\mathcal{J}_2 = (S, V, \mathcal{R}, \sim)$, where $V(p) = S$ and $\sim_a$ is the identity relation on S . In $\mathcal{R}$, we have one run r , where $r(t) = s_t$ for all $t \in \mathbb{N}$. We define $\mathcal{J}'_2 = (S, V, \mathcal{R}', \sim)$, where $\mathcal{R}'$ contains one run r' . Let $r'(t) = s_{t+1}$ for all $t \in \mathbb{N} - \{0, 1\}$ and let $r'(0) = r'(1) = s_0$.

The states coloured orange have p hold. The blue lines represent the equivalence relation $\sim_a$. It is clear that $\mathbf{A}\widehat{K}_a Yp$ holds at $(\mathcal{J}'_2, (r'_1, 0))$, but not $(\mathcal{J}_2, (r_1, 0))$. We also note that both structures are interpreted systems that have a unique first state, do not forget, and do not learn.

Lemma 5.14. *We have $(\mathcal{J}_2, (r, 0)) \equiv_{\infty}^{CTL^*CK} (\mathcal{J}'_2, (r', 0))$.*

Proof. This proof follows a simple argument, similar to that of Lemma 5.12. $\square$

Every class of asynchronous interpreted systems defined contains $\mathcal{I}_{ufs,nf,nl}$. We also note that $\mathbf{A}\widehat{K}_a Yp$ is expressible in Past LTLK, and thus the past counterparts of the other temporal epistemic logics. This shows that for all combinations of language and asynchronous systems, the inclusion of the past operator **S** does increase the expressibility of a temporal epistemic logic. This proposition fills out the first 8 columns of Figure 5.5.

The main idea behind the second set of interpreted systems is that one system is asynchronous. We cannot reuse this proof idea for the synchronous case. However, the first proposition uses

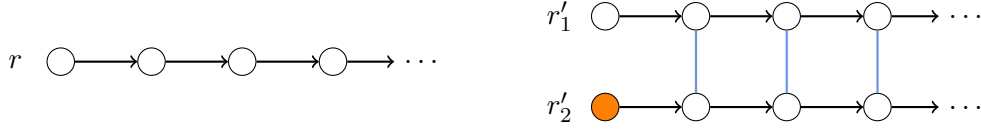


Figure 5.4: $\mathfrak{I}_3$ (Left) and $\mathfrak{I}'_3$ (Right)

	$\mathcal{I}$	$\mathcal{I}_{nl}$	$\mathcal{I}_{nf}$	$\mathcal{I}_{nl,nf}$	$\mathcal{I}_{ufs}$	$\mathcal{I}_{ufs,nl}$	$\mathcal{I}_{ufs,nf}$	$\mathcal{I}_{ufs,nl,nf}$	$\mathcal{I}_{syn}$	$\mathcal{I}_{nl,syn}$
LTLK	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
LTLCK	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CTL*K	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CTL*CK	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Figure 5.5: Does **S** add more expressibility?

the fact that one of the states is unreachable in the CTL*CK game. We can use the same idea for the final two columns of Figure 5.5.

Proposition 5.15. *The formula $\mathbf{AX}\widehat{K}_a\mathbf{Y}p$ is not expressible in CTL*CK over $\mathcal{I}_{nl,syn}$.*

We first define $\mathfrak{I}_1$ and $\mathfrak{I}'_1$, as seen in Figure 5.4. Let $A = \{a\}$, $P = \{p\}$ and $S = \{s_0, s_1, \dots\}$. We have $\mathfrak{I}_3 = (S, V, \mathcal{R}, \sim)$, where $V(p) = \{s_0\}$ and $\sim_a$ is the identity relation on S . In $\mathcal{R}$, we have one run r , where $r(t) = s_{t+1}$ for all $t \in \mathbb{N}$. We define $\mathfrak{I}'_3 = (S, V, \mathcal{R}', \sim)$, where $\mathcal{R}'$ contains two runs r'_1 and r'_2 . Let $r'_1(t) = s_{t+1}$ for all $t \in \mathbb{N}$. Let $r'_2(0) = s_0$ and $r'_2(t) = s_{t+1}$ for all $t \in \mathbb{N} - \{0\}$.

In Figure 5.4, the states coloured orange have p hold, while there are no atomic propositions true in uncoloured states. The blue lines represent the equivalence relation $\sim_a$. It is clear that $\mathbf{AX}\widehat{K}_a\mathbf{Y}p$ holds at $(\mathfrak{I}'_1, (r'_1, 0))$, but not $(\mathfrak{I}_1, (r_1, 0))$. We also note that both structures are synchronous interpreted systems that do not learn. The following lemma can be proved similarly to Lemma 5.12.

Lemma 5.16. *We have $(\mathfrak{I}_3, (r, 0)) \equiv_{\infty}^{CTL*CK} (\mathfrak{I}'_3, (r'_1, 0))$.*

We were unable to analyse the remaining temporal epistemic logics in this thesis. We leave this to be tackled in future works.

Chapter 6

Transitive Closure

This chapter aims to address the issue raised for the CEL EF game in Definition 3.25. That is, we want the computation of the transitive closure to be a part of the C_G move rather than an external calculation. To do so, we first look at first-order logic with transitive closure [Imm83] and introduce a novel associated EF game. This game serves as inspiration for a subgame that computes the transitive closure to improve the C_G move. We also examine temporal logics from the perspective of being defined through transitive closure. A by-product is the creation of an alternative game for unary temporal logic as well.

6.1 Extending First-order Logic

Many extensions of modal logic cannot be embedded into first-order logic. We have shown this to be true with common knowledge in Proposition 3.28. However, common knowledge is defined through transitive closure. By enriching first-order logic with transitive closure, we find that this classical logic expresses various extensions of modal logics in this thesis. We employ a characterisation of transitive closure defined through MSO [CE12].

Definition 6.1. First-order logic *with transitive closure* ($\text{FO} + \text{TC}^1$) extends FO by adding the following rule for defining the set of formulae over σ :

- If $\varphi(x, y, \bar{z})$ is a formula, then $\text{TC}_{x,y}[\varphi(x, y, \bar{z})](t_1, t_2)$ is a formula.

The free variables of $\text{TC}_{x,y}[\varphi(x, y, \bar{z})](t_1, t_2)$ are $\bar{z}$ and the free variables of t_1 and t_2 . Consider a formula $\varphi(x, y, \bar{z})$. The statement “ X is $\varphi(x, y, \bar{z})$ -closed”, where X is a unary predicate, is defined by the formula

$$\forall x \forall y (X(x) \wedge \varphi(x, y, \bar{z}) \rightarrow X(y)).$$

We write φ -closed if the arguments are clear. Let $\mathfrak{A}$ be a σ -structure and $\bar{a}$ be a tuple of elements in $\mathfrak{A}$. The semantics is defined as follows:

- $\mathfrak{A} \models \text{TC}_{x,y}[\varphi(x, y, \bar{a})](t_1, t_2)$ iff $\mathfrak{A} \models \forall X ("X \text{ is } \varphi(x, y, \bar{a})\text{-closed}" \wedge X(t_1) \rightarrow X(t_2))$.

Definition 6.2. We extend Definition 1.6 with $\text{qr}(\text{TC}_{x,y}[\varphi](t_1, t_2)) = 1 + \text{qr}(\varphi)$. The set of formulae in $\text{FO} + \text{TC}^1$ with quantifier rank at most n is denoted $\text{FO} + \text{TC}^1[n]$.

Example 6.3. We can express the connectivity of a graph $G = (V, E)$:

$$\forall x \forall y \text{TC}_{x,y}[E(x, y)](x, y).$$

The transitive closure allows us to talk about paths of arbitrary length. This sentence states that for any two vertices x, y , there exists a path from x to y .

Example 6.4. Consider a finite set A and $G \subseteq A$. We can extend the standard translation to express the common knowledge operator as follows:

$$\text{ST}_x(C_G \varphi) = \forall y \left(\text{TC}_{x,y} \left[\bigvee_{a \in G} x \sim_a y \right] (x, y) \rightarrow \text{ST}_y(\varphi) \right).$$

Example 6.5. A simple EF games argument can show that $\mathbf{F}$ cannot be expressed in $\text{FO}[s]$. However, $\text{FO}[s] + \text{TC}^1$ can. In fact, we can express $\mathbf{U}$. We extend the standard translation:

$$\text{ST}_x(\varphi \mathbf{U} \psi) = \exists y (\text{TC}_{x,y}[s(x, y) \wedge \text{ST}_y(\varphi)](x, y) \wedge \exists x (s(y, x) \wedge \text{ST}_x(\psi))).$$

This also yields a translation for $\mathbf{S}$. Thus, LTL can be embedded into $\text{FO}[s] + \text{TC}^1$. These two logics are not equivalent, as $\text{FO}[s] + \text{TC}^1$ can express that a proposition is true at every point in the future, which is an even number of points away. We can translate $\text{even}(\varphi)$ as follows:

$$\text{ST}_x(\text{even}(\varphi)) = \forall y (\text{TC}_{x,y}[\exists z (s(x, z) \wedge s(z, y))](x, y) \rightarrow \text{ST}_y(\varphi)).$$

Recall from Proposition 4.31 that even cannot be expressed in LTL.

We have presented a simplified fragment of first-order logic with transitive closure found in the literature, which we denote as $\text{FO} + \text{TC}$. In $\text{FO} + \text{TC}$, the transitive closure operator can be applied to $2k$ -ary relations, for all k . The “1” in $\text{FO} + \text{TC}^1$ refers to our transitive closure operator only taking binary relations, that is, $k = 1$. Immerman and Vardi give a non-trivial translation of CTL^* into $\text{FO} + \text{TC}$ [IV97], which inspired the translation of $\mathbf{U}$ found above.

6.1.1 The $FO + TC^1$ Game

We now present a novel EF game for $FO + TC^1$. We note that there is a trivial generalisation of the following game to an $FO + TC$ EF game.

Definition 6.6. Let σ have the constant symbols $c_1, \dots, c_l$. We extend the n -round FO EF game played between the structures $\mathfrak{A}, \mathfrak{B}$ to form the $FO + TC^1$ EF game. Suppose that k rounds have been played so far and $k + l \geq 2$. Let A be the universe of $\mathfrak{A}$ and B the universe of $\mathfrak{B}$. The pebbles in play are $a_1, \dots, a_k \in A$ and $b_1, \dots, b_k \in B$. Label $c_i^{\mathfrak{A}}$ as a_{k+i} and $c_i^{\mathfrak{B}}$ as b_{k+i} . At the start of the $(k + 1)$ th round, instead of playing the normal move, the spoiler has the option to play the TC move, which goes as follows:

TC : The spoiler picks distinct $i, j \in [1, k + l]$. Then, the spoiler chooses either $\mathfrak{A}$ or $\mathfrak{B}$. We detail the move for $\mathfrak{A}$ as the other case is symmetric. The spoiler chooses a set $X \subseteq A$ such that $a_i \in X$ and $a_j \notin X$. The duplicator responds with a set $Y \subseteq B$ such that $b_i \in Y$ and $b_j \notin Y$. The spoiler places a pebble on some $y \in Y$ and $y' \notin Y$ in $\mathfrak{B}$. The duplicator places a pebble on some $x \in X$ and $x' \notin X$ in $\mathfrak{A}$.

An n -round winning strategy for the duplicator is denoted $\mathfrak{A} \equiv_n^{TC} \mathfrak{B}$.

While our move for TC is novel, an EF game for $FO + TC^1$ exists [Grä92]. Grädel gives us the following move for TC :

TC : The spoiler picks distinct $i, j \in [1, k + l]$. Then, the spoiler chooses either $\mathfrak{A}$ or $\mathfrak{B}$. We detail the move for $\mathfrak{A}$ as the other case is symmetric. The spoiler selects a sequence $x_0, x_1, \dots, x_m \in A$ such that $x_0 = a_i$ and $x_m = a_j$. The duplicator selects a sequence $y_0, y_1, \dots, y_{m'} \in B$ with $y_0 = b_i$ and $y_{m'} = b_j$. The spoiler places pebbles on $y_{p'}$ and $y_{p'+1}$ for some p' . The duplicator responds with pebbles on x_p and x_{p+1} for some p .

While Grädel's move requires the selection of an ordered sequence of elements, our move only requires the selection of an unordered set of elements. This is a major improvement as our move requires less structure to be used.

Lemma 6.7. *Let $n \in \mathbb{N}$. There are finitely many formulae in $FO + TC^1[n]$, up to equivalence.*

Theorem 6.8. *Let $\mathfrak{A}$ and $\mathfrak{B}$ be σ -structures, where σ is finite and relational with constant symbols $c_1, \dots, c_l$. Then, for all $k \in \mathbb{N}$, the following are equivalent:*

1. *For all formulae $\Phi \in FO + TC^1[n]$, $\mathfrak{A}, a_1, \dots, a_k \models \Phi$ iff $\mathfrak{B}, b_1, \dots, b_k \models \Phi$.*
2. *$(\mathfrak{A}, a_1, \dots, a_k) \equiv_n^{TC} (\mathfrak{B}, b_1, \dots, b_k)$.*

Proof. We proceed with induction on n . The base case is trivial when $n = 0$ for all $k \in \mathbb{N}$.

Assume the statement is true for some $n \in \mathbb{N}$. Assume that the statement holds for some $n \in \mathbb{N}$. Suppose that k pairs of pebbles have been placed so far. Denote the pebbles on $\mathfrak{A}$ as $a_1, \dots, a_k$ and the pebbles on $\mathfrak{B}$ as $b_1, \dots, b_k$. Label $c_i^{\mathfrak{A}}$ as a_{k+i} and $c_i^{\mathfrak{B}}$ as b_{k+i} . We will prove that $(\mathfrak{A}, a_1, \dots, a_k) \equiv_{n+1}^{\text{TC}} (\mathfrak{B}, b_1, \dots, b_k)$ iff for all $\Phi \in \text{FO} + \text{TC}^1[n+1]$, $\mathfrak{A}, a_1, \dots, a_k \models \Phi \Leftrightarrow \mathfrak{B}, b_1, \dots, b_k \models \Phi$. In particular, we show that $\neg(1) \Leftrightarrow \neg(2)$.

Suppose that there exists $\Phi \in \text{FO} + \text{TC}^1[n+1]$, where $\mathfrak{A}, a_1, \dots, a_k \models \Phi$, but $\mathfrak{B}, b_1, \dots, b_k \not\models \Phi$. If Φ has the form $\exists x\psi$ or $\forall x\psi$, we can create a winning strategy of the spoiler following the proof of Theorem 2.4. We consider the case where Φ has the form $\text{TC}_{x,y}[\varphi(x, y, \bar{z})](x, y)$, where $\text{qr}(\varphi) = n$. There exist distinct i, j such that $\text{TC}_{x,y}[\varphi(x, y, \bar{z})](a_i, a_j)$ holds while $\text{TC}_{x,y}[\varphi(x, y, \bar{z})](b_i, b_j)$ does not. Have the spoiler first play a TC move and pick i, j . Then, have the spoiler choose a $\varphi(x, y, \bar{z})$ -closed set Y in $\mathfrak{B}$ which includes b_i . By definition, this set must not include b_j . The duplicator does the same in $\mathfrak{A}$ by choosing a set X where $a_i \in X$ and $a_j \notin X$. We know that (a_i, a_j) is in the transitive closure of $\varphi(x, y, \bar{z})$, and so X is not $\varphi(x, y, \bar{z})$ -closed. There exists $x \in X$ and $x' \notin X$ where $\varphi(x, x', \bar{z})$ is true. This is the x, x' that the spoiler chooses. Let the duplicator choose $y \in Y$ and $y' \notin Y$. Since Y is $\varphi(x, y, \bar{z})$ -closed, we find that $\varphi(y, y', \bar{z})$ is false. From our inductive hypothesis, the spoiler has a winning strategy for the remaining n rounds as $\varphi(x, x', \bar{z})$ holds and $\varphi(y, y', \bar{z})$ does not.

Conversely, suppose that $(\mathfrak{A}, a_1, \dots, a_k) \not\equiv_{n+1}^{\text{TC}} (\mathfrak{B}, b_1, \dots, b_k)$. If the first move is the move in the FO EF game, the proof is similar to that of Theorem 2.4. Suppose that the spoiler's first move is a TC move where distinct i, j are picked. Without loss of generality, the spoiler chooses $X \subseteq A$ such that $a_i \in X$ and $a_j \notin X$. Let $Y \subseteq B$ be the response by the duplicator with $b_i \in Y$ and $b_j \notin Y$. Now, the spoiler can choose $y \in Y$ and $y' \notin Y$ according to the winning strategy. For all $x \in X$ and $x' \notin X$ that the duplicator chooses, the spoiler has an n -round winning strategy. Thus, there exists a formula $\varphi_{x,x'}^Y$ with quantifier rank n which is false on (x, x') , but true on (y, y') . Then, we define

$$\varphi = \bigvee_{Y \subseteq B} \bigwedge_{\substack{x \in X \\ x' \notin X}} \varphi_{x,x'}^Y.$$

Since there are finitely many formulae in $\text{FO} + \text{TC}^1[n]$, up to equivalence, φ can be rewritten as a formula in $\text{FO} + \text{TC}^1[n]$. We show that $\text{TC}_{x,y}[\varphi(x, y)](a_i, a_j)$ does not hold. From our definition, it suffices to show that X is φ -closed while having $a_i \in X$ and $a_j \notin X$. For any $x \in X$ and $x' \notin X$, we know $\varphi(x, x')$ is false. From the definition of φ -closure, X is φ -closed. We now prove that $\text{TC}_{x,y}[\varphi(x, y)](b_i, b_j)$ holds. For any set $Y \subseteq B$ with $b_i \in Y$ and $b_j \notin Y$, we show that Y is not φ -closed. By construction, we have $y \in Y$ and $y' \notin Y$ such that $\varphi(y, y')$ holds. As $\text{TC}_{x,y}[\varphi(x, y)](x, y)$ has quantifier rank $n+1$, this completes our proof. $\square$

6.2 Extending Modal Logic

We present an extension of modal logic with transitive closure as defined in [DS12].

Definition 6.9. The syntax of *modal logic with transitive closure* (ML + TC) over modalities $\Box_1, \dots, \Box_m$ and atomic propositions P is

$$\varphi := p \mid \neg\varphi \mid \varphi \wedge \varphi \mid \Box_i \mid \Box_i^*,$$

where $p \in P$ and $i \in [1, m]$. Let $M = \langle S, R, V \rangle$ be a Kripke model. Then, let

$$M, s \models \Box_i^* \varphi \text{ iff for all } s' \in S \text{ such that } \text{TC}_{x,y}[x R_i y](s, s'), \text{ we have } M, s' \models \varphi.$$

We define $\Diamond_i^*$ as the dual of $\Box_i^*$. The standard translation can be extended as follows:

$$\begin{aligned} \text{ST}_x(\Box_i^* \varphi) &= \forall y (\text{TC}_{x,y}[x R_i y](x, y) \rightarrow \text{ST}_y(\varphi)), \\ \text{ST}_x(\Diamond_i^* \varphi) &= \exists y (\text{TC}_{x,y}[x R_i y](x, y) \wedge \text{ST}_y(\varphi)). \end{aligned}$$

Example 6.10. We find that $\text{LTL}(\mathbf{X}, \mathbf{Y}) + \text{TC}$ is equivalent to UTL. The operator $\mathbf{F}$ is equivalent to $\mathbf{XX}^*$ and the operator $\mathbf{P}$ is equivalent to $\mathbf{YY}^*$.

We can extend the ML EF game with a move for $\Box_i^*$ to make an ML + TC game. Suppose that the current configuration is (s, s') over Kripke models $M = \langle S, R, V \rangle$ and $M' = \langle S', R', V' \rangle$. The move can be defined as follows:

$\Box_i^*$: The spoiler either chooses $t \in S$ such that $\text{TC}_{x,y}[x R_i y](s, t)$ and the duplicator chooses $t' \in S'$ such that $\text{TC}_{x,y}[x R_i y](s', t')$, or the spoiler chooses $t' \in S'$ such that $\text{TC}_{x,y}[x R_i y](s', t')$ and the duplicator chooses $t \in S$ such that $\text{TC}_{x,y}[x R_i y](s, t)$.

This move does not follow the general philosophy of an EF game, as we are required to compute the transitive closure outside of the game. Instead, we present the following game to turn the computation of the transitive closure into a subgame.

Definition 6.11. Let $M = \langle S, R, V \rangle$ be a Kripke model with distinct $s, t \in S$. We define the *transitive closure check game* for R_i from s to t . The spoiler chooses $X \subseteq S$ such that $s, t \in X$. The duplicator chooses $Y \subseteq S$ such that $s \in Y$ and $t \notin Y$. The spoiler then picks $y \in Y$ and y' such that $y R_i y'$. If there is no such y' , the spoiler loses. The duplicator picks $x \in X$ and x' such that $x R_i x'$. The duplicator wins the transitive closure check game if $x' \in X \Leftrightarrow y' \in Y$. We denote the duplicator winning as $s \equiv^{R_i} t$.

Proposition 6.12. Let $M = \langle S, R, V \rangle$ be a Kripke model. Then, $\text{TC}_{x,y}[x R_i y](s, t)$ holds for distinct $s, t \in S$ iff the spoiler wins the transitive closure check game for R_i from s to t .

Proof. Suppose that $\text{TC}_{x,y}[x R_i y](s, t)$ holds. Then, the spoiler chooses an R_i -closed set X that contains s and t . Let $Y \subseteq S$ with $s \in Y$ and $t \notin Y$ be the duplicator's response. From our assumption, Y must not be R_i -closed. The spoiler chooses $y \in X - \{t\}$ such that there exists $y' \notin Y$ and $y R_i y'$. Now let the duplicator choose $x \in X$. Since X is R_i -closed, any choice by the duplicator x' has $x' \in X$. As $y' \notin Y$, the spoiler wins the transitive closure check game.

Suppose that $\text{TC}_{x,y}[x R_i y](s, t)$ does not hold and the spoiler chooses X such that $s, t \in X$. Let $Y' = \{s' : \text{TC}_{x,y}[x R_i y](s, s')\}$. The duplicator responds with $Y = X \cap Y'$. Note that $t \notin Y'$ so $t \notin Y$. Then, suppose the spoiler chooses $y \in Y$ and y' such that $y R_i y'$. Have the duplicator respond with $x = y$ and $x' = y'$. We now show that $y' \in X \Leftrightarrow y' \in Y$. If $y' \in Y$, then $y' \in X$ by construction. Consider if $y' \in X$. We have $y' \in Y'$ as $\text{TC}_{x,y}[x R_i y](s, y)$ and $y R_i y'$. Thus, $y' \in Y$. The duplicator wins the transitive closure check game. $\square$

Definition 6.13. The $\text{ML} + \text{TC}$ EF game extends Definition 3.9 with the following move:

$\square_i^+$: The spoiler either chooses $t \in S$ or $t' \in S'$. If the spoiler picks $t \in S$, we must have $s \not\equiv^{R_i} t$ or $s = t$. The duplicator then responds with $t' \in S'$ such that $s' \not\equiv^{R_i} t'$ or $s' = t'$. The other case is symmetric. The new configuration is (t, t') .

Example 6.14. Recall that s is the successor relation. Let the current configuration be (x, y) over linear models $\mathfrak{A}, \mathfrak{B}$. The revised **F** move is as follows:

F: The spoiler either chooses $x' \in A$ or $y' \in B$. If the spoiler picks $x' \in A$, we must have $x + 1 \not\equiv^s x'$. The duplicator then responds with $y' \in B$ such that $y + 1 \not\equiv^s y'$. The other case is symmetric. The new configuration is (x', y') .

The original **F** move is clear and easy to use due to the examples we have chosen. Our linear models are usually over the integers or the natural numbers, where the less-than relation is clear. This move is preferred when we do not have access to the less-than relation. In this case, the less-than relation would be defined as the transitive closure of the successor relation. This move makes the computation of the transitive closure an explicit part of the game.

Example 6.15. We finish off with our original inspiration for looking into transitive closure. The following common knowledge move aligns better with the properties that we want:

C_G : Let $\sim_{EG} = \bigcup_{a \in G} \sim_a$. The spoiler either chooses $t \in S$ or $t' \in S'$. If the spoiler picks $t \in S$, we must have $s \not\equiv^{\sim_{EG}} t$. The duplicator then responds with $t' \in S'$ such that $s' \not\equiv^{\sim_{EG}} t'$. The other case is symmetric. The new configuration is (t, t') .

The End

This thesis starts by outlining some known Ehrenfeucht-Fraïssé games. We then highlight how Ehrenfeucht-Fraïssé games can be used to rewrite properties in terms of game semantics. We have introduced new Ehrenfeucht-Fraïssé games for temporal epistemic logics and used them to prove novel expressibility results. In our final chapter, we provide improvements to games from the literature.

While we have reached the end of the thesis, this is not the end of the journey. Several interesting questions and directions have emerged, and we plan to explore some of them.

- Are there temporal epistemic logics where past operators do not increase expressibility?
- EF games are largely unexplored for branching time logics. Our game for CTL^* can be adapted to the branching time logic CTL. Are there EF games for other branching time logics or models of branching time?
- Is there a van Benthem-style characterisation for temporal epistemic logics?
- Is there a sensible measure of the complexity of a given move? One use case would be to compare the complexity of the two TC moves presented.
- Are there fixed-point logic games similar to the $\text{FO} + \text{TC}$ game?
- The transitive closure check game seems to be a special case of a model checking game. As far as we know, there is no such model-checking game.
- Some modal logics are equivalent to 2-variable first-order logic. We have proven this for UTL and $\text{LTL}(\mathbf{F}, \mathbf{P})$. Others, such as EL and $\text{LTL}(\mathbf{X}, \mathbf{Y})$, are not. Can we characterise a necessary and/or sufficient condition for a given modal logic being equivalent to the 2-variable counterpart?

Bibliography

Bibliography

- [Bar77] Jon Barwise. “On Moschovakis closure ordinals”. In: *Journal of Symbolic Logic* 42 (1977), pp. 292–296.
- [Ben76] Johan van Benthem. “Modal Correspondence Theory”. PhD thesis. Mathematisch Instituut & Instituut voor Grondslagenonderzoek, 1976.
- [BK08] Christel Baier and Joost-Pieter Katoen. *Principles of Model Checking*. MIT Press, 2008.
- [BMS98] Alexandru Baltag, Lawrence Mass, and Slawomir Solecki. “The Logic of Public Announcements and Common Knowledge and Private Suspicions”. In: *Proceedings of the 7th Conference on Theoretical Aspects of Rationality and Knowledge* (1998), pp. 43–56.
- [BRV01] Patrick Blackburn, Maarten de Rijke, and Yde Venema. *Modal Logic*. Cambridge Press, 2001.
- [CE12] Bruno Courcelle and Joost Engelfriet. *Graph Structure and Monadic Second-Order Logic, a Language Theoretic Approach*. Cambridge University Press, 2012.
- [DHK07] Hans Ditmarsch, Wiebe Hoek, and Barteld Kooi. *Dynamic Epistemic Logic*. Springer, 2007.
- [DS12] Christian Doczkal and Gert Smolka. “Constructive Completeness for Modal Logic with Transitive Closure”. In: *Certified Programs and Proofs 2012* (2012), pp. 224–239.
- [EH85] E. Allen Emerson and Joseph Y. Halpern. “Decision Procedures and Expressiveness in the Temporal Logic of Branching Time”. In: *Journal of Computer and Systems Science* 30 (1985), pp. 1–24.
- [Ehr61] Andrzej Ehrenfeucht. “An application of games to the completeness problem for formalized theories”. In: *Fundamenta Mathematicae* 49 (1961), pp. 129–141.
- [EVW02] Kousha Etessami, Moshe Y. Vardi, and Thomas Wilke. “First-Order Logic with Two Variables and Unary Temporal Logic”. In: *Information and Computation* (2002), pp. 279–295.

- [EW00] Kousha Etessami and Thomas Wilke. “An Until Hierarchy and Other Applications of an Ehrenfeucht–Fraïssé Game for Temporal Logic”. In: *Information and Computation* 160 (2000), pp. 88–108.
- [FH83] Marco Forti and Furio Honsell. “Set theory with free construction principles”. In: *Annali Scuola Normale Superiore, Pisa, Serie IV, X(3)* (1983).
- [Fra54] Roland Fraïssé. “Sur quelques classifications des systèmes de relations”. PhD thesis. University of Paris, 1954.
- [Fri69] Morris Friedell. “On the structure of shared awareness”. In: *Behavioral Science* 14 (1969), pp. 28–39.
- [Gab81] Dov M. Gabbay. “Expressive Functional Completeness in Tense Logic”. In: *Aspects of Philosophical Logic* 147 (1981), pp. 91–117.
- [Gab89] Dov M. Gabbay. “The Declarative Past and Imperative Future”. In: *Lecture Notes in Computer Science* 398 (1989), pp. 409–448.
- [Gai82] Haim Gaifman. “On Local and Non-Local Properties”. In: *Logic Journal of the IGPL* 7 (1982), pp. 103–124.
- [Grä92] Erich Grädel. “On Transitive Closure Logic”. In: *Lecture Notes in Computer Science* 626 (1992), pp. 146–163.
- [Hin62] Jaakko Hintikka. *Knowledge and Belief: An Introduction to the Logic of the Two Notions*. Cornell University Press, 1962.
- [HM85] Matthew Hennessey and Robin Milner. “Algebraic Laws for Nondeterminism and Concurrency”. In: *Journal of the Association for Computing Machinery* 32 (1985), pp. 137–161.
- [HT87] Thilo Hafer and Wolfgang Thomas. “Computation Tree Logic CTL* and Path Quantifiers in the Monadic Theory of the Binary Tree”. In: *International Colloquium on Automata, Languages, and Programming* (1987), pp. 269–279.
- [HV89] Joseph Y. Halpern and Moshe Y. Vardi. “The Complexity of Reasoning about Knowledge and Time. I. Lower Bounds”. In: *Journal of Computer and System Sciences* 38 (1989), pp. 195–237.
- [Imm82] Neil Immerman. “Upper and Lower Bounds for First Order Expressibility”. In: *Journal of Computer and System Sciences* 25 (1982), pp. 76–98.
- [Imm83] Neil Immerman. “Languages That Capture Complexity Classes”. In: *Proceedings of the fifteenth annual ACM Symposium on Theory of Computing* (1983), pp. 347–354.
- [IV97] Neil Immerman and Moshe Y. Vardi. “Model Checking and Transitive-Closure Logic”. In: *Lecture Notes in Computer Science* 1254 (1997), pp. 291–302.

- [Kam68] Johan Anthony Wilem Kamp. “Tense Logic and the Theory of Linear Order”. PhD thesis. University of California, Los Angeles, 1968.
- [KB04] Barteld Kooi and Johan van Benthem. “Reduction axioms for epistemic actions”. In: *Proceedings Advances in Model Logic* (2004), pp. 197–211.
- [KPV12] Orna Kupferman, Amir Pnueli, and Moshe Y. Vardi. “Once and For All”. In: *Journal of Computer and System Sciences* 78 (2012), pp. 981–996.
- [Kri63] Saul A. Kripke. “Semantical analysis of modal logic”. In: *Zeitschrift für Mathematische Logik und Grundlagen der Mathematik* 9 (1963), pp. 67–96.
- [Lew67] David Lewis. “Conventions of Language”. PhD thesis. Harvard University, 1967.
- [Lib04] Leonid Libkin. *Elements of Finite Model Theory*. Springer, 2004.
- [Mal36] Anatoly Malcev. “Untersuchungen aus dem Gebiete der mathematischen Logik”. In: *Matematicheskii Sbornik* (1936), pp. 323–336.
- [Mar10] David Marker. *Model Theory: An Introduction*. Springer, 2010.
- [MR99] Faron Moller and Alexander Rabinovich. “On the Expressive Power of CTL*”. In: *Proceedings. 14th Symposium on Logic in Computer Science* (1999), pp. 360–368.
- [Ott03] Martin Otto. *Elementary Proof of the van Benthem-Rosen Characterisation Theorem*. 2003.
- [Par81] David Park. “Concurrency and automata on infinite sequences”. In: *Lecture Notes in Computer Science* 104 (1981), pp. 167–183.
- [Pea89] Giuseppe Peano. “Arithmetices principia, nova methodo exposita”. In: *Fratres Bocca* (1889), pp. 83–97.
- [Pri57] Arthur Prior. *Time and Modality*. Oxford University Press, 1957.
- [Rab14] Alexander Rabinovich. “A Proof of Kamp’s Theorem”. In: *Logical Methods in Computer Science* (2014), pp. 1–16.
- [Sch72] Stephen Schiffer. *Meaning*. Oxford University Press, 1972.
- [Sti99] Colin Stirling. “Bisimulation, Modal Logic and Model Checking Games”. In: *Logic Journal of the IGPL* 7 (1999), pp. 103–124.
- [Tur84] György Turán. “On the Definability of Properties of Finite Graphs”. In: *Discrete Mathematics* 49 (1984), pp. 291–302.
- [Vää24] Jouko Väänänen. “Second-order and Higher-order Logic”. In: *The Stanford Encyclopedia of Philosophy*. Stanford University, 2024.
- [Zac25] Richard Zach. *Incompleteness and Computability*. Open Logic Project, 2025.

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